Revisiting DC Optimization: A Unified Notion of Criticality with Optimal Complexity

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Abstract

In this paper, we study a general class of nonconvex DC constrained problems , where the feasible set is possibly nonconvex, and the concave part of the DC objective is allowed to be nonsmooth. The fundamental challenge lies in maintaining feasibility for nonconvex constraints while achieving favorable oracle complexity.

Keywords: Nonsmooth nonconvex optimization, constrained optimization, DC optimization, penalty methods, smoothing approximation, momentum, oracle complexity.

1 INTRODUCTION

In this paper, we study a class of nonsmooth nonconvex constrained optimization problems formulated as

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) := \phi(\boldsymbol{x}) - g(\boldsymbol{x}), \tag{\textbf{P}}$$

where the objective $F = \phi - g$ has a *difference-of-convex* (DC) structure: $\phi : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ and $g : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ are proper closed convex functions, and g is allowed to be nonsmooth, so that F is in general both nonsmooth and nonconvex.

Two settings motivate our study of Problem (**P**). The first is deterministic constrained DC programming, in which both ϕ and g can be evaluated exactly. DC functions form a vector space and constitute a remarkably rich subclass of locally Lipschitz nonconvex functions [14, 20, 41]; the formulation subsumes smooth nonconvex minimization ($g \equiv 0$) as a trivial instance and covers a broad spectrum of classical applications, including nonconvex quadratic programming [2, 3], convex–concave programming [31, 39], kernel selection [4], and bilevel programming [21]. While the unconstrained DC objective is by now well understood, the nonlinear constraints $\boldsymbol{c}_{\mathcal{E}}$ and $\boldsymbol{c}_{\mathcal{I}}$, which arise naturally from structural or resource requirements such as energy budgets in deep network compression [11, 36] and safety constraints in control [30],introduce severe geometric complications of their own.

A second motivation for (**P**) is regularized empirical risk minimization in large-scale machine learning. The DC structure of the objective naturally captures the combination of a data-fitting risk and a nonconvex

regularizer. Specifically, any smooth nonconvex loss can be reformulated as a DC function by adding and subtracting a strongly convex quadratic term. Moreover, popular nonconvex regularizers such as the capped-ℓ_1 penalty [42] and SCAD [15] admit explicit DC decompositions $r = r_{\mathrm{cvx}} - g$. In large-scale training, the data-fitting component in ϕ is a population risk accessible only through stochastic gradients under data sampling, whereas the subtracted component g, which collects the concave part of the regularizer, remains a fixed deterministic function.

To ground our framework, we provide two concrete motivating examples that can be formulated as instances of Problem (**P**):

1.1 Motivating Examples

1. **Deterministic DC objective.** Losses given by a pointwise minimum of convex functions are intrinsically DC, via the identity $\min_{k\le K} a_k = \sum_{k=1}^{K} a_k - \max_{k\le K} \sum_{j\neq k} a_j$ for convex $a_1, \dots, a_K$. For instance, in piecewise-affine (clusterwise) regression, each observation is explained by the best of K candidate affine models,

$$\min_{\boldsymbol{w}_1,\dots,\boldsymbol{w}_K} \sum_{i=1}^{N} \min_{1\le k\le K} \big|y_i - \langle \boldsymbol{w}_k, \boldsymbol{x}_i\rangle\big|,$$

   and applying the identity samplewise yields $F = \phi - g$ with $\phi(\boldsymbol{w}) = \sum_{i,k} |y_i - \langle \boldsymbol{w}_k, \boldsymbol{x}_i\rangle|$ and $g(\boldsymbol{w}) = \sum_i \max_k \sum_{j\neq k} |y_i - \langle \boldsymbol{w}_j, \boldsymbol{x}_i\rangle|$. Both components are convex, piecewise affine, and nonsmooth, with subgradients available in closed form.

2. **Stochastic nonconvex regularized ERM.** Consider a convex data-fitting loss ℓ coupled with a capped-ℓ_1 penalty:

$$\min_{\boldsymbol{x}} \underbrace{\mathbb{E}_\xi[\ell(\boldsymbol{x},\xi)] + \lambda\|\boldsymbol{x}\|_1}_{\phi(\boldsymbol{x})} - \underbrace{\lambda \sum_i \max\{|x_i| - \theta, 0\}}_{g(\boldsymbol{x})}.$$

   Here, the data-fitting component ϕ is a population risk evaluated solely via stochastic gradients, whereas the subtracted penalty g is a fixed, deterministic, and nonsmooth convex function whose subgradients can be computed in closed form.

1.2 Related Work

Non-asymptotic rates for DC optimization. Removing differentiability of the subtracted component g has been a central theme in DC programming. Recent non-asymptotic analyses have made this theme quantitative, but their guarantees are expressed through different decomposition-dependent certificates. For DCA and CCCP-type methods, rates are usually stated in terms of objective decrease, step lengths, or critical-distance proxies under additional smoothness or curvature assumptions [1, 40]. Projection-free DC methods over compact convex domains instead use Frank–Wolfe-type linear minimization gaps [19, 26]. Stochastic DC methods use critical-distance criteria or one-sided gap functions, often with a smooth or variance-reduced stochastic component [22, 29, 35]. These certificates are value-typed or algorithm-dependent, while the exact DC residual

$$\operatorname{dist}\big(\mathbf{0}, \partial\phi(\boldsymbol{x}) - \partial g(\boldsymbol{x})\big)$$

is norm-typed and same-point. Passing from a gap bound to a residual bound therefore changes units, typically through a square-root relation. Existing finite-time results hence do not settle whether exact same-point ϵ-criticality is itself attainable for nonsmooth DC pairs. The first question is therefore:

Is ϵ-criticality finite-time attainable in DC optimization? If not, does a unified single notion exist that relaxes it while remaining both finite-time attainable and sound?

Non-asymptotic rates for unconstrained nonsmooth nonconvex optimization. For general Lipschitz objectives, ϵ-Clarke stationarity is unattainable in finite time [41], while the structure-blind Goldstein (δ, ϵ)-stationarity relaxation admits $O(\delta^{-1}\epsilon^{-3})$ guarantees [14, 41], one ϵ-factor away from the $O(\epsilon^{-2})$ smooth benchmark. These results treat the objective as a black box.

The second question is therefore:

Under an finite-sum attainable targat of DC criticality, can the DC structure be provably exploited to match the $O(\epsilon^{-2})$ rate of unconstrained smooth nonconvex optimization?

Constrained nonconvex optimization. The rates above are established either without constraints or over simple convex compact sets accessed through projections or linear minimization oracles [19, 26]; general nonlinear constraints are handled by penalty or augmented-Lagrangian schemes whose inner solves make the methods double-loop and couple the inner accuracy to the outer complexity [18, 24, 34, 38]. The third question is:

can a single-loop method, free of inner solvers and backtracking, retain the optimal $O(\epsilon^{-3})$ complexity under general nonlinear, possibly nonconvex, constraints? (ALM)

1.3 Contributions

We tackle the three questions for Problem (**P**) when both components of the DC decomposition may be nonsmooth. This class is rich enough to cover constrained DC programs and nonconvex-regularized empirical risk minimization. At first sight, the DC structure makes the search for stationary points easier on both sides of the oracle–target ledger: the natural target, DC criticality $\mathbf{0} \in \partial\phi(\boldsymbol{x}) - \partial g(\boldsymbol{x})$, is strictly weaker than Clarke stationarity, and the natural oracle, returning values and subgradients of ϕ and g separately, is strictly stronger than a Clarke subgradient oracle. Surprisingly, even with a weaker target and a stronger oracle, finding an ϵ-critical point, i.e., a point $\bar{\boldsymbol{x}}$ with $\operatorname{dist}(\mathbf{0}, \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})) \le \epsilon$, is intractable. In other words, no algorithm interacting with the component-wise DC oracle can guarantee an ϵ-critical point among its queried iterates within a finite number of oracle calls.This intractability suggests that, to obtain meaningful non-asymptotic results for DC problems, we need to refine the notion of criticality. We then introduce such a unified notion and base our analysis on it, leading to the following main contributions:

- We show that a traditional ϵ-critical point cannot be obtained in finite time, even under the strong component-wise DC oracle, returning values and subgradients of ϕ and g separately.
- We study $(\delta_\phi, \delta_g, \epsilon)$-critical points (Definition 3.1), which unify existing relaxations of DC criticality.For smooth DC functions, this notion reduces to the usual ϵ-stationarity by choosing $\delta_\phi = O(\epsilon^2/L_\phi)$ and $\delta_g = O(\epsilon^2/L_g)$.
- For unconstrained DC optimization, we show that proximal DCA returns a $(0, \delta_g, \epsilon)$-critical point within $\mathcal{O}(\max\{\epsilon^{-2}, \delta_g^{-1}\})$ subproblem solves. When g is additionally L_g-smooth, one may take $\delta_g = \mathcal{O}(\epsilon^2/L_g)$, so that the two-sided certificate collapses to the standard $\mathcal{O}(\epsilon^{-2})$ guarantee for ϵ-stationarity. Moreover, composing this certificate with an inner solver of per-subproblem cost $T_{\mathrm{sub}}(\omega)$ at accuracy

$$\omega = \mathcal{O}\left(\min\left\{\eta\epsilon^2,\ \frac{\delta_\phi^2}{\eta G_\phi^2},\ \frac{\delta_g^2}{\eta G_g^2}\right\}\right)$$

  yields a $(\delta_\phi, \delta_g, \mathcal{O}(\epsilon))$-critical point with total inner-iteration complexity

$$\mathcal{O}\left(\max\left\{\frac{\Delta}{\eta\epsilon^2},\ \frac{\eta\Delta G_g^2}{\delta_g^2}\right\} \cdot T_{\mathrm{sub}}(\omega)\right),$$

  by the same inexact bookkeeping as in the bare-DCA case (Theorem 4.2); here $T_{\mathrm{sub}}(\omega)$ is the number of inner iterations to solve one proximal subproblem to accuracy ω.

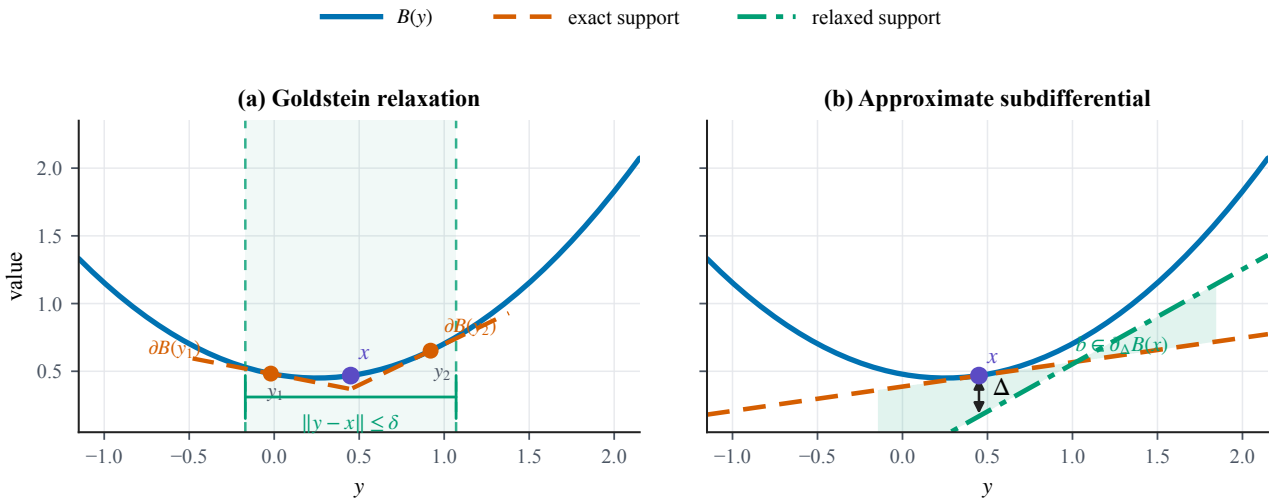


Figure 1: The Goldstein subdifferential ∂_δ^G and the δ-subdifferential ∂_δ relax exact stationarity along different axes. The former convexifies exact subgradients collected over the ball $\mathcal{B}_\delta(\bar{\boldsymbol{x}})$ (a spatial enlargement), while the latter fixes the base point and tolerates a Fenchel–Young gap of at most δ in the supporting inequality (a value relaxation).

- We extend the criticality notion to the constrained problem (**P**) and propose a single-loop method attaining a $(\delta_\phi, \delta_g, \epsilon)$-critical point in $O(\epsilon^{-2})$ iterations under general nonlinear, possibly nonconvex, constraints (see Section 4), matching the unconstrained rate above despite nonsmoothness in both components.

1.4 Notations

Throughout, $\|\cdot\|$ denotes the Euclidean norm and $\langle\cdot,\cdot\rangle$ the standard inner product on $\mathbb{R}^n$. For a proper closed convex function h, $\partial h(\boldsymbol{x})$ denotes its convex subdifferential, $\partial_\delta h(\boldsymbol{x})$ its δ-subdifferential, and h^* its Fenchel conjugate; for a locally Lipschitz F, $\partial_C F(\boldsymbol{x})$ denotes the Clarke subdifferential. We write $\operatorname{dist}(A,B) := \inf_{\boldsymbol{a}\in A,\,\boldsymbol{b}\in B} \|\boldsymbol{a}-\boldsymbol{b}\|$ for nonempty sets $A, B \subseteq \mathbb{R}^n$, and $\operatorname{dist}(\boldsymbol{v}, A) := \operatorname{dist}(\{\boldsymbol{v}\}, A)$. For two sets $A, B \subseteq \mathbb{R}^n$, we use the Minkowski difference $A - B$, also called the pairwise difference set, defined by

$$A - B := \{\boldsymbol{a} - \boldsymbol{b} : \boldsymbol{a} \in A,\ \boldsymbol{b} \in B\}.$$

Bold lowercase letters denote vectors and $\mathbf{0}$ the zero vector.

2 Preliminaries

We introduce the definitions of the Clarke subdifferential and various approximate Clarke stationarity points for general locally Lipschitz functions.

Let $F : \mathbb{R}^d \to \mathbb{R}$ be locally Lipschitz. The Clarke subdifferential of F at $\boldsymbol{x}$ is denoted by $\partial_C F(\boldsymbol{x})$ [12]. A point $\boldsymbol{x}$ is called an ε-Clarke stationary point if

$$\operatorname{dist}\big(\mathbf{0}, \partial_C F(\boldsymbol{x})\big) \le \varepsilon.$$

For general nonsmooth nonconvex functions, such a same-point certificate is not a finite-time target in the worst case. This motivates the Goldstein-type relaxation below [16, 41].

3 Convergence Criteria for Nonsmooth DC Optimization

3.1 Criticality and ε-criticality

A point $\bar{\boldsymbol{x}}$ is called a *critical point* of Problem (**P**) if

$$\mathbf{0} \in \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}}). \tag{3.1}$$

Equivalently, there exist $\boldsymbol{u} \in \partial\phi(\bar{\boldsymbol{x}})$ and $\boldsymbol{v} \in \partial g(\bar{\boldsymbol{x}})$ such that $\boldsymbol{u} = \boldsymbol{v}$. When both convex components are finite and locally Lipschitz around $\bar{\boldsymbol{x}}$, the Clarke calculus gives $\partial_C F(\bar{\boldsymbol{x}}) \subseteq \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})$. Hence, any local minimizer $\bar{\boldsymbol{x}}$ of $F = \phi - g$ satisfies

$$\mathbf{0} \in \partial_C F(\bar{\boldsymbol{x}}) \subseteq \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}}).$$

Exact same-point DC criticality is decomposition-dependent and may be weaker than Clarke stationarity of the difference function. For example, let

$$\phi(x) = |x|, \qquad g(x) = |x| + x.$$

Then $\Psi = \phi - g = -x$, and hence

$$\partial_C \Psi(0) = \{-1\},$$

so $0 \notin \partial_C \Psi(0)$. However,

$$\partial\phi(0) = [-1, 1], \qquad \partial g(0) = [0, 2],$$

and therefore

$$0 \in \partial\phi(0) - \partial g(0).$$

Thus $x = 0$ is DC-critical for this particular decomposition even though it is not Clarke-stationary for Ψ. This example shows that a finite-time lower bound for exact DC criticality cannot be obtained by directly invoking lower bounds for pointwise Clarke stationarity of Ψ. The lower-bound construction must instead be carried out at the level of the DC pair (ϕ, g) and the residual

$$\operatorname{dist}\big(0, \partial\phi(x) - \partial g(x)\big).$$

Given $\varepsilon > 0$, a natural finite-time goal is the ε-criticality condition:

$$\operatorname{dist}(\mathbf{0}, \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})) \le \varepsilon. \tag{3.2}$$

This equivalently requires the existence of $\boldsymbol{u} \in \partial\phi(\bar{\boldsymbol{x}})$ and $\boldsymbol{v} \in \partial g(\bar{\boldsymbol{x}})$ such that $\|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon$.

For comparison, the ε-stationarity condition for a locally Lipschitz nonsmooth nonconvex objective F is $\operatorname{dist}\big(\mathbf{0}, \partial_C F(\bar{\boldsymbol{x}})\big) \le \varepsilon$. Since $\partial_C F(\bar{\boldsymbol{x}}) \subseteq \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})$, we have

$$\operatorname{dist}(\mathbf{0}, \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})) \le \operatorname{dist}\big(\mathbf{0}, \partial_C F(\bar{\boldsymbol{x}})\big).$$

Thus, (3.2) serves as a decomposition-dependent DC analogue of Clarke ε-stationarity, and it ism weaker.

Zhang et al. [41] established that for a broad class of Lipschitz nonsmooth nonconvex functions, no finite-time first-order algorithm can guarantee finding a point satisfying $\operatorname{dist}\big(\mathbf{0}, \partial_C F(\boldsymbol{x})\big) \le \varepsilon$. For DC optimization, this limitation need to be re-evaluated for two reasons:

(Q1) *The DC oracle is stronger.* A DCA-type method may query $\partial\phi$ and ∂g *separately*, and may even observe $\phi(x)$ and $g(x)$ individually, whereas the obstruction of Zhang et al. [41, Theorem 5] grants only one generalized gradient of the whole objective. Since more information can only make the problem easier, the lower bound is not inherited automatically.

(Q2) *The DC measure is weaker.* The DC calculus gives $\partial_C(\phi - g)(x) \subseteq \partial\phi(x) - \partial g(x)$, hence

$$\operatorname{dist}\big(0, \partial\phi(x) - \partial g(x)\big) \le \operatorname{dist}\big(0, \partial_C(\phi - g)(x)\big).$$

Thus DC criticality is *easier* to satisfy than Clarke stationarity, so proving that even this weaker target is unattainable is a strictly stronger statement than Zhang et al. [41, Theorem 5], not a corollary of it. A construction whose DC residual and Clarke residual coincide everywhere – such as one built purely from disjoint component kinks – proves the Clarke-level statement but never actually tests (Q2), since the DC target set is never larger than the Clarke target set on such an instance. The construction below is designed so that (Q2) is tested directly, at an explicit point where the two residuals differ.

The relevant question is whether finite-time ε-criticality can be achieved under this stronger DC oracle. Unfornunately, the answer is negative, as we show next. We settle both on a single explicit instance which is different from the hard instance constructed in [41].

We formulate the lower bound under the setting where the pair (ϕ, g) is explicitly provided as problem data, granting the algorithm separate access to both convex components. Throughout this paper, we focus on the following subset of Lipschitz functions. For $\Delta, L, G > 0$ and a fixed initial point $\boldsymbol{x}^0$, we define:

$$\begin{aligned}\mathcal{D}(\Delta, L, G) := \Big\{ (\phi, g) : \phi, g \text{ are proper, closed, convex, and finite-valued,} \\ F := \phi - g \text{ is } L\text{-Lipschitz and Hadamard directionally differentiable,} \\ F(\boldsymbol{x}^0) - \inf_{\boldsymbol{x}} F(\boldsymbol{x}) \le \Delta, \\ \sup_{\boldsymbol{x}} \sup_{\boldsymbol{u}\in\partial\phi(\boldsymbol{x}),\, \boldsymbol{v}\in\partial g(\boldsymbol{x})} \max\{\|\boldsymbol{u}\|, \|\boldsymbol{v}\|\} \le G \Big\}.\end{aligned}$$

We then consider the *strong* DC first-order oracle, which returns both component values in addition to both component subgradients:

$$\mathcal{O}_{\mathrm{DC}}(\boldsymbol{x}) = \big(\phi(\boldsymbol{x}),\, g(\boldsymbol{x}),\, \boldsymbol{u},\, \boldsymbol{v}\big), \qquad \boldsymbol{u} \in \partial\phi(\boldsymbol{x}), \quad \boldsymbol{v} \in \partial g(\boldsymbol{x}).$$

This dominates the weaker oracle $\big(F(\boldsymbol{x}), \boldsymbol{u}, \boldsymbol{v}\big)$ used above, since $F(\boldsymbol{x}) = \phi(\boldsymbol{x}) - g(\boldsymbol{x})$ is recoverable from it; a lower bound against $\mathcal{O}_{\mathrm{DC}}$ is therefore the stronger statement required by (Q1). We make this claim precise as follows.

Theorem 3.1. *Fix $\Delta, L, G > 0$. There exists an absolute constant $c > 0$ such that for every query budget $K \le cG/L$, every $\varepsilon \in (0, L)$, and every algorithm $\mathcal{A}$ interacting with the oracle $\mathcal{O}_{\mathrm{DC}}$, there exists a pair $(\phi, g) \in \mathcal{D}(\Delta, L, G)$ such that, with probability at least $1/2$, none of the first K points $\{\boldsymbol{x}^k\}_{k=1}^K$ queried by $\mathcal{A}$ is ε-critical:*

$$\operatorname{dist}\big(\mathbf{0},\, \partial\phi(\boldsymbol{x}^k) - \partial g(\boldsymbol{x}^k)\big) > \varepsilon, \qquad k = 1, \dots, K.$$

Moreover, the constructed pair contains a point s with

$$\mathbf{0} \in \partial\phi(s) - \partial g(s), \qquad \mathbf{0} \notin \partial_C(\phi - g)(s).$$

Remark 3.1. *Equivalently, for every K the same construction gives a hard pair in $\mathcal{D}(\Delta, L, G_K)$ with $G_K = O(KL)$: each hidden background kink forces a slope jump in whichever convex component carries it, and monotonicity of a one-dimensional convex subgradient means these jumps accumulate rather than cancel. This K-dependence is specific to the one-dimensional construction; it is not present in the underlying Clarke-level obstruction of Zhang et al. [41], which does not need to keep two components separately bounded.*

The hard instance has two layers, detailed in Appendix A.

Proof sketch. We first construct a DC pair (ϕ_0, g_0) by decompositing the resisting-oracle construction in Zhang et al. [41].

A common-kink layer $c_s(x) = M|x - s| + \beta - Ms$ is then added identically to both components, $\phi_s = \phi_0 + c_s$ and $g_s = g_0 + c_s$. Because the same term is added to both sides, $F = \phi_s - g_s = \phi_0 - g_0$ is completely unaffected by s: its Lipschitz constant, its optimality gap, and its Clarke subdifferential everywhere are independent of s. Choosing the intercept $\beta - Ms$ so that c_s agrees with a fixed, s-independent affine function at every point to the

left of s makes the entire strong-oracle transcript – both component values and both component subgradients – identical at every queried point, for every admissible hidden location s beyond the largest queried point. At $x = s$ itself, the common kink enlarges $\partial\phi_s(s) - \partial g_s(s)$ into an interval containing zero while leaving $\partial_C(\phi_s - g_s)(s)$ unchanged, producing the point promised in the theorem.

A two-instance argument over locations $s \neq s'$, both consistent with the same K-query transcript, then shows that no algorithm can guarantee success against both simultaneously, giving the stated probability bound. □

3.2 A Unified Notion of Approximate Criticality

In this section, we propose a unified notion to measure approximate criticality. We first recall the standard δ-subdifferential of a proper closed convex function.

If $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a convex function and $\delta > 0$, then for $\bar{\boldsymbol{x}} \in \operatorname{dom} h$, the δ-subdifferential of h at $\bar{\boldsymbol{x}}$ is defined by

$$\partial_\delta h(\bar{\boldsymbol{x}}) := \{\boldsymbol{v} \in \mathbb{R}^n : h(\boldsymbol{z}) \geq h(\bar{\boldsymbol{x}}) + \langle \boldsymbol{v}, \boldsymbol{z} - \bar{\boldsymbol{x}} \rangle - \delta, \quad \forall \boldsymbol{z} \in \mathbb{R}^n\}.$$

When $\delta = 0$, it reduces to the standard convex subdifferential. Figure 2 illustrates that δ acts as a vertical relaxation of the supporting affine function in the geometric interpretation of the δ-subgradient.

We first record the Fenchel-gap characterization of the δ-subdifferential below.

Lemma 3.1. *Let $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex, let $\bar{\boldsymbol{x}} \in \operatorname{dom} h$, and let $\boldsymbol{v} \in \mathbb{R}^n$, $\delta \geq 0$. Then*

$$\boldsymbol{v} \in \partial_\delta h(\bar{\boldsymbol{x}}) \quad \Longleftrightarrow \quad \Phi_h(\bar{\boldsymbol{x}}, \boldsymbol{v}) \leq \delta.$$

Moreover, $\Phi_h(\bar{\boldsymbol{x}}, \boldsymbol{v}) \geq 0$ for every $\boldsymbol{v} \in \mathbb{R}^n$, and $\boldsymbol{v} \in \partial h(\bar{\boldsymbol{x}})$ if and only if $\Phi_h(\bar{\boldsymbol{x}}, \boldsymbol{v}) = 0$.

Proof. By definition, $\boldsymbol{v} \in \partial_\delta h(\bar{\boldsymbol{x}})$ if and only if

$$h(\boldsymbol{z}) \;\geq\; h(\bar{\boldsymbol{x}}) + \langle \boldsymbol{v}, \boldsymbol{z} - \bar{\boldsymbol{x}} \rangle - \delta \qquad \text{for all } \boldsymbol{z} \in \mathbb{R}^n, \tag{3.3}$$

which, upon rearrangement, is equivalent to

$$\langle \boldsymbol{v}, \boldsymbol{z} \rangle - h(\boldsymbol{z}) \;\leq\; \langle \boldsymbol{v}, \bar{\boldsymbol{x}} \rangle - h(\bar{\boldsymbol{x}}) + \delta \qquad \text{for all } \boldsymbol{z} \in \mathbb{R}^n.$$

Taking the supremum over $\boldsymbol{z} \in \mathbb{R}^n$ on the left-hand side yields $h^*(\boldsymbol{v}) \leq \langle \boldsymbol{v}, \bar{\boldsymbol{x}} \rangle - h(\bar{\boldsymbol{x}}) + \delta$, that is, $\Phi_h(\bar{\boldsymbol{x}}, \boldsymbol{v}) = h(\bar{\boldsymbol{x}}) + h^*(\boldsymbol{v}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}} \rangle \leq \delta$. Conversely, if $\Phi_h(\bar{\boldsymbol{x}}, \boldsymbol{v}) \leq \delta$, then for every $\boldsymbol{z} \in \mathbb{R}^n$,

$$\langle \boldsymbol{v}, \boldsymbol{z} \rangle - h(\boldsymbol{z}) \;\leq\; h^*(\boldsymbol{v}) \;\leq\; \langle \boldsymbol{v}, \bar{\boldsymbol{x}} \rangle - h(\bar{\boldsymbol{x}}) + \delta,$$

and rearranging recovers (3.3). The inequality $\Phi_h(\bar{\boldsymbol{x}}, \boldsymbol{v}) \geq 0$ is the Fenchel–Young inequality, and the case $\delta = 0$ gives the equality characterization of $\partial h(\bar{\boldsymbol{x}})$. □

Given the limitations established in Theorem 3.1, we relax the standard notion of criticality. By seeking a point in the proximity of an ε-critical point rather than an exact one, we are motivated to adopt the following unified definition of a $(\delta_\phi, \delta_g, \varepsilon)$-critical point.

Definition 3.1 ($(\delta_\phi, \delta_g, \varepsilon)$-critical point)**.** *Given $\delta_\phi, \delta_g \geq 0$ and $\varepsilon > 0$, a point $\bar{\boldsymbol{x}} \in \operatorname{dom} \phi \cap \operatorname{dom} g$ is called a $(\delta_\phi, \delta_g, \varepsilon)$-critical point of Problem* (**P**) *if*

$$\operatorname{dist}\left(\partial_{\delta_\phi} \phi(\bar{\boldsymbol{x}}), \partial_{\delta_g} g(\bar{\boldsymbol{x}})\right) \leq \varepsilon.$$

Based on this definition, we introduce the following gap functions to quantify the deviation from exact DC criticality:

$$\Phi_\phi(\boldsymbol{x}, \boldsymbol{u}) := \phi(\boldsymbol{x}) + \phi^*(\boldsymbol{u}) - \langle \boldsymbol{u}, \boldsymbol{x} \rangle,$$
$$\Phi_g(\boldsymbol{x}, \boldsymbol{v}) := g(\boldsymbol{x}) + g^*(\boldsymbol{v}) - \langle \boldsymbol{v}, \boldsymbol{x} \rangle.$$

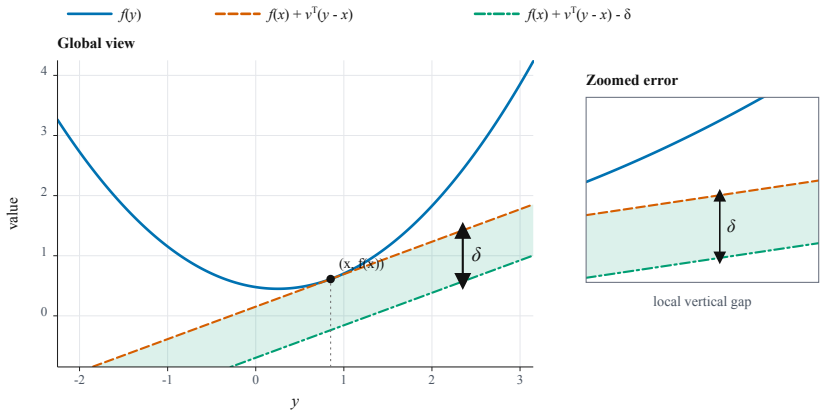


Figure 2: Geometric interpretation of the δ-subgradient. The green affine function is the ordinary supporting affine function shifted downward by δ.

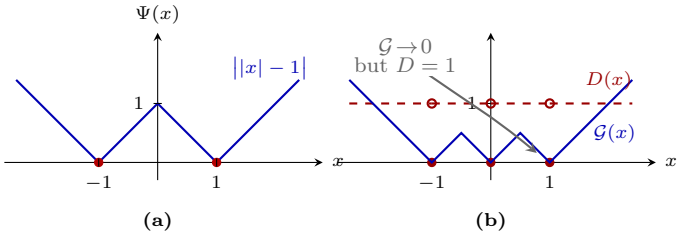


Figure 3: Exact critical-distance residual versus Fenchel-gap behavior in the one-dimensional example.

Observe that exact criticality is equivalent to the existence of a single dual vector $\boldsymbol{v}$ such that both gap functions vanish, i.e., $\exists \boldsymbol{v}$ such that $\Phi_\phi(\boldsymbol{x}, \boldsymbol{v}) = 0$ and $\Phi_g(\boldsymbol{x}, \boldsymbol{v}) = 0$. In contrast, the $(\delta_\phi, \delta_g, \varepsilon)$-critical point in Definition 3.1 allows for different approximate subgradients on the two convex components, alongside a bounded mismatch between these dual vectors:

$$\exists \boldsymbol{u}, \boldsymbol{v}, \qquad \Phi_\phi(\boldsymbol{x}, \boldsymbol{u}) \le \delta_\phi, \quad \Phi_g(\boldsymbol{x}, \boldsymbol{v}) \le \delta_g, \quad \|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon.$$

When $\delta_\phi = \delta_g = 0$, the $(0, 0, \varepsilon)$-critical point reduces to the classical ε-critical point. In fact, the $(\delta_\phi, \delta_g, \varepsilon)$-criticality serves as a more reliable diagnostic metric for approximate criticality. For nonsmooth DC functions, classical point-based stationarity residuals can remain stubbornly large ($\mathcal{O}(1)$) near a nonsmooth minimizer, whereas our proposed Fenchel-gap metric smoothly converges to zero. The following one-dimensional, closed-form example makes this diagnostic obstruction explicit.

Example 3.1 (Exact vs. Fenchel-gap certificate)**.** *Consider*

$$\Psi(x) = \big||x| - 1\big| = \phi(x) - g(x),$$

where $\phi(x) = 3 + 4(|x| - 1)_+$ *and* $g(x) = |x + 1| + |x| + |x - 1|$. *Both* ϕ *and* g *are convex, and* g *is nonsmooth at the global minimizers* $x = \pm 1$. *The exact same-point critical-distance residual is*

$$D(x) := \operatorname{dist}\big(0, \partial\phi(x) - \partial g(x)\big) = \begin{cases} 0, & x \in \{-1, 0, 1\}, \\ 1, & \textit{otherwise.} \end{cases}$$

Hence, even when x is arbitrarily close to the nonsmooth minimizer $x = 1$, the exact critical distance remains $D(x) = 1$.

For comparison, specialize the two-sided Fenchel certificate to the common-certificate scalar gap

$$\mathcal{G}(x) := \inf_{b\in\mathbb{R}} \{\Phi_\phi(x, b) + \Phi_g(x, b)\}.$$

This corresponds to the case $\boldsymbol{u} = \boldsymbol{v} = b$: the same scalar b must act as an approximate subgradient certificate for both convex components at the same point. A direct conjugate calculation gives

$$\phi^*(b) = \begin{cases} |b| - 3, & |b| \le 4, \\ +\infty, & |b| > 4, \end{cases}$$

$$g^*(b) = \begin{cases} -2, & |b| \le 1, \\ |b| - 3, & 1 \le |b| \le 3, \\ +\infty, & |b| > 3. \end{cases}$$

Consequently,

$$\mathcal{G}(x) = \begin{cases} |x|, & 0 \le |x| \le \frac{1}{2}, \\ 1 - |x|, & \frac{1}{2} \le |x| \le 1, \\ |x| - 1, & |x| \ge 1. \end{cases}$$

In particular, $\mathcal{G}(x) \to 0$ as $x \to \pm 1$, while $D(x) = 1$ for every $x \notin \{-1, 0, 1\}$. Equivalently, near $x = 1$, choosing $b = 1 \in \partial g(x)$ for $0 < x < 1$ gives

$$\Phi_g(x, 1) = 0, \qquad \Phi_\phi(x, 1) = 1 - x \to 0 \ \text{ as } x \uparrow 1,$$

and choosing $b = 3 \in \partial g(x)$ for $x > 1$ gives

$$\Phi_g(x, 3) = 0, \qquad \Phi_\phi(x, 3) = x - 1 \to 0 \ \text{ as } x \downarrow 1.$$

This is the behavior shown in Figure 3: the Fenchel gap accurately detects convergence toward a nonsmooth minimizer, whereas the exact critical distance remains an $O(1)$ step function around it.

Example 3.1 shows that the $(\delta_\phi, \delta_g, \varepsilon)$-critical point repairs the diagnostic failure of the exact residual, but a diagnostic that is easy to satisfy is only useful if it is also sound: an overly permissive relaxation could remain satisfiable at points bearing no relation to criticality at all. We close this section by ruling out this concern as follows. The next proposition shows that Definition 3.1 degenerates correctly, in the sense that driving all three tolerances to zero forces exact DC criticality at the same point. This properpsition depends on the monotonicity of the δ-subdifferential in δ together with lower semicontinuity of ϕ, ϕ^*, g, g^*;

Proposition 3.1. *Suppose ϕ and g are G-Lipschitz on $\mathbb{R}^n$. If $\bar{\boldsymbol{x}}$ is a $(\delta_\phi^{(k)}, \delta_g^{(k)}, \varepsilon^{(k)})$-critical point of Problem (**P**) for every $k \in \mathbb{N}$, with $\delta_\phi^{(k)}, \delta_g^{(k)}, \varepsilon^{(k)} \to 0$ as $k \to \infty$, then $\bar{\boldsymbol{x}}$ is a critical point of Problem (**P**), i.e., $\mathbf{0} \in \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})$.*

Proof. For each k, choose $\boldsymbol{u}_k \in \partial_{\delta_\phi^{(k)}}\phi(\bar{\boldsymbol{x}})$ and $\boldsymbol{v}_k \in \partial_{\delta_g^{(k)}} g(\bar{\boldsymbol{x}})$ with $\|\boldsymbol{u}_k - \boldsymbol{v}_k\| \le \varepsilon^{(k)}$. Since ϕ is G-Lipschitz on all of $\mathbb{R}^n$, every element of $\partial_\delta\phi(\bar{\boldsymbol{x}})$ has norm at most G, regardless of δ: for $\boldsymbol{v} \in \partial_\delta\phi(\bar{\boldsymbol{x}})$, the defining inequality $\phi(\bar{\boldsymbol{x}}) + G\|\boldsymbol{z} - \bar{\boldsymbol{x}}\| \ge \phi(\boldsymbol{z}) \ge \phi(\bar{\boldsymbol{x}}) + \langle \boldsymbol{v}, \boldsymbol{z} - \bar{\boldsymbol{x}}\rangle - \delta$ along $\boldsymbol{z} - \bar{\boldsymbol{x}} = t\boldsymbol{v}/\|\boldsymbol{v}\|$ forces $\|\boldsymbol{v}\| \le G + \delta/t$ for every $t > 0$, hence $\|\boldsymbol{v}\| \le G$ as $t \to \infty$. The same bound holds for $\partial_\delta g(\bar{\boldsymbol{x}})$. Thus $\{\boldsymbol{u}_k\}$ and $\{\boldsymbol{v}_k\}$ are bounded, and we may pass to a subsequence along which $\boldsymbol{u}_k \to \boldsymbol{u}^\star$ and $\boldsymbol{v}_k \to \boldsymbol{v}^\star$; since $\|\boldsymbol{u}_k - \boldsymbol{v}_k\| \to 0$, we get $\boldsymbol{u}^\star = \boldsymbol{v}^\star =: \boldsymbol{w}$.

It remains to show $\boldsymbol{w} \in \partial\phi(\bar{\boldsymbol{x}})$; the argument for $\boldsymbol{w} \in \partial g(\bar{\boldsymbol{x}})$ is identical. Since ϕ and ϕ^* are proper, closed, and convex, both are lower semicontinuous, so

$$\Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{w}) = \phi(\bar{\boldsymbol{x}}) + \phi^*(\boldsymbol{w}) - \langle \boldsymbol{w}, \bar{\boldsymbol{x}}\rangle \le \liminf_{k\to\infty} \Big[\phi(\bar{\boldsymbol{x}}) + \phi^*(\boldsymbol{u}_k) - \langle \boldsymbol{u}_k, \bar{\boldsymbol{x}}\rangle\Big] = \liminf_{k\to\infty} \Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{u}_k) \le \liminf_{k\to\infty} \delta_\phi^{(k)} = 0.$$

The Fenchel–Young inequality gives $\Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{w}) \geq 0$ unconditionally, so $\Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{w}) = 0$, i.e., $\boldsymbol{w} \in \partial\phi(\bar{\boldsymbol{x}})$. Hence $\boldsymbol{w} \in \partial\phi(\bar{\boldsymbol{x}}) \cap \partial g(\bar{\boldsymbol{x}})$, and $\mathbf{0} = \boldsymbol{w} - \boldsymbol{w} \in \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})$. □

3.3 Relationship between Relaxed DC Criticality Notions

We show that the $(\delta_\phi, \delta_g, \varepsilon)$-critical point in Definition 3.1 is a unifying value-relaxed formulation: every previously proposed relaxed DC criticality notion for Problem (**P**) embeds into Definition 3.1 for a suitable choice of $(\delta_\phi, \delta_g, \varepsilon)$, and this embedding admits a sharper, second-order accurate rate whenever the underlying component is smooth.

We recall three such notions, in increasing order of the spatial freedom they allow.

Definition 3.2 (ε-critical point [27, 33])**.** *Given $\varepsilon > 0$, a point $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ is an ε-critical point of Problem* (**P**) *if there exist $\bar{\boldsymbol{y}} \in \mathbb{R}^n$ and $\bar{\boldsymbol{u}} \in \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{y}})$ such that*

$$\max\{\|\bar{\boldsymbol{u}}\|, \|\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}\|\} \leq \varepsilon.$$

Definition 3.3 (Nearly ε-critical point [37])**.** *Given $\varepsilon > 0$, a point $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ is a nearly ε-critical point of Problem* (**P**) *if there exist $\bar{\boldsymbol{y}}_1, \bar{\boldsymbol{y}}_2 \in \mathbb{R}^n$ and $\bar{\boldsymbol{u}} \in \partial\phi(\bar{\boldsymbol{y}}_1) - \partial g(\bar{\boldsymbol{y}}_2)$ such that*

$$\max\{\|\bar{\boldsymbol{u}}\|, \|\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}_1\|, \|\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}_2\|\} \leq \varepsilon.$$

Definition 3.4 (Nearly ε-stationary point [35])**.** *Given $\varepsilon > 0$, a point $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ is a nearly ε-stationary point of Problem* (**P**) *if there exist $c > 0$ and $\boldsymbol{x}^* \in \mathbb{R}^n$ with $\|\bar{\boldsymbol{x}} - \boldsymbol{x}^*\| \leq c\varepsilon$ such that*

$$\operatorname{dist}\big(\partial\phi(\boldsymbol{x}^*), \partial g(\boldsymbol{x}^*)\big) \leq \varepsilon.$$

Definitions 3.2 and 3.3 relax criticality by letting ϕ and g be evaluated at points that may drift apart from each other and from $\bar{\boldsymbol{x}}$; Definition 3.4 instead forces ϕ and g to share a common witness point $\boldsymbol{x}^*$, which itself may drift from $\bar{\boldsymbol{x}}$. All three reduce to a single mechanism: converting a spatial offset into a Fenchel-gap slack.

Lemma 3.2. *Let $q : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, convex, and G-Lipschitz on a convex set containing $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} q$.*

(a) If $\boldsymbol{v} \in \partial q(\boldsymbol{y})$, then $\boldsymbol{v} \in \partial_{2G\|\boldsymbol{x}-\boldsymbol{y}\|} q(\boldsymbol{x})$.

(b) If, in addition, q is L-smooth on this set and $\boldsymbol{v} = \nabla q(\boldsymbol{y})$, then $\boldsymbol{v} \in \partial_{\frac{L}{2}\|\boldsymbol{x}-\boldsymbol{y}\|^2} q(\boldsymbol{x})$.

Proof. By the Fenchel–Young equality at a subgradient, $q^*(\boldsymbol{v}) = \langle \boldsymbol{v}, \boldsymbol{y} \rangle - q(\boldsymbol{y})$, so

$$\Phi_q(\boldsymbol{x}, \boldsymbol{v}) = q(\boldsymbol{x}) - q(\boldsymbol{y}) - \langle \boldsymbol{v}, \boldsymbol{x} - \boldsymbol{y} \rangle =: D_q(\boldsymbol{x}, \boldsymbol{y}).$$

(a) Convexity gives $D_q(\boldsymbol{x}, \boldsymbol{y}) \geq 0$; G-Lipschitzness gives $q(\boldsymbol{x}) - q(\boldsymbol{y}) \leq G\|\boldsymbol{x} - \boldsymbol{y}\|$ and $\|\boldsymbol{v}\| \leq G$, so $D_q(\boldsymbol{x}, \boldsymbol{y}) \leq 2G\|\boldsymbol{x} - \boldsymbol{y}\|$. (b) $D_q(\boldsymbol{x}, \boldsymbol{y})$ is the Bregman divergence of q at $(\boldsymbol{x}, \boldsymbol{y})$; for convex L-smooth q it satisfies $0 \leq D_q(\boldsymbol{x}, \boldsymbol{y}) \leq \frac{L}{2}\|\boldsymbol{x} - \boldsymbol{y}\|^2$ by the descent lemma. □

We then establish that the $(\delta_\phi, \delta_g, \varepsilon)$-critical point in Definition 3.1 recovers all three existing relaxed DC criticality notions

Proposition 3.2. *Suppose ϕ and g are, respectively, G_ϕ- and G_g-Lipschitz on the relevant neighborhoods below. Then:*

(i) If $\bar{\boldsymbol{x}}$ is an ε-critical point in the sense of Definition 3.2, then $\bar{\boldsymbol{x}}$ is a $(0,\, 2G_g\varepsilon,\, \varepsilon)$-critical point.

(ii) If $\bar{\boldsymbol{x}}$ is a nearly ε-critical point in the sense of Definition 3.3, then $\bar{\boldsymbol{x}}$ is a $(2G_\phi\varepsilon,\, 2G_g\varepsilon,\, \varepsilon)$-critical point.

(iii) If $\bar{\boldsymbol{x}}$ is a nearly ε-stationary point in the sense of Definition 3.4, then $\bar{\boldsymbol{x}}$ is a $(2G_\phi c\varepsilon,\, 2G_g c\varepsilon,\, \varepsilon)$-critical point.

Proof. (i) Write $\bar{\boldsymbol{u}} = \boldsymbol{p} - \boldsymbol{q}$ with $\boldsymbol{p} \in \partial\phi(\bar{\boldsymbol{x}}) = \partial_0\phi(\bar{\boldsymbol{x}})$ and $\boldsymbol{q} \in \partial g(\bar{\boldsymbol{y}})$. Lemma 3.2(a) with $q = g$, $\boldsymbol{x} = \bar{\boldsymbol{x}}$, $\boldsymbol{y} = \bar{\boldsymbol{y}}$ gives $\boldsymbol{q} \in \partial_{2G_g\|\bar{\boldsymbol{x}}-\bar{\boldsymbol{y}}\|} g(\bar{\boldsymbol{x}}) \subseteq \partial_{2G_g\varepsilon} g(\bar{\boldsymbol{x}})$, so $\operatorname{dist}(\partial_0\phi(\bar{\boldsymbol{x}}), \partial_{2G_g\varepsilon} g(\bar{\boldsymbol{x}})) \le \|\boldsymbol{p} - \boldsymbol{q}\| = \|\bar{\boldsymbol{u}}\| \le \varepsilon$.

(ii) Identical, applying Lemma 3.2(a) to both $\boldsymbol{p} \in \partial\phi(\bar{\boldsymbol{y}}_1)$ and $\boldsymbol{q} \in \partial g(\bar{\boldsymbol{y}}_2)$.

(iii) Take $\boldsymbol{p} \in \partial\phi(\boldsymbol{x}^*)$, $\boldsymbol{q} \in \partial g(\boldsymbol{x}^*)$ with $\|\boldsymbol{p} - \boldsymbol{q}\| \le \varepsilon$; this is exactly case (ii) with $\bar{\boldsymbol{y}}_1 = \bar{\boldsymbol{y}}_2 = \boldsymbol{x}^*$ and spatial offset $c\varepsilon$ in place of ε. □

Remark 3.2. *Case (iii) is not an independent construction: it is case (ii) specialized to a common witness point. The distinction between Definition 3.2/3.3 and Definition 3.4 is therefore not a difference in the relaxation mechanism, but only in whether the two witness points are allowed to coincide.*

The embedding above only uses Lipschitz continuity and is tight up to constants. When a component happens to be smooth, Lemma 3.2(b) applies in place of (a) and yields a second-order accurate slack on that side.

Proposition 3.3. *In the setting of Definition 3.4, suppose in addition that g is convex and L_g-smooth on a neighborhood of $\boldsymbol{x}^*$. Then $\bar{\boldsymbol{x}}$ is a*

$$\Big(2G_\phi c\varepsilon,\ \tfrac{L_g}{2}c^2\varepsilon^2,\ \varepsilon\Big)\text{-critical point.}$$

If ϕ is also convex and L_ϕ-smooth on a neighborhood of $\boldsymbol{x}^*$, the first slack sharpens to $\frac{L_\phi}{2}c^2\varepsilon^2$, so that $\bar{\boldsymbol{x}}$ is an $\big(\mathcal{O}(\varepsilon^2), \mathcal{O}(\varepsilon^2), \varepsilon\big)$-critical point.

Proof. Take $\boldsymbol{v}^* = \nabla g(\boldsymbol{x}^*)$ in place of an arbitrary $\boldsymbol{q} \in \partial g(\boldsymbol{x}^*)$ in the proof of Proposition 3.2(iii), and apply Lemma 3.2(b) instead of (a): $\boldsymbol{v}^* \in \partial_{\frac{L_g}{2}\|\bar{\boldsymbol{x}}-\boldsymbol{x}^*\|^2} g(\bar{\boldsymbol{x}}) \subseteq \partial_{\frac{L_g}{2}c^2\varepsilon^2} g(\bar{\boldsymbol{x}})$. The ϕ-side bound is unchanged from Proposition 3.2(iii). The refinement when ϕ is also smooth follows by the same substitution on the ϕ-side. □

Proposition 3.2 and Proposition 3.3 together show that Definition 3.1 strictly subsumes Definitions 3.2, 3.3, and 3.4: the value slack incurred is $\mathcal{O}(\varepsilon)$ under Lipschitz continuity alone, and tightens to $\mathcal{O}(\varepsilon^2)$ on any component that is additionally smooth. This also isolates precisely what smoothness of g buys in Xu et al. [35]: it is not needed for the embedding into Definition 3.1, which only requires Lipschitz continuity, but it tightens the resulting slack by one order in ε.

3.3.1 The Converse Embedding

Proposition 3.2 and Proposition 3.3 show that $(\delta_\phi, \delta_g, \varepsilon)$-criticality is no harder to satisfy than any previously proposed relaxed DC criticality notion. We now establish the converse direction: satisfying Definition 3.1 is also no easier, in the sense that it forces the same qualitative conclusion as Definition 3.3, up to an explicit loss of one order in the tolerance. The mechanism is the classical reachback theorem of Brøndsted and Rockafellar [6], which converts an approximate subgradient at $\bar{\boldsymbol{x}}$ into an exact subgradient at a nearby point, and is the exact converse of Lemma 3.2(a).

Lemma 3.3. *Let $q : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex, and let $\boldsymbol{v} \in \partial_\delta q(\bar{\boldsymbol{x}})$ for some $\delta > 0$. Then for every $\lambda > 0$ there exist $\hat{\boldsymbol{x}} \in \mathbb{R}^n$ and $\hat{\boldsymbol{v}} \in \partial q(\hat{\boldsymbol{x}})$ such that*

$$\|\hat{\boldsymbol{x}} - \bar{\boldsymbol{x}}\| \le \lambda, \qquad \|\hat{\boldsymbol{v}} - \boldsymbol{v}\| \le \frac{\delta}{\lambda}.$$

In particular, taking $\lambda = \sqrt{\delta}$ gives $\|\hat{\boldsymbol{x}} - \bar{\boldsymbol{x}}\| \le \sqrt{\delta}$ and $\|\hat{\boldsymbol{v}} - \boldsymbol{v}\| \le \sqrt{\delta}$.

Proposition 3.4 (Reverse embedding)**.** *If $\bar{\boldsymbol{x}}$ is a $(\delta_\phi, \delta_g, \varepsilon)$-critical point of Problem (**P**), then $\bar{\boldsymbol{x}}$ is a nearly ε'-critical point in the sense of Definition 3.3, with*

$$\varepsilon' = \varepsilon + \sqrt{\delta_\phi} + \sqrt{\delta_g}.$$

Proof. Let $\boldsymbol{u} \in \partial_{\delta_\phi}\phi(\bar{\boldsymbol{x}})$ and $\boldsymbol{v} \in \partial_{\delta_g} g(\bar{\boldsymbol{x}})$ with $\|\boldsymbol{u}-\boldsymbol{v}\| \le \varepsilon$. Apply Lemma 3.3 to $\boldsymbol{u}$ with $\lambda = \sqrt{\delta_\phi}$ and to $\boldsymbol{v}$ with $\lambda = \sqrt{\delta_g}$, obtaining $\bar{\boldsymbol{y}}_1$, $\hat{\boldsymbol{u}} \in \partial\phi(\bar{\boldsymbol{y}}_1)$ and $\bar{\boldsymbol{y}}_2$, $\hat{\boldsymbol{v}} \in \partial g(\bar{\boldsymbol{y}}_2)$ with $\|\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{x}}\| \le \sqrt{\delta_\phi}$, $\|\hat{\boldsymbol{u}} - \boldsymbol{u}\| \le \sqrt{\delta_\phi}$, and analogously for $\bar{\boldsymbol{y}}_2, \hat{\boldsymbol{v}}$. Set $\bar{\boldsymbol{u}} := \hat{\boldsymbol{u}} - \hat{\boldsymbol{v}} \in \partial\phi(\bar{\boldsymbol{y}}_1) - \partial g(\bar{\boldsymbol{y}}_2)$. The triangle inequality gives

$$\|\bar{\boldsymbol{u}}\| \le \|\hat{\boldsymbol{u}} - \boldsymbol{u}\| + \|\boldsymbol{u} - \boldsymbol{v}\| + \|\boldsymbol{v} - \hat{\boldsymbol{v}}\| \le \sqrt{\delta_\phi} + \varepsilon + \sqrt{\delta_g} = \varepsilon',$$

while $\|\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}_1\| \le \sqrt{\delta_\phi} \le \varepsilon'$ and $\|\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}_2\| \le \sqrt{\delta_g} \le \varepsilon'$, which verifies the defining condition of Definition 3.3 with ε' in place of ε. □

Proposition 3.5 (Same-point refinement under smoothness)**.** *In the setting of Proposition 3.4, suppose in addition that g is convex and L_g-smooth on a neighborhood of $\bar{\boldsymbol{x}}$. Then*

$$\operatorname{dist}\big(\nabla g(\bar{\boldsymbol{x}}),\, \partial_{\delta_\phi}\phi(\bar{\boldsymbol{x}})\big) \le \varepsilon + \sqrt{2L_g\delta_g}.$$

If ϕ is also convex and L_ϕ-smooth on a neighborhood of $\bar{\boldsymbol{x}}$, then

$$\big\|\nabla\phi(\bar{\boldsymbol{x}}) - \nabla g(\bar{\boldsymbol{x}})\big\| \le \varepsilon + \sqrt{2L_\phi\delta_\phi} + \sqrt{2L_g\delta_g},$$

so that $\bar{\boldsymbol{x}}$ is, up to this explicit slack, a same-point approximate stationary point of the smooth DC problem.

Proof. For L_g-smooth convex g, the conjugate g^* is $\frac{1}{L_g}$-strongly convex, so $\Phi_g(\bar{\boldsymbol{x}}, \boldsymbol{v}) \ge \frac{1}{2L_g}\|\boldsymbol{v} - \nabla g(\bar{\boldsymbol{x}})\|^2$ for every $\boldsymbol{v}$. Applied to $\boldsymbol{v} \in \partial_{\delta_g} g(\bar{\boldsymbol{x}})$, this gives $\|\boldsymbol{v} - \nabla g(\bar{\boldsymbol{x}})\| \le \sqrt{2L_g\delta_g}$; combined with $\boldsymbol{u} \in \partial_{\delta_\phi}\phi(\bar{\boldsymbol{x}})$ and $\|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon$, the triangle inequality gives the first bound. The second bound follows by the same argument applied to $\boldsymbol{u}$ once ϕ is also L_ϕ-smooth and convex. □

Remark 3.3 (Asymmetry between the two directions)**.** *The two directions are not symmetric. Embedding an existing notion into Definition 3.1 costs a value slack linear in the spatial tolerance under Lipschitz continuity alone (Lemma 3.2(a)) and quadratic under smoothness (Lemma 3.2(b)). Recovering an existing notion from Definition 3.1 costs a spatial displacement on the order of the square root of the value tolerance under Lipschitz continuity alone (Lemma 3.3); this square-root rate is the classical Brøndsted–Rockafellar rate and is not, in general, improvable to a linear one. Smoothness removes the asymmetry in the strongest sense: on a smooth component, the reverse direction also collapses to a same-point bound (Proposition 3.5), matching the same-point character of the forward direction under smoothness (Proposition 3.3), at the cost of a square-root rather than a quadratic rate.*

3.3.2 Relation to Goldstein Stationarity

We conclude by relating $(\delta_\phi, \delta_g, \varepsilon)$-criticality to a relaxed stationarity notion outside the DC literature: the Goldstein δ-subdifferential relaxation of Clarke stationarity [16, 41], introduced schematically in Figure 1. For a locally Lipschitz function $F : \mathbb{R}^n \to \mathbb{R}$, the Goldstein δ-subdifferential at $\bar{\boldsymbol{x}}$ is

$$\partial_\delta^G F(\bar{\boldsymbol{x}}) := \overline{\operatorname{conv}} \bigcup_{\boldsymbol{y} \in B_\delta(\bar{\boldsymbol{x}})} \partial_C F(\boldsymbol{y}),$$

and $\bar{\boldsymbol{x}}$ is a (δ, ε)-Goldstein stationary point of F if $\operatorname{dist}(\boldsymbol{0}, \partial_\delta^G F(\bar{\boldsymbol{x}})) \le \varepsilon$, where $B_\delta(\bar{\boldsymbol{x}}) := \{\boldsymbol{y} \in \mathbb{R}^n : \|\boldsymbol{y} - \bar{\boldsymbol{x}}\| \le \delta\}$. Applied to $F = \Psi = \phi - g$, this relaxes Clarke stationarity along the spatial axis: it collects exact generalized gradients over a δ-neighborhood of $\bar{\boldsymbol{x}}$ and convexifies, rather than relaxing the defining inequality at $\bar{\boldsymbol{x}}$ itself. Definition 3.1 relaxes along the orthogonal, value axis, and couples ϕ and g independently rather than through a common witness point in $B_\delta(\bar{\boldsymbol{x}})$. Table 1 summarizes the distinction.

Despite this structural difference, the Goldstein relaxation embeds into ours.

Table 1: Two relaxation mechanisms for the same base criterion $\mathbf{0} \in \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}})$.

	(δ, ε)-Goldstein	$(\delta_\phi, \delta_g, \varepsilon)$-critical
Relaxation axis	spatial, $B_\delta(\bar{\boldsymbol{x}})$	value, Fenchel–Young gap
Witness structure	common $\boldsymbol{y} \in B_\delta(\bar{\boldsymbol{x}})$	independent $\boldsymbol{u}, \boldsymbol{v}$ at $\bar{\boldsymbol{x}}$
Evaluation point	nearby $\boldsymbol{y}$	$\bar{\boldsymbol{x}}$ itself

Lemma 3.4 (Goldstein bridge)**.** *Let $q : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, convex, and G-Lipschitz on a convex set containing $B_\delta(\bar{\boldsymbol{x}})$. Then*

$$\bigcup_{\boldsymbol{y} \in B_\delta(\bar{\boldsymbol{x}})} \partial q(\boldsymbol{y}) \subseteq \partial_{2G\delta} q(\bar{\boldsymbol{x}}).$$

Proof. Immediate from Lemma 3.2(a): for $\boldsymbol{y} \in B_\delta(\bar{\boldsymbol{x}})$ and $\boldsymbol{v} \in \partial q(\boldsymbol{y})$, $\|\bar{\boldsymbol{x}} - \boldsymbol{y}\| \le \delta$ gives $\boldsymbol{v} \in \partial_{2G\delta} q(\bar{\boldsymbol{x}})$. □

Proposition 3.6 (Goldstein implies ours)**.** *Suppose ϕ, g are, respectively, G_ϕ- and G_g-Lipschitz on a convex set containing $B_\delta(\bar{\boldsymbol{x}})$. If $\bar{\boldsymbol{x}}$ is a (δ, ε)-Goldstein stationary point of $\Psi = \phi - g$, then $\bar{\boldsymbol{x}}$ is a $(2G_\phi\delta, 2G_g\delta, \varepsilon)$-critical point.*

Proof. Let $\boldsymbol{w} \in \partial_\delta^G \Psi(\bar{\boldsymbol{x}})$ with $\|\boldsymbol{w}\| \le \varepsilon$. By Carathéodory's theorem, $\boldsymbol{w} = \sum_i \lambda_i \boldsymbol{w}_i$ for finitely many $\boldsymbol{w}_i \in \partial_C \Psi(\boldsymbol{y}_i)$, $\boldsymbol{y}_i \in B_\delta(\bar{\boldsymbol{x}})$, $\lambda_i \ge 0$, $\sum_i \lambda_i = 1$. By the Clarke sum rule for the difference of two convex functions, $\boldsymbol{w}_i = \boldsymbol{u}_i - \boldsymbol{v}_i$ with $\boldsymbol{u}_i \in \partial\phi(\boldsymbol{y}_i)$, $\boldsymbol{v}_i \in \partial g(\boldsymbol{y}_i)$. Set $\bar{\boldsymbol{u}} := \sum_i \lambda_i \boldsymbol{u}_i$ and $\bar{\boldsymbol{v}} := \sum_i \lambda_i \boldsymbol{v}_i$. By Lemma 3.4, each $\boldsymbol{u}_i \in \partial_{2G_\phi\delta}\phi(\bar{\boldsymbol{x}})$; since $\partial_{2G_\phi\delta}\phi(\bar{\boldsymbol{x}})$ is an intersection of half-spaces, it is convex, so $\bar{\boldsymbol{u}} \in \partial_{2G_\phi\delta}\phi(\bar{\boldsymbol{x}})$. Likewise $\bar{\boldsymbol{v}} \in \partial_{2G_g\delta} g(\bar{\boldsymbol{x}})$. Since $\bar{\boldsymbol{u}} - \bar{\boldsymbol{v}} = \sum_i \lambda_i \boldsymbol{w}_i = \boldsymbol{w}$,

$$\operatorname{dist}\big(\partial_{2G_\phi\delta}\phi(\bar{\boldsymbol{x}}), \partial_{2G_g\delta} g(\bar{\boldsymbol{x}})\big) \le \|\bar{\boldsymbol{u}} - \bar{\boldsymbol{v}}\| = \|\boldsymbol{w}\| \le \varepsilon.$$ □

The converse fails in general, and the reason is structural rather than a looseness of constants: the DC decomposition can manufacture a critical point that Ψ itself does not have.

Example 3.2 (The converse fails)**.** *Let $\phi(x) = \max(x, 0)$ and $g(x) = \max(-x, 0)$ on $\mathbb{R}$, and consider $\bar{x} = 0$. Then $\partial\phi(0) = [0, 1]$ and $\partial g(0) = [-1, 0]$, so $\partial\phi(0) - \partial g(0) = [0, 2] \ni 0$: $\bar{x} = 0$ is an exact DC critical point, i.e., a $(0, 0, 0)$-critical point. However, $\Psi(x) = \phi(x) - g(x) = x$ for every x, so $\partial_C \Psi \equiv \{1\}$ everywhere, and $\partial_\delta^G \Psi(0) = \{1\}$ for every $\delta \ge 0$. Hence $\operatorname{dist}(0, \partial_\delta^G \Psi(0)) = 1$ for every $\delta \ge 0$, and $\bar{x} = 0$ is never a (δ, ε)-Goldstein stationary point for $\varepsilon < 1$.*

Remark 3.4 (Agreement under a faithful decomposition)**.** *If ϕ is differentiable at $\bar{\boldsymbol{x}}$, the Clarke sum rule holds with equality,*

$$\partial_C \Psi(\bar{\boldsymbol{x}}) = \nabla\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}}) = \partial\phi(\bar{\boldsymbol{x}}) - \partial g(\bar{\boldsymbol{x}}),$$

and symmetrically if g is differentiable at $\bar{\boldsymbol{x}}$. We call such a decomposition faithful *at $\bar{\boldsymbol{x}}$: exact DC criticality and exact Clarke stationarity of Ψ coincide there, so the inflation exhibited in Example 3.2 cannot occur. That example is unfaithful precisely because ϕ and g are simultaneously nonsmooth at $\bar{x} = 0$; nonsmoothness on at most one side already rules it out.*

Taken together, Propositions 3.2 to 3.6 position $(\delta_\phi, \delta_g, \varepsilon)$-criticality against every relaxed stationarity notion available in the DC and nonsmooth nonconvex literature: it is implied by each of them under Lipschitz continuity alone, it implies a matching notion in return at the cost of a square-root loss inherent to reachback from a value relaxation, and both directions tighten to a same-point bound whenever the relevant component is smooth. Example 3.2 shows that the reverse implication from Goldstein stationarity cannot be strengthened to an equivalence without restricting to faithful decompositions, an intrinsic limitation of relaxing criticality at the level of a DC pair rather than at the level of Ψ itself.

3.4 Decomposition Dependence and the Frank–Wolfe DC Gap

For DC programs posed over a compact convex set $D \subseteq \mathbb{R}^n$, Maskan et al. [26] measure progress by the Frank–Wolfe-type DC gap

$$\mathrm{gap}_{\mathrm{DC}}(\bar{\boldsymbol{x}}) := \max_{\boldsymbol{z}\in D} \min_{\boldsymbol{v}\in\partial g(\bar{\boldsymbol{x}})} \big\{\phi(\bar{\boldsymbol{x}}) - \phi(\boldsymbol{z}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}} - \boldsymbol{z}\rangle\big\}, \tag{3.4}$$

the largest descent certified by the convex surrogate that keeps ϕ exact and linearizes the concave part $-g$ at $\bar{\boldsymbol{x}}$. They observe that this measure is *decomposition-dependent*: the choice of DC decomposition determines the notion of stationarity captured by the gap, and trades computational efficiency against the strength of that notion. The observation is supported by an example in which two natural decompositions of the same objective produce qualitatively different gap landscapes [26, Figure 1]; how readings taken under different decompositions relate to one another is left open there. The $(\delta_\phi, \delta_g, \varepsilon)$-framework supplies the missing common scale. First, the Frank–Wolfe gap is exactly a one-parameter slice of Definition 3.1: it presses δ_g and ε to zero and reads the entire gap off the δ_ϕ dial, with the constraint absorbed into the convex component via the indicator function ι_D.

Lemma 3.5 (The Frank–Wolfe DC gap is a $(\delta, 0, 0)$-slice)**.** *Let $\phi, g : \mathbb{R}^n \to \mathbb{R}$ be convex, let $D \subseteq \mathbb{R}^n$ be compact and convex, and let $\bar{\boldsymbol{x}} \in D$, $\delta \geq 0$.*

1. *If $\partial g(\bar{\boldsymbol{x}}) \cap \partial_\delta(\phi + \iota_D)(\bar{\boldsymbol{x}}) \neq \emptyset$, then $\mathrm{gap}_{\mathrm{DC}}(\bar{\boldsymbol{x}}) \leq \delta$.*
2. *If g is differentiable at $\bar{\boldsymbol{x}}$, then*

$$\mathrm{gap}_{\mathrm{DC}}(\bar{\boldsymbol{x}}) = \min\big\{\delta \geq 0 : \nabla g(\bar{\boldsymbol{x}}) \in \partial_\delta(\phi + \iota_D)(\bar{\boldsymbol{x}})\big\};$$

that is, $\mathrm{gap}_{\mathrm{DC}}(\bar{\boldsymbol{x}}) \leq \delta$ if and only if $\bar{\boldsymbol{x}}$ is a $(\delta, 0, \varepsilon)$-critical point of the pair $(\phi + \iota_D, g)$ for every $\varepsilon > 0$.

Proof. Fix $\boldsymbol{v} \in \partial g(\bar{\boldsymbol{x}})$ and set $h_{\boldsymbol{v}}(\boldsymbol{z}) := \phi(\boldsymbol{z}) + \iota_D(\boldsymbol{z}) - \langle \boldsymbol{v}, \boldsymbol{z}\rangle$, a proper closed convex function. Then

$$\max_{\boldsymbol{z}\in D}\big\{\phi(\bar{\boldsymbol{x}}) - \phi(\boldsymbol{z}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}} - \boldsymbol{z}\rangle\big\} = h_{\boldsymbol{v}}(\bar{\boldsymbol{x}}) - \inf h_{\boldsymbol{v}},$$

and $h_{\boldsymbol{v}}(\bar{\boldsymbol{x}}) - \inf h_{\boldsymbol{v}} \leq \delta$ if and only if $\mathbf{0} \in \partial_\delta h_{\boldsymbol{v}}(\bar{\boldsymbol{x}}) = \partial_\delta(\phi + \iota_D)(\bar{\boldsymbol{x}}) - \boldsymbol{v}$. For (i), if $\boldsymbol{v} \in \partial g(\bar{\boldsymbol{x}}) \cap \partial_\delta(\phi + \iota_D)(\bar{\boldsymbol{x}})$, then for every $\boldsymbol{z} \in D$ the inner minimum in (3.4) is bounded above by the bracket evaluated at this fixed $\boldsymbol{v}$, so $\mathrm{gap}_{\mathrm{DC}}(\bar{\boldsymbol{x}}) \leq h_{\boldsymbol{v}}(\bar{\boldsymbol{x}}) - \inf h_{\boldsymbol{v}} \leq \delta$. For (ii), when $\partial g(\bar{\boldsymbol{x}}) = \{\nabla g(\bar{\boldsymbol{x}})\}$ the inner minimization is redundant, and $\mathrm{gap}_{\mathrm{DC}}(\bar{\boldsymbol{x}}) = h_{\nabla g(\bar{\boldsymbol{x}})}(\bar{\boldsymbol{x}}) - \inf h_{\nabla g(\bar{\boldsymbol{x}})}$, which is at most δ precisely when $\nabla g(\bar{\boldsymbol{x}}) \in \partial_\delta(\phi + \iota_D)(\bar{\boldsymbol{x}})$. The $(\delta, 0, \varepsilon)$-reformulation follows since $\partial g(\bar{\boldsymbol{x}})$ is a singleton and $\partial_\delta(\phi + \iota_D)(\bar{\boldsymbol{x}})$ is closed. □

Once every gap reading is a point on the same $(\delta_\phi, \delta_g, \varepsilon)$ scale, decomposition dependence becomes a question about how certificates transfer between decompositions of the same objective. Any two DC decompositions $F = \phi_1 - g_1 = \phi_2 - g_2$ differ by a single function $h := \phi_2 - \phi_1 = g_2 - g_1$ added to both components. When h is convex, the transfer is lossless.

Lemma 3.6 (Zero-loss transfer under convex shifts)**.** *Let $F = \phi_1 - g_1 = \phi_2 - g_2$, where $\phi_1, g_1, \phi_2, g_2 : \mathbb{R}^n \to \mathbb{R}$ are convex and $h := \phi_2 - \phi_1 = g_2 - g_1$ is convex. If $\bar{\boldsymbol{x}}$ is a $(\delta_\phi, \delta_g, \varepsilon)$-critical point of the pair (ϕ_1, g_1), then $\bar{\boldsymbol{x}}$ is a $(\delta_\phi, \delta_g, \varepsilon)$-critical point of (ϕ_2, g_2), with the same parameters.*

Proof. Take $\boldsymbol{u} \in \partial_{\delta_\phi}\phi_1(\bar{\boldsymbol{x}})$ and $\boldsymbol{v} \in \partial_{\delta_g} g_1(\bar{\boldsymbol{x}})$ with $\|\boldsymbol{u} - \boldsymbol{v}\| \leq \varepsilon$, and any $\boldsymbol{w} \in \partial h(\bar{\boldsymbol{x}})$, which is nonempty because h is convex and finite-valued. Adding the subgradient inequality of $\boldsymbol{w}$ to the δ-subgradient inequality of $\boldsymbol{u}$ gives $\boldsymbol{u} + \boldsymbol{w} \in \partial_{\delta_\phi}(\phi_1 + h)(\bar{\boldsymbol{x}}) = \partial_{\delta_\phi}\phi_2(\bar{\boldsymbol{x}})$, and likewise $\boldsymbol{v} + \boldsymbol{w} \in \partial_{\delta_g} g_2(\bar{\boldsymbol{x}})$. Since $\|(\boldsymbol{u} + \boldsymbol{w}) - (\boldsymbol{v} + \boldsymbol{w})\| = \|\boldsymbol{u} - \boldsymbol{v}\| \leq \varepsilon$, the claim follows. □

The shift $\boldsymbol{w}$ cancels inside the mismatch, which is what makes the transfer free; no analogue exists for $\mathrm{gap}_{\mathrm{DC}}$, whose value genuinely changes under the same shift. A one-dimensional closed form makes the contrast concrete: for $F = -|x|$ with the decompositions $(\phi_1, g_1) = (0, |x|)$ and $(\phi_2, g_2) = (|x|, 2|x|)$, a direct

computation gives $\operatorname{dist}\big(\partial_\delta\phi_i(x), \partial_\delta g_i(x)\big) = \max\{0,\, 1-\delta/|x|\}$ for both $i = 1, 2$ and all $x \neq 0$: the two decompositions produce identical readings at every point. The two decompositions compared by Maskan et al. [26] are not related by a convex shift: in our notation they are $(\phi_1, g_1) = (Q,\, Q - F)$ and $(\phi_2, g_2) = (F + Q,\, Q)$ with $Q = \frac{L}{2}\|\cdot\|^2$, whose difference $h = \pm F$ is nonconvex in both directions, so Lemma 3.6 applies in neither direction. This is precisely the regime in which their Figure 1 exhibits qualitatively different gap landscapes. Even there, the mismatch induced by Definition 3.1 remains stable across decompositions whenever the components are smooth: its decomposition dependence is $O(\sqrt{\lambda\delta})$ and vanishes as the dial closes, whereas the two gaps differ by a fixed, dial-free amount.

Proposition 3.7 (Vanishing decomposition dependence of the mismatch)**.** *Let $F = \phi_i - g_i$ for $i = 1, 2$, where all four components are convex with λ-Lipschitz gradients, and define the tightest mismatch at level $\delta \geq 0$ by*

$$\varepsilon^*_{i,\delta}(\bar{\boldsymbol{x}}) := \operatorname{dist}\big(\partial_\delta\phi_i(\bar{\boldsymbol{x}}), \partial_\delta g_i(\bar{\boldsymbol{x}})\big),$$

so that $\bar{\boldsymbol{x}}$ is $(\delta, \delta, \varepsilon)$-critical for (ϕ_i, g_i) precisely when $\varepsilon \geq \varepsilon^*_{i,\delta}(\bar{\boldsymbol{x}})$. Then for every $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ and $i \in \{1, 2\}$,

$$\max\big\{0,\ \|\nabla F(\bar{\boldsymbol{x}})\| - 2\sqrt{2\lambda\delta}\big\} \ \leq\ \varepsilon^*_{i,\delta}(\bar{\boldsymbol{x}}) \ \leq\ \|\nabla F(\bar{\boldsymbol{x}})\|,$$

and consequently $\sup_{\bar{\boldsymbol{x}}} \big|\varepsilon^*_{1,\delta}(\bar{\boldsymbol{x}}) - \varepsilon^*_{2,\delta}(\bar{\boldsymbol{x}})\big| \leq 2\sqrt{2\lambda\delta}$.

Proof. The upper bound follows from the feasible pair $\boldsymbol{u} = \nabla\phi_i(\bar{\boldsymbol{x}})$, $\boldsymbol{v} = \nabla g_i(\bar{\boldsymbol{x}})$, whose mismatch is $\|\nabla\phi_i(\bar{\boldsymbol{x}}) - \nabla g_i(\bar{\boldsymbol{x}})\| = \|\nabla F(\bar{\boldsymbol{x}})\|$. For the lower bound, let h be convex with λ-Lipschitz gradient and let $\boldsymbol{v} \in \partial_\delta h(\bar{\boldsymbol{x}})$. Combining $h(\boldsymbol{z}) \geq h(\bar{\boldsymbol{x}}) + \langle \boldsymbol{v}, \boldsymbol{z} - \bar{\boldsymbol{x}}\rangle - \delta$ with $h(\boldsymbol{z}) \leq h(\bar{\boldsymbol{x}}) + \langle \nabla h(\bar{\boldsymbol{x}}), \boldsymbol{z} - \bar{\boldsymbol{x}}\rangle + \frac{\lambda}{2}\|\boldsymbol{z} - \bar{\boldsymbol{x}}\|^2$ yields $\langle \boldsymbol{v} - \nabla h(\bar{\boldsymbol{x}}), \boldsymbol{z} - \bar{\boldsymbol{x}}\rangle \leq \frac{\lambda}{2}\|\boldsymbol{z} - \bar{\boldsymbol{x}}\|^2 + \delta$ for all $\boldsymbol{z}$; taking $\boldsymbol{z} = \bar{\boldsymbol{x}} + t\,(\boldsymbol{v} - \nabla h(\bar{\boldsymbol{x}}))/\|\boldsymbol{v} - \nabla h(\bar{\boldsymbol{x}})\|$ with $t = \sqrt{2\delta/\lambda}$ gives $\|\boldsymbol{v} - \nabla h(\bar{\boldsymbol{x}})\| \leq \sqrt{2\lambda\delta}$. Hence $\partial_\delta\phi_i(\bar{\boldsymbol{x}}) \subseteq \mathcal{B}_{\sqrt{2\lambda\delta}}\big(\nabla\phi_i(\bar{\boldsymbol{x}})\big)$ and likewise for g_i, so $\varepsilon^*_{i,\delta}(\bar{\boldsymbol{x}}) \geq \|\nabla F(\bar{\boldsymbol{x}})\| - 2\sqrt{2\lambda\delta}$. The final claim follows because both $\varepsilon^*_{1,\delta}(\bar{\boldsymbol{x}})$ and $\varepsilon^*_{2,\delta}(\bar{\boldsymbol{x}})$ lie in an interval of length at most $2\sqrt{2\lambda\delta}$. □

Figure 4 reads the same example on the (δ_ϕ, δ_g) scale. At every probe point the two decompositions trace different Pareto frontiers, yet each is pinned by the same landmarks: its $\delta_g = 0$ intercept is the decomposition's Frank–Wolfe DC gap and its $\delta_\phi = \delta_g$ section is the symmetric mismatch reading. The Frank–Wolfe intercepts reproduce the observation of Maskan et al. [26]—curvature preserving is cheaper near the minimum and dearer near the maximum—while the whole frontier shrinks to the origin as $\bar{\boldsymbol{x}}$ approaches the stationary point, uniformly in the decomposition, in line with Proposition 3.7. In summary, the trade-off identified by Maskan et al. [26] is real but lives on the achievability side, namely which certificates a given decomposition renders cheap to compute. The certificates themselves, expressed in the $(\delta_\phi, \delta_g, \varepsilon)$ coordinates, subsume the Frank–Wolfe gap as a $(\delta, 0, 0)$-slice (Lemma 3.5), transfer without loss under convex shifts (Lemma 3.6), and degrade in a controlled $O(\sqrt{\lambda\delta})$ fashion between smooth decompositions (Proposition 3.7).

4 Complexity Analysis of Deterministic DCA

This section establishes the iteration complexity of DCA-type methods for reaching a $(\delta_\phi, \delta_g, \varepsilon)$-critical point of $F = \phi - g$, without imposing smoothness on g. We analyze three algorithmic templates against this single certificate: (1) exact and inexact DCA, which solve the strongly convex subproblem to full or controlled accuracy; (2) proximal DCA, which replaces that solve by a single proximal-subgradient step; and (3) concave-side smoothing. shown to reproduce the same certificate the other two attain.

We use the unified DCA summarized in Algorithm 1. At the k-th iteration, DCA computes a subgradient $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$ and solves a γ-strongly convex subproblem to obtain $\boldsymbol{x}^{k+1}$ (Option I). Option II is an abstract inexact inner-solver interface: Algorithm 2 gives one realization, in which the convex subproblem is solved to value accuracy by an inner first-order method, and other inner-loop algorithms can be used as long as their output satisfies the inexact certificate required in the analysis below.

Before analyzing the complexity of DCA, we first introduce the main assumptions used throughout the section.

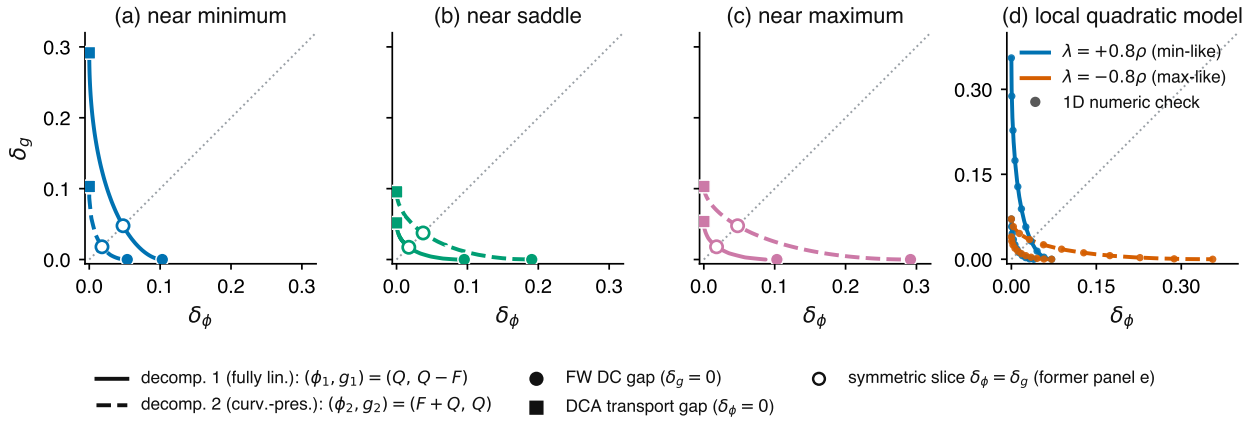


Figure 4: Decomposition dependence read on the (δ_ϕ, δ_g) scale, on the example of Maskan et al. [26, Figure 1]: $F(\boldsymbol{x}) = \sin(\pi x_1)\cos(\pi x_2)$, $L = \pi^2$, with $Q = \frac{L}{2}\|\cdot\|^2$ and the two convex decompositions $(\phi_1, g_1) = (Q, Q - F)$ (fully linearized, solid) and $(\phi_2, g_2) = (F + Q, Q)$ (curvature preserving, dashed). A certificate $\boldsymbol{v}$ lies in $\partial_\delta h(\bar{\boldsymbol{x}})$ exactly when its Fenchel gap $h(\bar{\boldsymbol{x}}) + h^*(\boldsymbol{v}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}} \rangle \le \delta$; each curve is the lower-left (Pareto) boundary of the slack pairs (δ_ϕ, δ_g) for which a *single* certificate $\boldsymbol{v}$ renders $\bar{\boldsymbol{x}}$ a $(\delta_\phi, \delta_g, 0)$-critical point of (ϕ_i, g_i). **a**–**c**: frontiers at three probe points $\bar{\boldsymbol{x}} = \boldsymbol{x}^* + d\,\boldsymbol{u}$ ($d = 0.15$, $\boldsymbol{u}$ the leading Hessian eigenvector of F at the stationary point $\boldsymbol{x}^*$) near the global minimum, a saddle, and the global maximum. On each curve the $\delta_g = 0$ intercept (•) is the Frank–Wolfe DC gap of that decomposition (Lemma 3.5), the $\delta_\phi = 0$ intercept (■) is the DCA transport gap, and the diagonal $\delta_\phi = \delta_g$ section (○) recovers the symmetric single-dial reading of the mismatch ε^*_δ. The two decompositions trace genuinely different frontiers, and their $\delta_g = 0$ intercepts swap order between the minimum (curvature preserving smaller) and the maximum (fully linearized smaller): the qualitative effect reported by Maskan et al. [26], localized here to an intercept property. Every frontier contracts to the origin as $\bar{\boldsymbol{x}} \to \boldsymbol{x}^*$ at rate $O(d^2)$. **d**: local quadratic model. Along an eigendirection of curvature λ, writing $\rho = L$, the frontiers are $\sqrt{2(\rho + \lambda)\delta_\phi} + \sqrt{2\rho\,\delta_g} = |\lambda d|$ (curvature preserving) and $\sqrt{2\rho\,\delta_\phi} + \sqrt{2(\rho - \lambda)\delta_g} = |\lambda d|$ (fully linearized): moving F across the decomposition transfers the curvature coefficient from the δ_ϕ scale to the δ_g scale. Dots: matching one-dimensional numeric frontiers on $F(t) = \frac{\lambda}{2}t^2$. Conjugates of the nonquadratic components are evaluated on a grid.

4.1 Main Assumption

Assume that F is bounded below on $\operatorname{dom}\phi$, and denote

$$F^\star := \inf_x F(x) > -\infty, \qquad \Delta_0 := F(\boldsymbol{x}^0) - F^\star > 0.$$

For any $\gamma > 0$, F admits the reformulation

$$F = \Big(\phi + \tfrac{\gamma}{2}\|\boldsymbol{x}\|^2\Big) - \Big(\tfrac{\gamma}{2}\|\boldsymbol{x}\|^2 + g\Big),$$

so WLOG we take ϕ and g to be γ-strongly convex, formalized as follows.

Assumption 4.1. *The functions ϕ and g are strongly convex for some constant $\gamma > 0$, and g has bounded subgradients:* $\sup_{\boldsymbol{x}} \sup_{\boldsymbol{v} \in \partial g(\boldsymbol{x})} \|\boldsymbol{v}\| \le G$.

Our goal is to characterize the iteration complexity of DCA-type methods, without imposing smoothness on g, for reaching a $(\delta_\phi, \delta_g, \varepsilon)$-critical point. As discussed below, the three parameters play distinct roles in this analysis: ε controls the residual mismatch between the approximate dual vectors of ϕ and g, while δ_ϕ and

Algorithm 1: (In)exact DCA

Input: Initial point $\boldsymbol{x}^0 \in \mathrm{dom}\phi$, inner solver tolerance $\{\epsilon_k\}_{k=0}^{K-1}$, stepsize η.
for $k = 0, 1, \ldots, K-1$ **do**
  Compute $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$.
  Solve inner strongly convex subproblem:
  ▷ **Option I:** exact subproblem solve

$$\boldsymbol{x}^{k+1} = \operatorname*{argmin}_{\boldsymbol{x}\in\mathbb{R}^n}\{\phi(\boldsymbol{x}) - \langle \boldsymbol{v}^k, \boldsymbol{x}\rangle\} \tag{4.1}$$

  ▷ **Option II:** inexact inner solve

$$\boldsymbol{x}^{k+1} = \mathcal{S}_{\mathcal{A}_k}(\boldsymbol{x}^k, \boldsymbol{v}^k; \Theta_k), \tag{4.2}$$

  where $\mathcal{S}_{\mathcal{A}_k}$ denotes the inner solver selected by $\mathcal{A}_k$, and Θ_k collects its parameters and stopping tolerance.
end for
Output: $\hat{\boldsymbol{x}}$, chosen uniformly from $\{\boldsymbol{x}^k\}_{k=0}^{K-1}$.

δ_g control how loosely each convex component's Fenchel–Young gap is allowed to vanish.

4.2 Common Auxiliary Lemmas

The next lemma is the pivot of the analysis, which shows that the ϕ-side Fenchel gap Φ_ϕ is at any candidate point equals the value suboptimality of that point for the current convex subproblem. Consequently, the parameter δ_ϕ of Definition 3.1 coincides, exactly and without slack, with the accuracy to which the subproblem is solved.

Lemma 4.1. *Let Assumption 4.1 hold, $\bar{\boldsymbol{x}} \in \mathrm{dom}\phi$, $\boldsymbol{v} \in \partial g(\bar{\boldsymbol{x}})$, and define $P(\boldsymbol{x}) := \phi(\boldsymbol{x}) - \langle \boldsymbol{v}, \boldsymbol{x}\rangle$. Then the following statements hold.*

(i) $\inf_{\boldsymbol{x}\in\mathbb{R}^n} P(\boldsymbol{x}) \geq F^* + g(\bar{\boldsymbol{x}}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}}\rangle > -\infty$, *and* $\phi^*(\boldsymbol{v}) = -\inf_{\boldsymbol{x}\in\mathbb{R}^n} P(\boldsymbol{x}) \in \mathbb{R}$.

(ii) For every $\boldsymbol{y} \in \mathrm{dom}\phi$, the following holds

$$\Phi_\phi(\boldsymbol{y}, \boldsymbol{v}) \;=\; P(\boldsymbol{y}) - \inf_{\boldsymbol{x}\in\mathbb{R}^n} P(\boldsymbol{x}).$$

(iii) For every $\sigma \geq 0$ and every $\boldsymbol{y} \in \mathrm{dom}\phi$, the following three statements are equivalent:

$$\textit{(a)}\ P(\boldsymbol{y}) \leq \inf_{\boldsymbol{x}\in\mathbb{R}^n} P(\boldsymbol{x}) + \sigma; \qquad \textit{(b)}\ \Phi_\phi(\boldsymbol{y}, \boldsymbol{v}) \leq \sigma; \qquad \textit{(c)}\ \boldsymbol{v} \in \partial_\sigma \phi(\boldsymbol{y}).$$

In particular, $\boldsymbol{y} \in \operatorname*{argmin}_{\boldsymbol{x}\in\mathbb{R}^n} P(\boldsymbol{x})$ if and only if $\boldsymbol{v} \in \partial\phi(\boldsymbol{y})$.

Proof. (i) Since $\bar{\boldsymbol{x}} \in \mathrm{dom}\phi \subseteq \mathrm{dom}g$ and $\boldsymbol{v} \in \partial g(\bar{\boldsymbol{x}})$, the subgradient inequality gives

$$g(\boldsymbol{x}) \;\geq\; g(\bar{\boldsymbol{x}}) + \langle \boldsymbol{v}, \boldsymbol{x} - \bar{\boldsymbol{x}}\rangle \qquad \text{for all } \boldsymbol{x} \in \mathbb{R}^n, \tag{4.3}$$

equivalently $\langle \boldsymbol{v}, \boldsymbol{x}\rangle \leq g(\boldsymbol{x}) - g(\bar{\boldsymbol{x}}) + \langle \boldsymbol{v}, \bar{\boldsymbol{x}}\rangle$ for all $\boldsymbol{x} \in \mathbb{R}^n$. Hence, for every $\boldsymbol{x} \in \mathrm{dom}\phi \subseteq \mathrm{dom}g$,

$$P(\boldsymbol{x}) \;=\; \phi(\boldsymbol{x}) - \langle \boldsymbol{v}, \boldsymbol{x}\rangle \;\geq\; \phi(\boldsymbol{x}) - g(\boldsymbol{x}) + g(\bar{\boldsymbol{x}}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}}\rangle \;=\; F(\boldsymbol{x}) + g(\bar{\boldsymbol{x}}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}}\rangle \;\geq\; F^* + g(\bar{\boldsymbol{x}}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}}\rangle,$$

while $P(\boldsymbol{x}) = +\infty$ for $\boldsymbol{x} \notin \mathrm{dom}\phi$. Since $g(\bar{\boldsymbol{x}}) \in \mathbb{R}$, the claimed lower bound follows, and it is finite. Furthermore, $P(\bar{\boldsymbol{x}}) = \phi(\bar{\boldsymbol{x}}) - \langle \boldsymbol{v}, \bar{\boldsymbol{x}} \rangle < +\infty$, so $\inf P \in \mathbb{R}$. By the definition of the Fenchel conjugate,

$$\phi^*(\boldsymbol{v}) \;=\; \sup_{\boldsymbol{x} \in \mathbb{R}^n} \big\{ \langle \boldsymbol{v}, \boldsymbol{x} \rangle - \phi(\boldsymbol{x}) \big\} \;=\; -\inf_{\boldsymbol{x} \in \mathbb{R}^n} \big\{ \phi(\boldsymbol{x}) - \langle \boldsymbol{v}, \boldsymbol{x} \rangle \big\} \;=\; -\inf_{\boldsymbol{x} \in \mathbb{R}^n} P(\boldsymbol{x}) \;\in\; \mathbb{R}.$$

(ii) For $\boldsymbol{y} \in \mathrm{dom}\phi$, using (i),

$$\Phi_\phi(\boldsymbol{y}, \boldsymbol{v}) \;=\; \phi(\boldsymbol{y}) + \phi^*(\boldsymbol{v}) - \langle \boldsymbol{v}, \boldsymbol{y} \rangle \;=\; P(\boldsymbol{y}) + \phi^*(\boldsymbol{v}) \;=\; P(\boldsymbol{y}) - \inf_{\boldsymbol{x} \in \mathbb{R}^n} P(\boldsymbol{x}).$$

(iii) The equivalence of (a) and (b) is immediate from (ii). The equivalence of (b) and (c) is Lemma 3.1 applied to $h = \phi$. The final claim is the case $\sigma = 0$. □

The following identity decomposes the per-iteration decrease of F into three nonnegative quantities that together constitute a $(\delta_\phi, \delta_g, \varepsilon)$-point. It holds for an arbitrary candidate $\boldsymbol{x}^{k+1}$, regardless of how it is produced, and therefore serves as the common engine underlying both the exact and the inexact analyses.

Lemma 4.2 (One-step decomposition identity)**.** *Suppose Assumption 4.1 holds. For any $\boldsymbol{x}^k, \boldsymbol{x}^{k+1} \in \mathrm{dom}\phi$ and any $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$, it holds that*

$$F(\boldsymbol{x}^{k+1}) - F(\boldsymbol{x}^k) \;=\; \Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;-\; \Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) \;-\; \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k). \tag{4.4}$$

Proof. By the Fenchel–Young equality, i.e., for any

$$g^*(\boldsymbol{v}^k) \;=\; \langle \boldsymbol{v}^k, \boldsymbol{x}^k \rangle - g(\boldsymbol{x}^k) \;\in\; \mathbb{R}, \tag{4.5}$$

and ssubstituting (4.5) into the definition of Φ_g gives

$$\Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;=\; g(\boldsymbol{x}^{k+1}) + g^*(\boldsymbol{v}^k) - \langle \boldsymbol{v}^k, \boldsymbol{x}^{k+1} \rangle \;=\; g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) - \langle \boldsymbol{v}^k, \boldsymbol{x}^{k+1} - \boldsymbol{x}^k \rangle. \tag{4.6}$$

On the other hand, rearranging the terms in the definition of Φ_ϕ yields

$$\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) - \Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;=\; \big[\phi(\boldsymbol{x}^k) - \langle \boldsymbol{v}^k, \boldsymbol{x}^k \rangle\big] - \big[\phi(\boldsymbol{x}^{k+1}) - \langle \boldsymbol{v}^k, \boldsymbol{x}^{k+1} \rangle\big].$$

Adding (4.6) to the above equality yields

$$\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) - \Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;=\; \big[\phi(\boldsymbol{x}^k) - g(\boldsymbol{x}^k)\big] - \big[\phi(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^{k+1})\big] \;=\; F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}),$$

which completes the proof. □

By Lemma 4.1 (ii)–(iii), the quantity $\Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)$ is the value inaccuracy of the subproblem solve and populates the δ_ϕ-slot of the certificate at $\boldsymbol{x}^{k+1}$; the quantity $\Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)$ populates the δ_g-slot at $\boldsymbol{x}^{k+1}$; and the quantity $\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k)$ populates the δ_ϕ-slot of a second certificate anchored at $\boldsymbol{x}^k$, where the g-side is exact because $\Phi_g(\boldsymbol{x}^k, \boldsymbol{v}^k) = 0$. The objective decrease per iteration equals the sum of the two certificate budgets minus the injected inexactness. In the exact regime the injected inexactness vanishes and the identity strengthens the classical monotone descent property of DCA [1, 21] into an exact accounting of where the descent goes.

4.3 Iteration Complexity of DCA-type Methods

We translate the one-step identity above into finite-time complexity bounds. We first treat exact DCA, where each convex subproblem is solved exactly. We then analyze inexact DCA by separating the outer certificate bound from the inner first-order cost.

Complexity analysis of Exact DCA. We first analyze the complexity of DCA in Algorithm 1 with *Option I*, meaning that each subproblem is solved exactly. By Lemma 4.1 (ii)–(iii), the update (4.1) is equivalent to

$$\Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) = 0, \qquad \text{that is,} \qquad \boldsymbol{v}^k \in \partial\phi(\boldsymbol{x}^{k+1}). \tag{4.7}$$

Theorem 4.1 (Iteration Complexity of Exact DCA)**.** *Suppose Assumption 4.1 holds. Let* $\boldsymbol{x}^0 \in \mathrm{dom}\phi$ *and* $\{\boldsymbol{x}^k\}_{k\geq 0}$ *be generated by Algorithm 1 with* Option I. *Given* $\delta_g > 0$*, DCA returns a point* $\bar{\boldsymbol{x}} \in \{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^K\}$ *satisfying* $(0, \delta_g, 0)$*-critical within* $K = \mathcal{O}(\Delta_0 \delta_g^{-1})$ *iterations.*

Proof. Substituting the exact optimality condition (4.7) into the one-step identity (4.4) gives, for every $k \geq 0$,

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \;=\; \Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;\geq\; 0, \tag{4.8}$$

Summing (4.8) over $k = 0, \ldots, K-1$ and using $F(\boldsymbol{x}^K) \geq F^\star$, we obtain

$$\frac{1}{K}\sum_{k=0}^{K-1}\Big[\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)\Big] \;\leq\; \frac{\Delta_0}{K}.$$

where defining

$$k^* := \operatorname*{argmin}_{0\leq k\leq K-1}\Big[\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)\Big]$$

yields

$$\Phi_\phi(\boldsymbol{x}^{k^*}, \boldsymbol{v}^{k^*}) + \Phi_g(\boldsymbol{x}^{k^*+1}, \boldsymbol{v}^{k^*}) \;\leq\; \frac{\Delta_0}{K}.$$

Because both summands are nonnegative, each is individually bounded by their sum, implying

$$\Phi_g(\boldsymbol{x}^{k^*+1}, \boldsymbol{v}^{k^*}) \;\leq\; \frac{\Delta_0}{K}. \tag{4.9}$$

Then, we set $\boldsymbol{u} = \boldsymbol{v} := \boldsymbol{v}^{k^*}$. Combining (4.7) with the optimality condition that $\boldsymbol{v}^{k^*} \in \partial\phi(\boldsymbol{x}^{k^*+1}) \cap \partial g(\boldsymbol{x}^k)$, we obtain $\boldsymbol{u} \in \partial\phi(\boldsymbol{x}^{k^*+1})$. By (4.9) together with Lemma 3.1, $\boldsymbol{v} \in \partial_{\Delta_0/K} g(\boldsymbol{x}^{k^*+1})$. Hence the following holds

$$\mathrm{dist}\Big(\partial_0\phi(\boldsymbol{x}^{k^*+1}),\, \partial_{\Delta_0/K}\, g(\boldsymbol{x}^{k^*+1})\Big) \;\leq\; \|\boldsymbol{u} - \boldsymbol{v}\| \;=\; 0,$$

i.e., $\boldsymbol{x}^{k^*+1}$ is $(0, \Delta_0/K, 0)$-critical. Choosing $K = \lceil \Delta_0/\delta_g \rceil$ gives $\Delta_0/K \leq \delta_g$. Taking $\bar{\boldsymbol{x}} := \boldsymbol{x}^{k^*+1}$ completes the proof. □

Remark 4.1 (Tightness under smooth oracle)**.**

- Matching lower bound. *The rate of Theorem 4.1 is tight up to the condition number* L/γ*: embedding the smooth hard instance of Carmon et al. [7] as the DC pair* $\phi = \hat{F} + \frac{L}{2}\|\cdot\|^2$ *and* $g = \frac{L}{2}\|\cdot\|^2$ *renders the component-wise DC oracle informationally equivalent to a standard first-order oracle, and yields a lower bound of* $\Omega(\gamma L^{-1}\Delta\delta^{-1})$ *oracle calls for reaching a* $(0, \delta_g, \varepsilon)$*-critical point with* $\delta_g = O(\gamma\varepsilon^2/L^2)$*; see Appendix B.*

- Relation to classical measures. *When* g *is* L_g*-smooth,* δ_g *and the same-point residual* ε *are explicitly interconvertible with constant* L_g^2/γ*, so* $O(\Delta\delta^{-1})$ *recovers the classical* $O(\varepsilon^{-2})$ *rate; when* g *is nonsmooth this equivalence provably fails, as the classical target is finite-time unattainable (Theorem 3.1).*

Complexity Analysis of Inexact DCA We now analyze the complexity of DCA with inexact subproblem solves, as formalized in Algorithm 2. This setting models the practical situation in which the convex subproblem is solved by an iterative method and terminated early [37]. When $\sigma_k = 0$, condition (4.10) reduces to (4.1).

Algorithm 2: Inexact DCA

1: **Input:** Starting point $\boldsymbol{x}^0 \in \mathrm{dom}\phi$, tolerances $\{\sigma_k\}_{k\geq 0}$.
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Choose $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$.
4: Define the convex subproblem

$$P_k(\boldsymbol{x}) := \phi(\boldsymbol{x}) - \langle \boldsymbol{v}^k, \boldsymbol{x}\rangle.$$

5: Starting from $\boldsymbol{y}^0 = \boldsymbol{x}^k$, run an inner first-order method to find $\boldsymbol{y}^{J_k}$ satisfying

$$P_k(\boldsymbol{y}^{J_k}) \leq \inf_{\boldsymbol{x}\in\mathbb{R}^n} P_k(\boldsymbol{x}) + \sigma_k. \tag{4.10}$$

6: Set $\boldsymbol{x}^{k+1} := \boldsymbol{y}^{J_k}$.
7: **end for**
8: **Output:** $\{\boldsymbol{x}^k\}_{k\geq 0}$.

Lemma 4.3. *Suppose Assumption 4.1 holds. Let $\{\boldsymbol{x}^k\}_{k\geq 0}$ be generated by DCA with tolerances $\{\sigma_k\}_{k\geq 0} \subset [0, +\infty)$. Then the following statements hold.*

(i) For every $k \geq 0$, it holds that

$$\Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;\leq\; \sigma_k, \qquad \textit{equivalently,} \qquad \boldsymbol{v}^k \in \partial_{\sigma_k}\phi(\boldsymbol{x}^{k+1}).$$

(ii) For every integer $K \geq 1$, it holds that

$$\sum_{k=0}^{K-1}\Big[\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)\Big] \;\leq\; \Delta_0 + \sum_{k=0}^{K-1}\sigma_k. \tag{4.11}$$

Moreover, if $\sigma_k \equiv \sigma \geq 0$ for all k, then there exists $k^* \in \{0, 1, \ldots, K-1\}$ such that

$$\Phi_\phi(\boldsymbol{x}^{k^*}, \boldsymbol{v}^{k^*}) + \Phi_g(\boldsymbol{x}^{k^*+1}, \boldsymbol{v}^{k^*}) \;\leq\; \frac{\Delta_0}{K} + \sigma, \tag{4.12}$$

i.e., $\boldsymbol{x}^{k^*+1}$ is a $\big(\sigma,\ \Delta_0/K + \sigma,\ 0\big)$-critical point.

Proof. (i) Fix $k \geq 0$. By Algorithm 2, the iterate $\boldsymbol{x}^{k+1}$ satisfies the value-inexactness condition (4.10) for the subproblem $P_k(\boldsymbol{x}) = \phi(\boldsymbol{x}) - \langle \boldsymbol{v}^k, \boldsymbol{x}\rangle$. Applying Lemma 4.1 (iii) with $\bar{\boldsymbol{x}} = \boldsymbol{x}^k$, $\boldsymbol{v} = \boldsymbol{v}^k$, $\boldsymbol{y} = \boldsymbol{x}^{k+1}$, and $\sigma = \sigma_k$ gives

$$\Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \leq \sigma_k \qquad \Longleftrightarrow \qquad \boldsymbol{v}^k \in \partial_{\sigma_k}\phi(\boldsymbol{x}^{k+1}).$$

This proves (i).

(ii) Substituting the bound in (i) into the one-step identity (4.4) yields, for every $k \geq 0$,

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \;=\; \Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) - \Phi_\phi(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \;\geq\; \Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) - \sigma_k. \tag{4.13}$$

Summing (4.13) over $k = 0, \ldots, K-1$ and using $F(\boldsymbol{x}^K) \geq F^\star$, we obtain

$$\sum_{k=0}^{K-1}\Big[\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)\Big] \;\leq\; F(\boldsymbol{x}^0) - F(\boldsymbol{x}^K) + \sum_{k=0}^{K-1}\sigma_k \;\leq\; \Delta_0 + \sum_{k=0}^{K-1}\sigma_k,$$

which proves (4.11).

If $\sigma_k \equiv \sigma$, then the right-hand side of (4.11) equals $\Delta_0 + K\sigma$. Define

$$k^* := \operatorname*{argmin}_{0\le k\le K-1} \Big[\Phi_\phi(\boldsymbol{x}^k, \boldsymbol{v}^k) + \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)\Big].$$

Then

$$\Phi_\phi(\boldsymbol{x}^{k^*}, \boldsymbol{v}^{k^*}) + \Phi_g(\boldsymbol{x}^{k^*+1}, \boldsymbol{v}^{k^*}) \;\le\; \frac{\Delta_0}{K} + \sigma,$$

which is (4.12). Since $\Phi_\phi(\boldsymbol{x}^{k^*}, \boldsymbol{v}^{k^*}) \ge 0$, the same bound gives

$$\Phi_g(\boldsymbol{x}^{k^*+1}, \boldsymbol{v}^{k^*}) \;\le\; \frac{\Delta_0}{K} + \sigma.$$

Part (i) gives $\boldsymbol{v}^{k^*} \in \partial_\sigma \phi(\boldsymbol{x}^{k^*+1})$, while Lemma 3.1 gives

$$\boldsymbol{v}^{k^*} \in \partial_{\Delta_0/K+\sigma} g(\boldsymbol{x}^{k^*+1}).$$

Taking $\boldsymbol{u} = \boldsymbol{v} = \boldsymbol{v}^{k^*}$, we obtain

$$\operatorname{dist}\Big(\partial_\sigma \phi(\boldsymbol{x}^{k^*+1}),\, \partial_{\Delta_0/K+\sigma} g(\boldsymbol{x}^{k^*+1})\Big) \;\le\; \|\boldsymbol{u} - \boldsymbol{v}\| \;=\; 0.$$

Hence $\boldsymbol{x}^{k^*+1}$ is $\big(\sigma, \Delta_0/K + \sigma, 0\big)$-critical, completing the proof. □

Theorem 4.2 (First-order Complexity of Inexact DCA)**.** *Suppose Assumption 4.1 holds. Let $\{\boldsymbol{x}^k\}_{k\ge 0}$ be generated by Algorithm 2 with the constant tolerance*

$$\sigma_k \equiv \sigma_\star := \min\{\delta_\phi, \theta\delta_g\}, \qquad \theta \in (0,1),$$

where $\delta_\phi, \delta_g > 0$. Then DCA returns a $(\delta_\phi, \delta_g, 0)$-critical point $\bar{\boldsymbol{x}} \in \{\boldsymbol{x}^1, \dots, \boldsymbol{x}^K\}$ within $K = \big(\Delta_0(\delta_g - \sigma_\star)^{-1}\big)$ outer iterations. Moreover, if each inner subproblem can be solved to value accuracy $\sigma_\star$ using at most $\mathcal{N}_{\rm in}(\sigma_\star)$ inner first-order oracle calls, then the total number of first-order oracle calls is at most $\big(\Delta_0(\delta_g - \sigma_\star)^{-1}\mathcal{N}_{\rm in}(\sigma_\star)\big)$.

Proof. Applying Lemma 4.3 (ii) with $\sigma = \sigma_\star$ gives an index $k^* \in \{0, \dots, K-1\}$ such that $\boldsymbol{x}^{k^*+1}$ is $\big(\sigma_\star, \Delta_0/K + \sigma_\star, 0\big)$-critical. Since $\sigma_\star \le \delta_\phi$ and $K \ge \Delta_0/(\delta_g - \sigma_\star)$, we have $\Delta_0/K + \sigma_\star \le \delta_g$. Hence $\boldsymbol{x}^{k^*+1}$ is $(\delta_\phi, \delta_g, 0)$-critical. The oracle bound follows by multiplying the number of outer iterations by the worst-case inner cost per subproblem. □

Remark 4.2. *We now specialize the total first-order complexity bound in Theorem 4.2 by instantiating the inexact inner solver with several standard first-order methods:*

subproblem structure	$\mathcal{N}_{\rm in}(\sigma_\star)$	*total first-order complexity*
convex, subgradient	$O(\sigma_\star^{-2})$	$O\Big(\dfrac{\Delta_0}{\delta_g}\,\sigma_\star^{-2}\Big)$
strongly convex, subgradient	$O(\gamma^{-1}\sigma_\star^{-1})$	$O\Big(\dfrac{\Delta_0}{\delta_g}\,\gamma^{-1}\sigma_\star^{-1}\Big)$
smooth strongly convex, GD	$O(\kappa\log(1/\sigma_\star))$	$O\Big(\dfrac{\Delta_0}{\delta_g}\,\kappa\log(1/\sigma_\star)\Big)$
smooth strongly convex, APG	$O(\sqrt{\kappa}\log(1/\sigma_\star))$	$O\Big(\dfrac{\Delta_0}{\delta_g}\,\sqrt{\kappa}\log(1/\sigma_\star)\Big)$

The inner-solver bounds in the second column are the classical deterministic first-order complexity estimates for subgradient, gradient, and accelerated gradient methods; see, e.g., Nesterov [28, Chs. 3–4] and Beck [5, Chs. 3–5]. Here γ is the strong convexity modulus of the subproblem P_k, $\kappa = L_P/\gamma$ is its condition number

when P_k is L_P-smooth, and the subgradient rows additionally assume the usual bounded-subgradient constants, absorbed into $O(\cdot)$.

*We highlight the insight of this result about the structural and computational contributions to the total oracle complexity: the outer factor $O(\Delta_0/\delta_g)$ is invariant under any choice of inner solver, while the structure of the subproblem affects only the multiplicative inner factor, which improves from $O(\sigma_\star^{-2})$ to $O(\sqrt{\kappa}\log(1/\sigma_\star))$. Taking $\delta_\phi \to 0$, so that $\mathcal{N}_{\mathrm{in}}(\sigma_\star)$ diverges, leaves the outer factor unchanged: even an exact subproblem solver only certifies a $(0, \delta_g, 0)$-critical point; see Theorem 4.1. This residual δ_g therefore originates from the DCA mechanism itself, namely the mismatch between the subgradient $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$ fixed at the old iterate and the certificate evaluated at the new iterate $\boldsymbol{x}^{k+1}$, rather than from any deficiency of the inner solver. This reflects, at the level of oracle complexity, the asymmetry of Definition 3.1: δ_g is a structural cost of the nonsmooth component g, whereas δ_ϕ is a computational cost that enters only through $\mathcal{N}_{\mathrm{in}}$. This decomposition constitutes an algorithmic rationale for adopting two-sided tolerances in Definition 3.1: merging δ_ϕ and δ_g into a single parameter forces the inner accuracy to match the outer structural tolerance, eliminating the freedom to decouple the two costs that the schedule $\sigma_k \equiv \sigma_\star = \min\{\delta_\phi, \theta\delta_g\}$ in Theorem 4.2 exploits. We emphasize that this invariance is established within the DCA linearization framework and the two-point certificate structure of Definition 3.1; whether it persists as a lower bound over a broader class of first-order algorithms for Problem (**P**) is left open.*

4.4 Iteration Complexity of Proximal DCA-type Methods

We now analyze proximal DCA-type updates. A proximal term shifts the ϕ-side certificate from $\boldsymbol{v}^k$ to a transported vector of the form $\boldsymbol{u}^{k+1} = \boldsymbol{v}^k - \gamma^{-1}(\boldsymbol{x}^{k+1} - \boldsymbol{x}^k)$; the resulting step-proportional residual $r\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|$ is what activates the mismatch slot ε, whereas bare DCA, which has no proximal term, produces no shift and activates only the Fenchel-gap slots δ_ϕ, δ_g. The two deterministic updates—the single-loop proximal-subgradient DCA (Algorithm 3) and the exact proximal DCA as its special case—share this mechanism exactly, so we isolate their common estimates as Condition 4.1, prove the complexity bound once at that level, and then verify the condition for each. The stochastic variant of PS-DCA follows the same template but adds a first-order gradient-error term to the residual; it is treated separately in Section 5. Throughout this subsection, for $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$ and $s^k := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$, define the transported Fenchel gap of g

$$\delta_g^{k+1} := \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) = g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) - \langle \boldsymbol{v}^k, \boldsymbol{x}^{k+1} - \boldsymbol{x}^k \rangle. \tag{4.14}$$

The equality uses $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$ and the Fenchel–Young identity. In particular,

$$\boldsymbol{v}^k \in \partial_{\delta_g^{k+1}} g(\boldsymbol{x}^{k+1})$$

by Lemma 3.1: the subgradient $\boldsymbol{v}^k$ that is exact at $\boldsymbol{x}^k$ becomes a δ_g^{k+1}-subgradient once transported to $\boldsymbol{x}^{k+1}$.

Condition 4.1. *Given $\boldsymbol{x}^0 \in \mathrm{dom}\phi$ and constants $a > 0$, $r > 0$, the sequences $\{\boldsymbol{x}^{k+1}\}$ and $\{\boldsymbol{u}^{k+1}\}$, together with a choice $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$, satisfy the following: for every k there exists $\sigma_k \geq 0$ such that the ϕ-side certificate*

$$\boldsymbol{u}^{k+1} \in \partial_{\sigma_k} \phi(\boldsymbol{x}^{k+1}), \tag{4.15}$$

$$\|\boldsymbol{u}^{k+1} - \boldsymbol{v}^k\| \leq r\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|, \tag{4.16}$$

and the descent estimate

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \geq \delta_g^{k+1} + a\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 - \sigma_k \tag{4.17}$$

hold.

The next lemma is the abstract accounting device behind the deterministic results in this subsection. It separates the two effects in the certificate: the ϕ-side Fenchel tolerance and the transported g-side Fenchel tolerance.

Lemma 4.4. *Suppose $F^\star > -\infty$ and let $\Delta_0 := F(\boldsymbol{x}^0) - F^\star$. Let $K \geq 1$, and suppose Condition 4.1 holds for $k = 0, \ldots, K-1$ with*

$$0 \leq \sigma_k \leq \bar{\sigma}.$$

Define

$$\hat{k} \in \operatorname*{argmin}_{0 \leq k \leq K-1} \left\{ \delta_g^{k+1} + a\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \right\}, \qquad \bar{\boldsymbol{x}} := \boldsymbol{x}^{\hat{k}+1}.$$

Then $\bar{\boldsymbol{x}}$ satisfies

$$\delta_g^{\hat{k}+1} \leq \frac{\Delta_0}{K} + \bar{\sigma}, \tag{4.18}$$

$$\|\boldsymbol{x}^{\hat{k}+1} - \boldsymbol{x}^{\hat{k}}\| \leq \sqrt{\frac{1}{a}\left(\frac{\Delta_0}{K} + \bar{\sigma}\right)}, \tag{4.19}$$

and hence

$$\operatorname{dist}\left(\partial_{\bar{\sigma}}\phi(\bar{\boldsymbol{x}}), \partial_{\frac{\Delta_0}{K}+\bar{\sigma}} g(\bar{\boldsymbol{x}})\right) \leq r\sqrt{\frac{1}{a}\left(\frac{\Delta_0}{K} + \bar{\sigma}\right)}. \tag{4.20}$$

Consequently, $\bar{\boldsymbol{x}}$ is a

$$\left(\bar{\sigma},\ \frac{\Delta_0}{K} + \bar{\sigma},\ r\sqrt{\frac{1}{a}\left(\frac{\Delta_0}{K} + \bar{\sigma}\right)}\right)\text{-critical point.}$$

Proof. Summing the descent estimate (4.17) over $k = 0, \ldots, K-1$ gives

$$\sum_{k=0}^{K-1} \left[\delta_g^{k+1} + a\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\right] \leq F(\boldsymbol{x}^0) - F(\boldsymbol{x}^K) + \sum_{k=0}^{K-1} \sigma_k.$$

Since $F(\boldsymbol{x}^K) \geq F^\star$ and $\sigma_k \leq \bar{\sigma}$, we obtain

$$\sum_{k=0}^{K-1} \left[\delta_g^{k+1} + a\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\right] \leq \Delta_0 + K\bar{\sigma}. \tag{4.21}$$

By the definition of $\hat{k}$,

$$\delta_g^{\hat{k}+1} + a\|\boldsymbol{x}^{\hat{k}+1} - \boldsymbol{x}^{\hat{k}}\|^2 \leq \frac{\Delta_0}{K} + \bar{\sigma}.$$

Both terms on the left-hand side are nonnegative, and hence (4.18) and (4.19) follow.

It remains to verify the same-point certificate. By (4.15) and $\sigma_{\hat{k}} \leq \bar{\sigma}$,

$$\boldsymbol{u}^{\hat{k}+1} \in \partial_{\sigma_{\hat{k}}}\phi(\boldsymbol{x}^{\hat{k}+1}) \subseteq \partial_{\bar{\sigma}}\phi(\boldsymbol{x}^{\hat{k}+1}).$$

By the definition of $\delta_g^{\hat{k}+1}$ and (4.18),

$$\boldsymbol{v}^{\hat{k}} \in \partial_{\delta_g^{\hat{k}+1}} g(\boldsymbol{x}^{\hat{k}+1}) \subseteq \partial_{\frac{\Delta_0}{K}+\bar{\sigma}} g(\boldsymbol{x}^{\hat{k}+1}).$$

Therefore

$$\begin{aligned}\operatorname{dist}\left(\partial_{\bar{\sigma}}\phi(\boldsymbol{x}^{\hat{k}+1}), \partial_{\frac{\Delta_0}{K}+\bar{\sigma}} g(\boldsymbol{x}^{\hat{k}+1})\right) &\leq \|\boldsymbol{u}^{\hat{k}+1} - \boldsymbol{v}^{\hat{k}}\| \\ &\leq r\|\boldsymbol{x}^{\hat{k}+1} - \boldsymbol{x}^{\hat{k}}\|.\end{aligned}$$

Combining this inequality with (4.19) proves (4.20). □

The selection of $\hat{k}$ uses only computable quantities: δ_g^{k+1} is evaluated from function values of g and the chosen $\boldsymbol{v}^k$, and $\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|$ is observed, so $\hat{k}$ is implementable without knowledge of the certificate tolerance σ_k.

Theorem 4.3. *Suppose the assumptions of Lemma 4.4 hold. Let target tolerances $\delta_\phi, \delta_g, \varepsilon > 0$ be given, and choose $\bar{\sigma}$ such that*

$$0 \le \bar{\sigma} \le \delta_\phi, \qquad \bar{\sigma} < \min\left\{\delta_g,\ \frac{a\varepsilon^2}{r^2}\right\}. \tag{4.22}$$

If

$$K \ge \left\lceil \Delta_0 \max\left\{\frac{1}{\delta_g - \bar{\sigma}},\ \frac{1}{\frac{a\varepsilon^2}{r^2} - \bar{\sigma}}\right\}\right\rceil, \tag{4.23}$$

then the output $\bar{\boldsymbol{x}}$ of Lemma 4.4 is a $(\delta_\phi, \delta_g, \varepsilon)$-critical point.

Proof. The first lower bound in (4.23) gives

$$\frac{\Delta_0}{K} + \bar{\sigma} \le \delta_g.$$

The second lower bound gives

$$\frac{\Delta_0}{K} + \bar{\sigma} \le \frac{a\varepsilon^2}{r^2}.$$

Substituting these two inequalities into (4.20) yields

$$\operatorname{dist}\left(\partial_{\bar{\sigma}}\phi(\bar{\boldsymbol{x}}), \partial_{\delta_g} g(\bar{\boldsymbol{x}})\right) \le r\sqrt{\frac{1}{a}\cdot\frac{a\varepsilon^2}{r^2}} = \varepsilon.$$

Finally, $\bar{\sigma} \le \delta_\phi$ implies $\partial_{\bar{\sigma}}\phi(\bar{\boldsymbol{x}}) \subseteq \partial_{\delta_\phi}\phi(\bar{\boldsymbol{x}})$, so $\bar{\boldsymbol{x}}$ is $(\delta_\phi, \delta_g, \varepsilon)$-critical. □

We now verify Condition 4.1 for concrete updates. We consider the composite case

$$F(\boldsymbol{x}) = f(\boldsymbol{x}) + h(\boldsymbol{x}) - g(\boldsymbol{x}), \qquad \phi := f + h, \tag{4.24}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is convex and L_f-smooth, and h, g are proper closed convex functions. We assume that the proximal mapping of h is easy to compute. This setting gives a single-loop realization of the preceding certificate theorem.

4.4.1 Proximal-Subgradient DCA

For a stepsize $\eta \in (0, 2/L_f)$, the proximal-subgradient DCA (PS-DCA) update is

$$\boldsymbol{x}^{k+1} = \operatorname{prox}_{\eta h}\left(\boldsymbol{x}^k - \eta(\nabla f(\boldsymbol{x}^k) - \boldsymbol{v}^k)\right), \qquad \boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k). \tag{4.25}$$

Equivalently,

$$\boldsymbol{x}^{k+1} = \operatorname*{argmin}_{\boldsymbol{x}\in\mathbb{R}^n}\left\{h(\boldsymbol{x}) + \langle \nabla f(\boldsymbol{x}^k) - \boldsymbol{v}^k, \boldsymbol{x}\rangle + \frac{1}{2\eta}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2\right\}.$$

Define

$$a_\eta := \frac{1}{\eta} - \frac{L_f}{2} > 0, \qquad r_\eta := \frac{1}{\eta} + L_f.$$

Lemma 4.5 (PS-DCA realization)**.** *Assume* (4.24) *and let $\{\boldsymbol{x}^k\}_{k\ge 0}$ be generated by* (4.25)*, with a_η, r_η defined above. Then Condition 4.1 holds with*

$$a = a_\eta, \qquad r = r_\eta, \qquad \sigma_k = 0.$$

Algorithm 3: Proximal-Subgradient DCA (PS-DCA)

Input: $\boldsymbol{x}^0 \in \mathrm{dom}\phi$, stepsize $\eta \in (0, 2/L_f)$, iteration number $K \geq 1$.
for $k = 0, 1, \ldots, K-1$ **do**
  Choose $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$.
  $\boldsymbol{x}^{k+1} = \mathrm{prox}_{\eta h}\big(\boldsymbol{x}^k - \eta(\nabla f(\boldsymbol{x}^k) - \boldsymbol{v}^k)\big)$.
end for
Choose $\hat{k} \in \mathrm{argmin}_{0\leq k\leq K-1}\big\{\delta_g^{k+1} + a_\eta\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\big\}$, with δ_g^{k+1} as in (4.14).
Output: $\bar{\boldsymbol{x}} = \boldsymbol{x}^{\hat{k}+1}$.

More precisely, there exists $\boldsymbol{u}^{k+1} \in \partial\phi(\boldsymbol{x}^{k+1})$ such that

$$\|\boldsymbol{u}^{k+1} - \boldsymbol{v}^k\| \leq r_\eta\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|, \tag{4.26}$$

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \geq \delta_g^{k+1} + a_\eta\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2. \tag{4.27}$$

Proof. Let $s^k := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$. The optimality condition of (4.25) gives

$$\boldsymbol{0} \in \partial h(\boldsymbol{x}^{k+1}) + \nabla f(\boldsymbol{x}^k) - \boldsymbol{v}^k + \frac{1}{\eta}s^k.$$

Hence there exists $\boldsymbol{w}^{k+1} \in \partial h(\boldsymbol{x}^{k+1})$ such that

$$\boldsymbol{w}^{k+1} = \boldsymbol{v}^k - \nabla f(\boldsymbol{x}^k) - \frac{1}{\eta}s^k.$$

Since f is convex and differentiable,

$$\boldsymbol{u}^{k+1} := \nabla f(\boldsymbol{x}^{k+1}) + \boldsymbol{w}^{k+1} \in \partial(f+h)(\boldsymbol{x}^{k+1}) = \partial\phi(\boldsymbol{x}^{k+1}).$$

Moreover,

$$\begin{aligned}\|\boldsymbol{u}^{k+1} - \boldsymbol{v}^k\| &= \left\|\nabla f(\boldsymbol{x}^{k+1}) - \nabla f(\boldsymbol{x}^k) - \frac{1}{\eta}s^k\right\| \\ &\leq L_f\|s^k\| + \frac{1}{\eta}\|s^k\| = r_\eta\|s^k\|,\end{aligned}$$

which proves (4.26).

It remains to prove the descent estimate. Define

$$Q_k(\boldsymbol{x}) := h(\boldsymbol{x}) + \langle\nabla f(\boldsymbol{x}^k) - \boldsymbol{v}^k, \boldsymbol{x}\rangle + \frac{1}{2\eta}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2.$$

The function Q_k is η^{-1}-strongly convex, and $\boldsymbol{x}^{k+1}$ minimizes Q_k. Hence

$$Q_k(\boldsymbol{x}^k) - Q_k(\boldsymbol{x}^{k+1}) \geq \frac{1}{2\eta}\|s^k\|^2.$$

Expanding this inequality gives

$$h(\boldsymbol{x}^k) - h(\boldsymbol{x}^{k+1}) + \langle\boldsymbol{v}^k - \nabla f(\boldsymbol{x}^k), s^k\rangle \geq \frac{1}{\eta}\|s^k\|^2. \tag{4.28}$$

By L_f-smoothness of f,

$$f(\boldsymbol{x}^{k+1}) \leq f(\boldsymbol{x}^k) + \langle\nabla f(\boldsymbol{x}^k), s^k\rangle + \frac{L_f}{2}\|s^k\|^2,$$

or equivalently

$$f(\boldsymbol{x}^k) - f(\boldsymbol{x}^{k+1}) \geq -\langle \nabla f(\boldsymbol{x}^k), s^k \rangle - \frac{L_f}{2}\|s^k\|^2. \tag{4.29}$$

Combining (4.28) and (4.29), and using

$$\delta_g^{k+1} = g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) - \langle \boldsymbol{v}^k, s^k \rangle,$$

we obtain

$$\begin{aligned} F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) &= f(\boldsymbol{x}^k) - f(\boldsymbol{x}^{k+1}) + h(\boldsymbol{x}^k) - h(\boldsymbol{x}^{k+1}) + g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) \\ &= f(\boldsymbol{x}^k) - f(\boldsymbol{x}^{k+1}) + h(\boldsymbol{x}^k) - h(\boldsymbol{x}^{k+1}) + \langle \boldsymbol{v}^k, s^k \rangle + \delta_g^{k+1} \\ &\geq \left(\frac{1}{\eta} - \frac{L_f}{2}\right)\|s^k\|^2 + \delta_g^{k+1}. \end{aligned}$$

This proves (4.27). □

Corollary 4.1 (Iteration complexity of PS-DCA)**.** *Suppose $F^\star > -\infty$ and the assumptions of Lemma 4.5 hold. Let $\eta \in (0, 2/L_f)$ and let $\{\boldsymbol{x}^k\}_{k\geq 0}$ be generated by (4.25). Then for every $K \geq 1$, there exists $\hat{k} \in \{0, \ldots, K-1\}$ such that $\bar{\boldsymbol{x}} := \boldsymbol{x}^{\hat{k}+1}$ is a*

$$\left(0,\ \frac{\Delta_0}{K},\ r_\eta\sqrt{\frac{\Delta_0}{a_\eta K}}\right)\text{-critical point.} \tag{4.30}$$

Consequently, to obtain a $(0, \delta_g, \varepsilon)$-critical point, it suffices to take

$$K \geq \left\lceil \max\left\{ \frac{\Delta_0}{\delta_g},\ \frac{r_\eta^2 \Delta_0}{a_\eta \varepsilon^2} \right\} \right\rceil. \tag{4.31}$$

In particular, for the standard choice $\eta = 1/L_f$,

$$a_\eta = \frac{L_f}{2}, \qquad r_\eta = 2L_f,$$

and (4.31) becomes

$$K \geq \left\lceil \max\left\{ \frac{\Delta_0}{\delta_g},\ \frac{8L_f \Delta_0}{\varepsilon^2} \right\} \right\rceil. \tag{4.32}$$

The constant in the ε^{-2} term is minimized within the bound (4.31) by taking $\eta = 1/(2L_f)$, which gives

$$\frac{r_\eta^2}{a_\eta} = 6L_f.$$

Proof. Apply Lemma 4.4 with $a = a_\eta$, $r = r_\eta$, and $\bar{\sigma} = 0$, using Lemma 4.5. This gives

$$\delta_g^{\hat{k}+1} \leq \frac{\Delta_0}{K}, \qquad \|\boldsymbol{u}^{\hat{k}+1} - \boldsymbol{v}^{\hat{k}}\| \leq r_\eta\sqrt{\frac{\Delta_0}{a_\eta K}},$$

which is exactly (4.30). The target bound (4.31) follows by requiring $\Delta_0/K \leq \delta_g$ and $r_\eta\sqrt{\Delta_0/(a_\eta K)} \leq \varepsilon$.

For $\eta = 1/L_f$,

$$\frac{r_\eta^2}{a_\eta} = \frac{(2L_f)^2}{L_f/2} = 8L_f.$$

For the last claim, set $a = 1/\eta$ and minimize

$$\frac{(a + L_f)^2}{a - L_f/2} \qquad \text{over } a > L_f/2.$$

The derivative vanishes at $a = 2L_f$, i.e., at $\eta = 1/(2L_f)$, and the corresponding value is

$$\frac{(3L_f)^2}{3L_f/2} = 6L_f.$$

□

4.4.2 Exact Proximal DCA as a Special Case

The classical exact proximal DCA update with parameter $\gamma > 0$,

$$\boldsymbol{x}^{k+1} = \operatorname*{argmin}_{\boldsymbol{x}\in\mathbb{R}^n} \left\{ \phi(\boldsymbol{x}) - \langle \boldsymbol{v}^k, \boldsymbol{x}\rangle + \frac{1}{2\gamma}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2 \right\}, \qquad \boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k), \tag{4.33}$$

is the instance of (4.25) obtained by taking $f \equiv 0$ and $h = \phi$ with stepsize $\eta = \gamma$: the smooth part vanishes, so $L_f = 0$, every stepsize $\gamma > 0$ is admissible, and the update reduces to $\boldsymbol{x}^{k+1} = \operatorname{prox}_{\gamma\phi}(\boldsymbol{x}^k + \gamma\boldsymbol{v}^k)$, which is (4.33). The quadratic term makes the subproblem strongly convex, so the minimizer is unique whenever ϕ is proper closed convex; computing it is exactly the proximal oracle for the full convex part ϕ. Lemma 4.5 applies with $L_f = 0$ and yields the realization constants

$$a = \frac{1}{\gamma}, \qquad r = \frac{1}{\gamma}, \qquad \sigma_k = 0,$$

with the shifted dual certificate $\boldsymbol{u}^{k+1} = \boldsymbol{v}^k - \gamma^{-1}(\boldsymbol{x}^{k+1} - \boldsymbol{x}^k) \in \partial\phi(\boldsymbol{x}^{k+1})$.

Corollary 4.2 (Iteration complexity of exact proximal DCA)**.** *Suppose $F^\star > -\infty$, let ϕ and g be proper closed convex functions, and let $\{\boldsymbol{x}^k\}_{k\geq 0}$ be generated by* (4.33)*. Then for every $K \geq 1$, there exists $\hat{k} \in \{0, \dots, K-1\}$ such that $\bar{\boldsymbol{x}} := \boldsymbol{x}^{\hat{k}+1}$ is a*

$$\left(0,\ \frac{\Delta_0}{K},\ \sqrt{\frac{\Delta_0}{\gamma K}}\right) \text{-critical point.} \tag{4.34}$$

Consequently, to obtain a $(0, \delta_g, \varepsilon)$-critical point, it suffices to take

$$K \geq \left\lceil \max\left\{ \frac{\Delta_0}{\delta_g},\ \frac{\Delta_0}{\gamma\varepsilon^2} \right\} \right\rceil. \tag{4.35}$$

In particular, if $K \geq \lceil \Delta_0/(\gamma\varepsilon^2) \rceil$, then the same output is a $(0, \gamma\varepsilon^2, \varepsilon)$-critical point.

Proof. Apply Lemma 4.4 with the constants $a = r = \gamma^{-1}$ and $\bar{\sigma} = 0$ supplied by Lemma 4.5 with $f \equiv 0$ and $\eta = \gamma$. Since $r\sqrt{\Delta_0/(aK)} = \sqrt{\Delta_0/(\gamma K)}$, the abstract bound (4.20) gives (4.34), and (4.35) follows by imposing $\Delta_0/K \leq \delta_g$ and $\sqrt{\Delta_0/(\gamma K)} \leq \varepsilon$. □

Remark 4.3 (Realizations and activated slots)**.** *The two realizations above fit the same abstract certificate template:*

method	a	r	δ_ϕ*-slot*	ε*-slot*
exact proximal DCA	γ^{-1}	γ^{-1}	0	$\gamma^{-1}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|$
PS-DCA for $\phi = f + h$	$\eta^{-1} - L_f/2$	$\eta^{-1} + L_f$	0	$(\eta^{-1} + L_f)\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|$

Thus exact proximal DCA and PS-DCA both activate the ε-slot, whereas classical bare DCA activates only the Fenchel-gap slots δ_ϕ, δ_g.

Remark 4.4 (Value-gap inexactness for the proximal subproblem)**.** *The parameter σ_k in* (4.15)–(4.17) *is a certificate/decrease tolerance. A pure value-gap condition for the proximalized subproblem,*

$$M_k(\boldsymbol{x}^{k+1}) \le \inf_{\boldsymbol{x}} M_k(\boldsymbol{x}) + \sigma_k, \qquad M_k(\boldsymbol{x}) := \phi(\boldsymbol{x}) - \langle \boldsymbol{v}^k, \boldsymbol{x}\rangle + \frac{1}{2\gamma}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2,$$

does not by itself imply

$$\boldsymbol{v}^k - \gamma^{-1}(\boldsymbol{x}^{k+1} - \boldsymbol{x}^k) \in \partial_{\sigma_k}\phi(\boldsymbol{x}^{k+1}).$$

Indeed, expanding the value-gap inequality around $\boldsymbol{x}^{k+1}$ leaves an additional term

$$-\frac{1}{2\gamma}\|\boldsymbol{z} - \boldsymbol{x}^{k+1}\|^2$$

in the supporting inequality for ϕ, which is unbounded below as a function of $\boldsymbol{z}$. For this reason, the unified theorem is stated in terms of the verifiable certificate (4.15) *and the descent estimate* (4.17)*. Exact proximal DCA and PS-DCA satisfy these conditions with $\sigma_k = 0$.*

Remark 4.5 (Coupling between δ_g and ε)**.** *Corollary 4.2 gives the same output point with*

$$\delta_g \le \frac{\Delta_0}{K}, \qquad \varepsilon \le \sqrt{\frac{\Delta_0}{\gamma K}}.$$

Therefore, choosing $K = \lceil \Delta_0/(\gamma\varepsilon^2)\rceil$ gives

$$\delta_g \le \gamma\varepsilon^2.$$

This is the precise proximal DCA coupling between the transported Fenchel error and the dual certificate mismatch. It explains why the proximal analysis is the natural source of an $O(\Delta_0/(\gamma\varepsilon^2))$ upper bound for the ε-slot, while the bare DCA analysis gives $\varepsilon = 0$ and only measures δ_g.

4.5 Smoothing and the Two-Sided Certificate

Smoothing the concave part is a third, and popular, route to nonsmooth DC problems. Rather than propose a new method, we use it to make one point: the $(\delta_\phi, \delta_g, \varepsilon)$-criticality of Definition 3.1 is exactly the object such smoothing produces, so existing smoothing guarantees are *relabelings* of our certificate. Fix $\mu > 0$ and recall the Moreau envelope of g,

$$g_\mu(\boldsymbol{x}) := \min_{\boldsymbol{y}\in\mathbb{R}^n}\Big\{g(\boldsymbol{y}) + \tfrac{1}{2\mu}\|\boldsymbol{y} - \boldsymbol{x}\|^2\Big\}, \qquad \tilde{\boldsymbol{x}} := \operatorname{prox}_{\mu g}(\boldsymbol{x}) = \operatorname*{argmin}_{\boldsymbol{y}\in\mathbb{R}^n}\Big\{g(\boldsymbol{y}) + \tfrac{1}{2\mu}\|\boldsymbol{y} - \boldsymbol{x}\|^2\Big\}, \tag{4.36}$$

where the minimizer $\tilde{\boldsymbol{x}}$ is unique because the inner objective is $\frac{1}{\mu}$-strongly convex. Throughout this subsection we take $(\phi, g) \in \mathcal{D}(\Delta, L, G)$, so both convex components are finite-valued with uniformly bounded subgradients; accordingly we write

$$G_g := \sup_{\boldsymbol{x}} \sup_{\boldsymbol{v}\in\partial g(\boldsymbol{x})} \|\boldsymbol{v}\| \le G, \qquad G_\phi := \sup_{\boldsymbol{x}} \sup_{\boldsymbol{u}\in\partial\phi(\boldsymbol{x})} \|\boldsymbol{u}\| \le G.$$

Replacing g by g_μ renders the concave part smooth, and ε-criticality of the smoothed problem $\phi - g_\mu$ reads $\operatorname{dist}\big(\partial\phi(\boldsymbol{x}), \{\nabla g_\mu(\boldsymbol{x})\}\big) \le \varepsilon$. The next lemma shows that this smoothed certificate is, verbatim, a $(\delta_\phi, \delta_g, \varepsilon)$-certificate for the original nonsmooth problem, with an explicit δ_g that is linear in μ.

Lemma 4.6 (One-sided Moreau reduction)**.** *Let $(\phi, g) \in \mathcal{D}(\Delta, L, G)$, fix $\mu > 0$, and suppose $\boldsymbol{x} \in \mathbb{R}^n$ satisfies the ε-criticality condition for the smoothed problem $\phi - g_\mu$,*

$$\operatorname{dist}\big(\partial\phi(\boldsymbol{x}), \{\nabla g_\mu(\boldsymbol{x})\}\big) \le \varepsilon;$$

equivalently, there exists $\boldsymbol{u} \in \partial\phi(\boldsymbol{x})$ with $\|\boldsymbol{u} - \nabla g_\mu(\boldsymbol{x})\| \le \varepsilon$. Then $\boldsymbol{x}$ is a $\big(0,\, 2G_g^2\mu,\, \varepsilon\big)$-critical point of the original problem $(\mathbf{P})$ in the sense of Definition 3.1.

Proof. Write $\boldsymbol{v} := \nabla g_\mu(\boldsymbol{x})$ and $\tilde{\boldsymbol{x}} := \operatorname{prox}_{\mu g}(\boldsymbol{x})$.

Step 1 (Moreau identities). The optimality condition for $\tilde{\boldsymbol{x}}$ in (4.36) is $\mathbf{0} \in \partial g(\tilde{\boldsymbol{x}}) + \frac{1}{\mu}(\tilde{\boldsymbol{x}} - \boldsymbol{x})$, hence

$$\boldsymbol{v} = \nabla g_\mu(\boldsymbol{x}) = \frac{\boldsymbol{x} - \tilde{\boldsymbol{x}}}{\mu} \in \partial g(\tilde{\boldsymbol{x}}). \tag{4.37}$$

Since $\boldsymbol{v} \in \partial g(\tilde{\boldsymbol{x}})$ and the subgradients of g are bounded by G_g, taking norms in (4.37) gives

$$\|\boldsymbol{x} - \tilde{\boldsymbol{x}}\| = \mu\|\boldsymbol{v}\| \le \mu G_g. \tag{4.38}$$

Step 2 (Transported Fenchel gap). Because $\boldsymbol{v} \in \partial g(\tilde{\boldsymbol{x}})$, the Fenchel–Young equality of Lemma 3.1 at $\tilde{\boldsymbol{x}}$ yields $g^*(\boldsymbol{v}) = \langle \boldsymbol{v}, \tilde{\boldsymbol{x}}\rangle - g(\tilde{\boldsymbol{x}})$. Substituting this into the gap function evaluated at the base point $\boldsymbol{x}$,

$$\Phi_g(\boldsymbol{x}, \boldsymbol{v}) = g(\boldsymbol{x}) + g^*(\boldsymbol{v}) - \langle \boldsymbol{v}, \boldsymbol{x}\rangle = g(\boldsymbol{x}) - g(\tilde{\boldsymbol{x}}) - \langle \boldsymbol{v}, \boldsymbol{x} - \tilde{\boldsymbol{x}}\rangle. \tag{4.39}$$

Step 3 (Bounding the gap). Pick any $\boldsymbol{s}_{\boldsymbol{x}} \in \partial g(\boldsymbol{x})$, so $\|\boldsymbol{s}_{\boldsymbol{x}}\| \le G_g$; convexity of g gives the subgradient inequality $g(\boldsymbol{x}) - g(\tilde{\boldsymbol{x}}) \le \langle \boldsymbol{s}_{\boldsymbol{x}}, \boldsymbol{x} - \tilde{\boldsymbol{x}}\rangle$. Combining this with (4.39), the Cauchy–Schwarz and triangle inequalities, (4.37), and (4.38),

$$\begin{aligned}\Phi_g(\boldsymbol{x}, \boldsymbol{v}) &\le \langle \boldsymbol{s}_{\boldsymbol{x}} - \boldsymbol{v},\, \boldsymbol{x} - \tilde{\boldsymbol{x}}\rangle \le \|\boldsymbol{s}_{\boldsymbol{x}} - \boldsymbol{v}\|\, \|\boldsymbol{x} - \tilde{\boldsymbol{x}}\| \le \big(\|\boldsymbol{s}_{\boldsymbol{x}}\| + \|\boldsymbol{v}\|\big)\, \|\boldsymbol{x} - \tilde{\boldsymbol{x}}\| \\ &\le 2G_g\, \|\boldsymbol{x} - \tilde{\boldsymbol{x}}\| \le 2G_g^2\mu,\end{aligned} \tag{4.40}$$

where the last two inequalities use $\|\boldsymbol{s}_{\boldsymbol{x}}\| \le G_g$, $\|\boldsymbol{v}\| \le G_g$, and (4.38).

Step 4 (Assembling the certificate). By (4.40) and Lemma 3.1, $\boldsymbol{v} \in \partial_{2G_g^2\mu}\, g(\boldsymbol{x})$. The hypothesis furnishes $\boldsymbol{u} \in \partial\phi(\boldsymbol{x}) = \partial_0\phi(\boldsymbol{x})$, so $\Phi_\phi(\boldsymbol{x}, \boldsymbol{u}) = 0$, and $\|\boldsymbol{u} - \boldsymbol{v}\| = \|\boldsymbol{u} - \nabla g_\mu(\boldsymbol{x})\| \le \varepsilon$. The pair $(\boldsymbol{u}, \boldsymbol{v})$ therefore witnesses

$$\Phi_\phi(\boldsymbol{x}, \boldsymbol{u}) \le 0, \qquad \Phi_g(\boldsymbol{x}, \boldsymbol{v}) \le 2G_g^2\mu, \qquad \|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon,$$

which is exactly the Fenchel-gap form of a $(0, 2G_g^2\mu, \varepsilon)$-critical point in Definition 3.1. □

Corollary 4.3 (Two-sided DME reduction)**.** *Let $(\phi, g) \in \mathcal{D}(\Delta, L, G)$ and fix $\mu_\phi, \mu_g > 0$. Consider the difference-of-Moreau-envelopes surrogate $\phi_{\mu_\phi} - g_{\mu_g}$, where ϕ_{μ_ϕ} and g_{μ_g} are the Moreau envelopes of ϕ and g. If $\boldsymbol{x}$ is an ε-critical point of this smooth surrogate, i.e. $\|\nabla\phi_{\mu_\phi}(\boldsymbol{x}) - \nabla g_{\mu_g}(\boldsymbol{x})\| \le \varepsilon$, then $\boldsymbol{x}$ is a $\big(2G_\phi^2\mu_\phi,\, 2G_g^2\mu_g,\, \varepsilon\big)$-critical point of the original problem $(\mathbf{P})$.*

Proof. Apply the argument of Lemma 4.6 to each component. Put $\boldsymbol{u} := \nabla\phi_{\mu_\phi}(\boldsymbol{x})$ and $\boldsymbol{v} := \nabla g_{\mu_g}(\boldsymbol{x})$. Steps 1–3, run on g with parameter μ_g, give $\boldsymbol{v} \in \partial_{2G_g^2\mu_g}\, g(\boldsymbol{x})$; the same steps, run symmetrically on ϕ with parameter μ_ϕ (with $\operatorname{prox}_{\mu_\phi\phi}$ in place of $\operatorname{prox}_{\mu_g g}$ and G_ϕ in place of G_g), give $\boldsymbol{u} \in \partial_{2G_\phi^2\mu_\phi}\, \phi(\boldsymbol{x})$. The surrogate stationarity $\|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon$ then closes the certificate through Definition 3.1. □

Remark 4.6 (Smoothing relabels the target, not the rate)**.** *Lemma 4.6 and Corollary 4.3 place existing smoothing methods on the two-sided certificate and separate what their guarantees keep from what they merely relabel. Driving $\mu \to 0$ sends $\delta_g = 2G_g^2\mu \to 0$, so smoothed stationarity recovers exact DC criticality in the limit, while any fixed $\mu > 0$ delivers a genuine same-point certificate with an explicit, tunable $\delta_g = O(\mu)$.*

What is unchanged—the iteration exponent. *In the deterministic regime, the difference-of-Moreau-envelopes scheme of Sun and Sun [33] reaches an ε-stationary point of the surrogate $\phi_{\mu_\phi} - g_{\mu_g}$ in $O(\varepsilon^{-2})$ proximal evaluations; composing with Corollary 4.3 turns this into a $(2G_\phi^2\mu_\phi, 2G_g^2\mu_g, \varepsilon)$-certificate for the original problem at the* same *$O(\varepsilon^{-2})$ rate reached here without any smoothing (Corollary 4.2). In the stochastic regime, single-loop methods that optimize a smoothed objective—for instance Chayti and Jaggi [10] and Hu et al. [17]—reach ε-stationarity in $O(\varepsilon^{-4})$ iterations; since the reduction only relabels the point already produced, this exponent is inherited unchanged on the two-sided certificate, the smoothing parameter surfacing as the δ_g budget.*

What changes—the target. *A single ε-stationarity condition on a smooth surrogate becomes the three-slot $(\delta_\phi, \delta_g, \varepsilon)$-certificate of the original nonsmooth problem, in which the smoothing parameters appear as the explicit, separately controllable budgets $\delta_\phi = O(\mu_\phi)$ and $\delta_g = O(\mu_g)$. The bias hidden inside the surrogate—the discrepancy between the smoothed and the original objective—is thereby exposed as a certificate slot rather than left implicit. Smoothing is consequently dispensable in the deterministic regime, where the transported Fenchel gap δ_g already absorbs the nonsmoothness of g directly; it re-enters only as an algorithmic device that renders a component differentiable, which is precisely the structure a stochastic variance-reduced analysis can exploit.*

5 Oracle Complexity of Stochastic Case

We consider the stochastic composite DC problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) := f(\boldsymbol{x}) + h(\boldsymbol{x}) - g(\boldsymbol{x}), \qquad f(\boldsymbol{x}) := \mathbb{E}_\xi\big[f(\boldsymbol{x};\xi)\big], \tag{5.1}$$

where ξ denotes a random sample, the smooth component $f : \mathbb{R}^n \to \mathbb{R}$ is L_f-smooth and convex, and h, g are proper closed convex functions with g possibly nonsmooth, so that F is generally nonconvex and nonsmooth. Only f is stochastic: we access ∇f through unbiased samples $\nabla f(\boldsymbol{x};\xi)$ with $\mathbb{E}_\xi[\nabla f(\boldsymbol{x};\xi)] = \nabla f(\boldsymbol{x})$ and variance at most σ^2, while h and g are deterministic. From these samples we form a recursive-momentum (STORM) estimator $\boldsymbol{d}^k$ of $\nabla f(\boldsymbol{x}^k)$, which is variance-reduced but *not* conditionally unbiased. When f is smooth but nonconvex, the convex-side certificate below is unavailable, and one reverts to the concave-side decomposition (5.2), where the estimator error inflates the negative-side tolerance instead; we keep f convex for the cleaner two-sided statement.

Our analysis is organized around a single device, the same-point two-sided criticality certificate $(\delta_\phi, \delta_g, \varepsilon)$ of Definition 3.1, and asks where sampling noise enters this certificate and what removing it costs. Along the way, the two DC decompositions we use, concave-side and convex-side, generate *identical iterates* and carry equivalent certificate information: they reallocate the iterate movement $\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|$ between the tolerance and mismatch slots rather than favoring either. We develop the convex-side certificate because it keeps the original decomposition (ϕ, g), and hence the notation and target of the deterministic analysis, whereas the concave-side certificate is what remains available when f is nonconvex. The results rest on three building blocks: (i) a realization lemma (Lemma 5.1), which places the single-loop prox-subgradient iterate *pathwise* into the abstract template of Condition 4.1 and shows that the positive-side tolerance δ_ϕ vanishes exactly; (ii) a modular certificate (Corollary 5.1), which collapses the triple $(\delta_\phi, \delta_g, \varepsilon)$ onto a single averaged error $\bar{V}$; and (iii) two oracle instantiations (Corollary 5.3 and Proposition 5.1), which meet the resulting budget at cost $O(\varepsilon^{-4})$ and $O(\varepsilon^{-3})$, respectively. A recurring theme is that the estimator error enters the certificate at *two different powers*, so that variance reduction, and not a smaller step size, is what buys a small mismatch slot.

5.1 Stochastic DCA with momentum

Fix any $\rho > L_f$. By the L_f-smoothness of f, the function $\frac{L_f}{2}\|\boldsymbol{x}\|^2 - f(\boldsymbol{x})$ is convex. Consequently, $\frac{\rho}{2}\|\boldsymbol{x}\|^2 - f(\boldsymbol{x})$ is $(\rho - L_f)$-strongly convex, which yields the following DC decomposition:

$$F(\boldsymbol{x}) = \Phi(\boldsymbol{x}) - G(\boldsymbol{x}), \qquad \Phi(\boldsymbol{x}) := h(\boldsymbol{x}) + \frac{\rho}{2}\|\boldsymbol{x}\|^2, \qquad G(\boldsymbol{x}) := \frac{\rho}{2}\|\boldsymbol{x}\|^2 - f(\boldsymbol{x}) + g(\boldsymbol{x}). \tag{5.2}$$

Algorithm 4: Stochastic DCA with momentum (concave-side linearization)

1: **Input:** Initial point $\boldsymbol{x}^0 \in \mathrm{dom}\Phi$, iteration number $K \geq 1$, curvature parameter $\rho > L_f$.
2: **for** $k = 0, 1, \ldots, K-1$ **do**
3: Compute the STORM (recursive momentum) estimator $\boldsymbol{d}^k$ of $\nabla f(\boldsymbol{x}^k)$.
4: Choose $\boldsymbol{w}^k \in \partial g(\boldsymbol{x}^k)$.
5: Form the estimated subgradient of the concave part $-G$:

$$\zeta^k := \rho \boldsymbol{x}^k - \boldsymbol{d}^k + \boldsymbol{w}^k.$$

6: Solve the ρ-strongly convex subproblem

$$\boldsymbol{x}^{k+1} = \operatorname*{argmin}_{\boldsymbol{x}} \big\{ \Phi(\boldsymbol{x}) - \langle \zeta^k, \boldsymbol{x} \rangle \big\} = \mathrm{prox}_{h/\rho}\left(\boldsymbol{x}^k - \frac{1}{\rho}(\boldsymbol{d}^k - \boldsymbol{w}^k) \right). \tag{5.3}$$

7: **end for**
8: **Output:** $\boldsymbol{x}^K$.

Here, Φ is ρ-strongly convex and G is $(\rho - L_f)$-strongly convex. Classical DCA linearizes the concave part $-G$ at $\boldsymbol{x}^k$ using an estimated subgradient ζ^k of G, replacing the unavailable $\nabla f(\boldsymbol{x}^k)$ with the stochastic estimator $\boldsymbol{d}^k$; the resulting iteration is summarized in Algorithm 4. Because the $\frac{\rho}{2}\|\boldsymbol{x}\|^2$ term makes Φ ρ-strongly convex, the subproblem has a unique minimizer, given in closed form by the proximal step (5.3).

Strategy 2: convex-side linearization (PS-DCA). Keeping the original decomposition $\phi := f + h$, g unchanged, we linearize only the smooth f (proximal-gradient style) and use an exact subgradient of g, exactly as in (4.25), with $\nabla f(\boldsymbol{x}^k)$ replaced by $\boldsymbol{d}^k$:

$$\boldsymbol{x}^{k+1} = \mathrm{prox}_{\eta h}\big(\boldsymbol{x}^k - \eta(\boldsymbol{d}^k - \boldsymbol{v}^k)\big), \qquad \boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k). \tag{5.4}$$

Remark 5.1 (The two updates coincide)**.** *Setting $\eta = 1/\rho$ in* (5.4) *and $\boldsymbol{v}^k = \boldsymbol{w}^k$ recovers exactly* (5.3)*. The two strategies produce the* same iterate sequence*; they differ only in which DC decomposition, (Φ, G) or (ϕ, g), is used to certify criticality of that sequence.*

5.2 The two certificates of a single iterate

The iterate (5.3) solves both DC decompositions at once, so it can be certified against either (ϕ, g) or (Φ, G). We record both certificates and their exact relationship. The two carry equivalent information; they differ only in how they split the iterate movement $s^k := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$ between the negative-side tolerance and the subgradient mismatch.

Convex-side certificate (ϕ, g)**.** With f convex, the vector $\boldsymbol{u}^{k+1} := \nabla f(\boldsymbol{x}^{k+1}) + \boldsymbol{w}^{k+1}$ satisfies $\boldsymbol{u}^{k+1} \in \partial\phi(\boldsymbol{x}^{k+1})$ *exactly*, so $\delta_\phi = 0$ pathwise (Lemma 5.1). The estimator error and the movement both surface in the mismatch slot, $\|\boldsymbol{u}^{k+1} - \boldsymbol{v}^k\| \leq r_\eta \|s^k\| + \|e^k\|$, while the negative-side tolerance $\delta_g^{k+1} = \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k)$ keeps its deterministic definition and telescopes to $O(\Delta_0/K + \eta\bar{V})$ (Corollary 5.1).

Concave-side certificate (Φ, G)**.** The certificate vector on the G-side is a *free* choice, and the right one is the exact subgradient $\xi^k_{\mathrm{true}} := \rho\boldsymbol{x}^k - \nabla f(\boldsymbol{x}^k) + \boldsymbol{w}^k \in \partial G(\boldsymbol{x}^k)$, not the iterate-generating vector $\zeta^k = \rho\boldsymbol{x}^k - \boldsymbol{d}^k + \boldsymbol{w}^k$; the two differ by exactly the estimator error, $\zeta^k - \xi^k_{\mathrm{true}} = e^k$. Transporting ξ^k_{true} to $\boldsymbol{x}^{k+1}$ places, on the same point, $\zeta^k \in \partial\Phi(\boldsymbol{x}^{k+1})$ (so $\delta_\Phi = 0$), $\xi^k_{\mathrm{true}} \in \partial_{\delta_G} G(\boldsymbol{x}^{k+1})$, and a mismatch of exactly $\|\zeta^k - \xi^k_{\mathrm{true}}\| = \|e^k\|$, with

no $\|s^k\|$ term. By ρ-strong convexity of Φ,

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \;\geq\; \tfrac{\rho}{2}\|s^k\|^2 + \delta_G^{k+1} - \langle e^k, s^k\rangle,$$

so δ_G telescopes to $O(\Delta_0/K + \bar{V})$. Certifying with ζ^k itself is not merely looser but ill-posed: the Fenchel gap $\Phi_G(\boldsymbol{x}^k, \zeta^k)$ is $+\infty$ whenever $\zeta^k \notin \mathrm{dom} G^*$, which happens for any $e^k \neq \mathbf{0}$ when G has bounded subgradients (e.g. $f = \frac{\rho}{2}\|\cdot\|^2$, $g = \|\cdot\|_1$).

Exact relationship, and why neither dominates. The two negative-side tolerances differ by the Bregman divergence of the smooth part $G_s := \frac{\rho}{2}\|\cdot\|^2 - f$ of G,

$$\delta_G^{k+1} = \delta_g^{k+1} + D_{G_s}(\boldsymbol{x}^{k+1}, \boldsymbol{x}^k), \qquad 0 \leq D_{G_s}(\boldsymbol{x}^{k+1}, \boldsymbol{x}^k) \leq \tfrac{\rho}{2}\|s^k\|^2. \tag{5.5}$$

The movement $\|s^k\|$ thus appears in *both* certificates: linearly in the convex-side mismatch, and quadratically inside the concave-side tolerance. On the natural scale $\delta \sim \varepsilon^2$, where $\frac{1}{K}\sum_k \|s^k\|^2 = O(1/K)$ and $\frac{1}{K}\sum_k \|s^k\| = O(1/\sqrt{K})$, the two allocations coincide: both certificates reach an ε-slot in $K = O(\varepsilon^{-2})$ iterations and $O(\varepsilon^{-3})$ variance-reduced oracle calls (Proposition 5.1). Neither certificate dominates the other. Indeed no decomposition-free ordering exists: δ_G and δ_g are Fenchel gaps of *different* convex functions, so comparing $(\delta_\Phi, \delta_G, \varepsilon)$ with $(\delta_\phi, \delta_g, \varepsilon)$ is not defined until both are mapped to a common, decomposition-independent target such as Goldstein stationarity. The one intrinsic asymmetry is scope: the convex-side certificate requires f convex, whereas the concave-side certificate remains valid for nonconvex f.

	Convex-side (ϕ, g)	Concave-side (Φ, G)
δ_ϕ / δ_Φ	0 (exact)	0 (exact)
negative-side tolerance	$\delta_g = O(\Delta_0/K + \eta\bar{V})$	$\delta_G = \delta_g + D_{G_s} = O(\Delta_0/K + \bar{V})$
mismatch slot ε	$r_\eta\|s^k\| + \|e^k\|$	$\|e^k\|$ only
where $\|s^k\|$ sits	ε slot (linear)	δ_G (quadratic)
requires f convex	yes	no
oracle complexity (VR)	$O(\varepsilon^{-3})$	$O(\varepsilon^{-3})$

Table 2: The same iterate certified two ways. The certificates carry equivalent information, reallocating the movement $\|s^k\|$ between the mismatch and tolerance slots; on the scale $\delta \sim \varepsilon^2$ they yield the same rates. The one intrinsic difference is that the convex-side form requires f convex.

Remark 5.2 (Relation to intrinsic DC complexity)**.** *This is a concrete instance of decomposition-dependence: the same minimizing sequence yields different (δ, ε) certificates under different valid DC splittings of F. It motivates treating the decomposition itself as a free parameter and taking the infimum over valid splittings, as in the intrinsic DC complexity program.*

5.3 Oracle Complexity of stochastic DCA

We now instantiate the abstract certificate of Condition 4.1 for the stochastic prox-subgradient iterate and read its complexity off a single scalar, the averaged estimator error. The iterate uses stepsize $\eta \in (0, 1/L_f)$ and $s^k := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$. The subtracted subgradient $\boldsymbol{v}^k \in \partial g(\boldsymbol{x}^k)$ and, later, $\boldsymbol{w}^{k+1} \in \partial h(\boldsymbol{x}^{k+1})$ remain exact; only the smooth gradient is estimated. We keep the estimator abstract and control it solely through its mean-squared error,

$$e^k := \nabla f(\boldsymbol{x}^k) - d^k, \qquad V_k := \mathbb{E}\|e^k\|^2, \qquad \bar{V} := \frac{1}{K}\sum_{k=0}^{K-1} V_k, \tag{5.6}$$

so that V_k is the *total* error (bias plus variance); we do not assume $\mathbb{E}[d^k \mid \mathcal{F}_k] = \nabla f(\boldsymbol{x}^k)$. Every statement below is in terms of $\{V_k\}$ and $\bar{V}$; a concrete reduced-variance oracle only fixes the size of $\bar{V}$.

Lemma 5.1 (Stochastic PS-DCA realization)**.** *Assume* (5.1)*, let* $\eta \in (0, 1/L_f)$*, and let* $\{\boldsymbol{x}^k\}_{k\geq 0}$ *be generated by* (5.4)*. Define*

$$a'_\eta := \frac{1}{2\eta} - \frac{L_f}{2} = a_\eta - \frac{1}{2\eta}, \qquad r_\eta := \frac{1}{\eta} + L_f,$$

where $a_\eta = \eta^{-1} - L_f/2$ *is the deterministic curvature of Lemma 4.5. Then* $a'_\eta > 0$*. The* ϕ*-side certificate* (4.15) *and descent* (4.17) *of Condition 4.1 hold* pathwise *with*

$$a = a'_\eta, \qquad \sigma_k = \frac{\eta}{2}\|e^k\|^2,$$

while the dual residual is the deterministic bound $r_\eta\|s^k\|$ *augmented by a first-order error term* $\|e^k\|$*. Concretely, set* $\boldsymbol{w}^{k+1} := \boldsymbol{v}^k - d^k - \eta^{-1}s^k$ *and* $\boldsymbol{u}^{k+1} := \nabla f(\boldsymbol{x}^{k+1}) + \boldsymbol{w}^{k+1}$*. Then, for every realization,* (i) (exact ϕ-side certificate) $\boldsymbol{w}^{k+1} \in \partial h(\boldsymbol{x}^{k+1})$ *and*

$$\boldsymbol{u}^{k+1} \in \partial\phi(\boldsymbol{x}^{k+1}) \subseteq \partial_{\sigma_k}\phi(\boldsymbol{x}^{k+1}),$$

so (4.15) *holds with subgradient slack* 0*;* (ii) (dual residual — first-power error)

$$\|\boldsymbol{u}^{k+1} - \boldsymbol{v}^k\| = \|\nabla f(\boldsymbol{x}^{k+1}) - d^k - \eta^{-1}s^k\| \leq r_\eta\|s^k\| + \|e^k\|, \tag{5.7}$$

i.e. the deterministic residual (4.16) *plus the error term* $\|e^k\|$*, whose expectation obeys* $\mathbb{E}\|e^k\| \leq \sqrt{V_k}$*;* (iii) (descent — second-power error)

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \geq \delta_g^{k+1} + a'_\eta\|s^k\|^2 - \frac{\eta}{2}\|e^k\|^2, \tag{5.8}$$

i.e. (4.17) *with* $\sigma_k = \frac{\eta}{2}\|e^k\|^2$*, whence* $\mathbb{E}[\sigma_k] = \frac{\eta}{2}V_k$*.*

Proof. The iterate (5.4) is the minimizer of the η^{-1}-strongly convex map $Q_k(\boldsymbol{x}) := h(\boldsymbol{x}) + \langle d^k - \boldsymbol{v}^k, \boldsymbol{x}\rangle + \frac{1}{2\eta}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2$, with optimality condition $\mathbf{0} \in \partial h(\boldsymbol{x}^{k+1}) + (d^k - \boldsymbol{v}^k) + \eta^{-1}s^k$.

(i) The optimality condition gives $\boldsymbol{w}^{k+1} = \boldsymbol{v}^k - d^k - \eta^{-1}s^k \in \partial h(\boldsymbol{x}^{k+1})$. Since f is convex and differentiable, $\boldsymbol{u}^{k+1} = \nabla f(\boldsymbol{x}^{k+1}) + \boldsymbol{w}^{k+1} \in \partial f(\boldsymbol{x}^{k+1}) + \partial h(\boldsymbol{x}^{k+1}) = \partial\phi(\boldsymbol{x}^{k+1})$, an *exact* inclusion; hence $\boldsymbol{u}^{k+1} \in \partial_{\sigma_k}\phi(\boldsymbol{x}^{k+1})$ for the σ_k of part (iii), $\sigma_k \geq 0$.

(ii) Substituting $\boldsymbol{w}^{k+1}$,

$$\boldsymbol{u}^{k+1} - \boldsymbol{v}^k = \nabla f(\boldsymbol{x}^{k+1}) - d^k - \eta^{-1}s^k = \big(\nabla f(\boldsymbol{x}^{k+1}) - \nabla f(\boldsymbol{x}^k) - \eta^{-1}s^k\big) + e^k.$$

The bracket is bounded by $(L_f + \eta^{-1})\|s^k\| = r_\eta\|s^k\|$ exactly as in Lemma 4.5, and the triangle inequality yields (5.7). By Jensen, $\mathbb{E}\|e^k\| \leq (\mathbb{E}\|e^k\|^2)^{1/2} = \sqrt{V_k}$.

(iii) Strong convexity of Q_k at its minimizer gives $Q_k(\boldsymbol{x}^k) - Q_k(\boldsymbol{x}^{k+1}) \geq \frac{1}{2\eta}\|s^k\|^2$, which upon expansion reads

$$h(\boldsymbol{x}^k) - h(\boldsymbol{x}^{k+1}) + \langle \boldsymbol{v}^k - d^k, s^k\rangle \geq \eta^{-1}\|s^k\|^2. \tag{5.9}$$

Using L_f-smoothness, $f(\boldsymbol{x}^k) - f(\boldsymbol{x}^{k+1}) \geq -\langle\nabla f(\boldsymbol{x}^k), s^k\rangle - \frac{L_f}{2}\|s^k\|^2$, the identity $\delta_g^{k+1} = g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) - \langle \boldsymbol{v}^k, s^k\rangle = \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \geq 0$, and (5.9),

$$\begin{aligned} F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) &= \big[f(\boldsymbol{x}^k) - f(\boldsymbol{x}^{k+1})\big] + \big[h(\boldsymbol{x}^k) - h(\boldsymbol{x}^{k+1})\big] + \langle \boldsymbol{v}^k, s^k\rangle + \delta_g^{k+1} \\ &\geq a_\eta\|s^k\|^2 + \langle d^k - \nabla f(\boldsymbol{x}^k), s^k\rangle + \delta_g^{k+1} \\ &= a_\eta\|s^k\|^2 - \langle e^k, s^k\rangle + \delta_g^{k+1}, \end{aligned}$$

which is the deterministic bound of Lemma 4.5 plus the sole new object, the error cross term $-\langle e^k, s^k\rangle$.

Splitting it by Young's inequality, $\langle e^k, s^k\rangle \le \frac{\eta}{2}\|e^k\|^2 + \frac{1}{2\eta}\|s^k\|^2$, gives

$$F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1}) \ge \Big(a_\eta - \frac{1}{2\eta}\Big)\|s^k\|^2 + \delta_g^{k+1} - \frac{\eta}{2}\|e^k\|^2,$$

which is (5.8) with $a'_\eta = a_\eta - \frac{1}{2\eta} = \frac{1}{2\eta} - \frac{L_f}{2}$. Finally $a'_\eta > 0 \iff \eta < 1/L_f$, and $\mathbb{E}[\sigma_k] = \frac{\eta}{2}V_k$. □

Remark 5.3 (Two powers of one error)**.** *The estimator error e^k enters the two sides of the certificate at* different *powers. In the dual residual* (5.7) *it appears* linearly*, as $\|e^k\|$ (hence $\sqrt{V_k}$ in expectation), because it sits additively inside a norm. In the descent* (5.8) *it appears* quadratically*, as $\frac{\eta}{2}\|e^k\|^2$ (hence $\frac{\eta}{2}V_k$), because the only error-dependent term, the cross term $\langle e^k, s^k\rangle$, must be Young-split against the curvature $\|s^k\|^2$. Through the abstract accounting the linear error $\|e^k\|$ feeds only the additive part of the ε-mismatch, while the quadratic σ_k feeds δ_ϕ, δ_g, and the $\sqrt{\cdot}$ part of ε. Consequently the same $\bar V$ surfaces as $\sqrt{\bar V}$ in one slot and as $\eta\bar V$ in the others; this is precisely why variance reduction ($\bar V \to 0$) is what buys a small ε-slot, and why a fixed noise level leaves an irreducible $\sqrt{\bar V}$ floor.*

We call $\bar{\boldsymbol{x}}$ a $(\delta_\phi, \delta_g, \varepsilon)$-*critical point in expectation* if there exist (random) dual vectors $\boldsymbol{u}, \boldsymbol{v}$ with $\mathbb{E}[\Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{u})] \le \delta_\phi$, $\mathbb{E}[\Phi_g(\bar{\boldsymbol{x}}, \boldsymbol{v})] \le \delta_g$, and $\mathbb{E}\|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon$ — the in-expectation lift of the Fenchel-gap form of Definition 3.1. Unlike the deterministic analysis, which returns the best iterate through $\hat k \in \operatorname{argmin}_k\{\delta_g^{k+1} + a\|s^k\|^2\}$, the stochastic result draws $\hat k$ *uniformly at random*: the argmin index is correlated with the realized errors $\{e^k\}$, so the selected-index residual $\mathbb{E}\|e^{\hat k}\|$ cannot be controlled by the second-moment budget alone (it may grow like $\sqrt{K\bar V}$), whereas averaging over a uniform, run-independent $\hat k$ restores $\mathbb{E}[Z_{\hat k}] = \frac{1}{K}\sum_k \mathbb{E}[Z_k]$ and closes the certificate.

Corollary 5.1 (Abstract stochastic complexity of PS-DCA)**.** *Suppose $F^\star > -\infty$ and the assumptions of Lemma 5.1 hold. Fix $K \ge 1$, let the output index $\hat k$ be drawn uniformly at random from $\{0, \dots, K-1\}$ independently of $\{d^k\}$, and set $\Delta_0 := F(\boldsymbol{x}^0) - F^\star$ and $\bar{\boldsymbol{x}} := \boldsymbol{x}^{\hat k+1}$. Then, with $\boldsymbol{u} := \boldsymbol{u}^{\hat k+1}$ and $\boldsymbol{v} := \boldsymbol{v}^{\hat k}$ from Lemma 5.1,*

$$\mathbb{E}[\Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{u})] \le \frac{\eta}{2}\,\bar V =: \delta_\phi, \tag{5.10}$$

$$\mathbb{E}[\Phi_g(\bar{\boldsymbol{x}}, \boldsymbol{v})] \le \frac{\Delta_0}{K} + \frac{\eta}{2}\,\bar V =: \delta_g, \tag{5.11}$$

$$\mathbb{E}\|\boldsymbol{u} - \boldsymbol{v}\| \le r_\eta\sqrt{\frac{1}{a'_\eta}\Big(\frac{\Delta_0}{K} + \frac{\eta}{2}\,\bar V\Big)} + \sqrt{\bar V} =: \varepsilon, \tag{5.12}$$

so $\bar{\boldsymbol{x}}$ is a $(\delta_\phi, \delta_g, \varepsilon)$-critical point in expectation. The bound (5.10) *is in fact sharp with $\Phi_\phi(\bar{\boldsymbol{x}}, \boldsymbol{u}) = 0$ pathwise: the value $\frac{\eta}{2}\bar V$ is the tolerance the template assigns through the shared slack σ_k, not a genuine ϕ-side inexactness.*

Proof. Summing (5.8) over $k = 0, \dots, K-1$ and using $F(\boldsymbol{x}^K) \ge F^\star$,

$$\sum_{k=0}^{K-1}\big[\delta_g^{k+1} + a'_\eta\|s^k\|^2\big] \le \Delta_0 + \frac{\eta}{2}\sum_{k=0}^{K-1}\|e^k\|^2 \qquad \text{(pathwise)}.$$

Taking full expectation and using (5.6),

$$\sum_{k=0}^{K-1}\mathbb{E}\big[\delta_g^{k+1} + a'_\eta\|s^k\|^2\big] \le \Delta_0 + \frac{\eta}{2}K\bar V. \tag{5.13}$$

Both summands on the left are nonnegative ($\delta_g^{k+1} = \Phi_g(\boldsymbol{x}^{k+1}, \boldsymbol{v}^k) \ge 0$, $a'_\eta > 0$). Since $\hat k$ is uniform on $\{0, \dots, K-1\}$ and independent of the run, $\mathbb{E}[Z_{\hat k}] = \frac{1}{K}\sum_k \mathbb{E}[Z_k]$ for any adapted nonnegative $\{Z_k\}$. Dropping

the $a'_\eta\|s^k\|^2$ sum in (5.13) and using $\Phi_g(\bar{\boldsymbol{x}},\boldsymbol{v}) = \delta_g^{\hat{k}+1}$ gives (5.11); dropping instead the δ_g^{k+1} sum gives

$$\mathbb{E}\|s^{\hat{k}}\|^2 = \frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|s^k\|^2 \le \frac{1}{a'_\eta}\Big(\frac{\Delta_0}{K} + \frac{\eta}{2}\bar{V}\Big).$$

For the ϕ-side, $\Phi_\phi(\bar{\boldsymbol{x}},\boldsymbol{u}) = 0$ by Lemma 5.1(i), and $\mathbb{E}[\sigma_{\hat{k}}] = \frac{\eta}{2}\cdot\frac{1}{K}\sum_k V_k = \frac{\eta}{2}\bar{V}$, which is (5.10). Finally, by (5.7), Jensen, and concavity of $\sqrt{\cdot}$,

$$\mathbb{E}\|\boldsymbol{u}-\boldsymbol{v}\| \le r_\eta\,\mathbb{E}\|s^{\hat{k}}\| + \mathbb{E}\|e^{\hat{k}}\| \le r_\eta\sqrt{\mathbb{E}\|s^{\hat{k}}\|^2} + \frac{1}{K}\sum_{k=0}^{K-1}\sqrt{V_k} \le r_\eta\sqrt{\frac{1}{a'_\eta}\Big(\frac{\Delta_0}{K}+\frac{\eta}{2}\bar{V}\Big)} + \sqrt{\bar{V}},$$

using $\frac{1}{K}\sum_k\sqrt{V_k} \le \sqrt{\frac{1}{K}\sum_k V_k} = \sqrt{\bar{V}}$. This is (5.12). □

Remark 5.4 (Instance, limits, and the noise floor)**.** *For $\eta = 1/(2L_f)$ one has $a'_\eta = L_f/2$ and $r_\eta = 3L_f$, so Corollary 5.1 reads*

$$\delta_\phi = \frac{\bar{V}}{4L_f},\quad \delta_g = \frac{\Delta_0}{K} + \frac{\bar{V}}{4L_f},\quad \varepsilon \le \sqrt{\frac{18L_f\Delta_0}{K} + \frac{9}{2}\bar{V}} + \sqrt{\bar{V}}.$$

Two regimes follow. (a) Variance reduction: *as $\bar{V}\to 0$ the triple collapses to $\big(0,\ \Delta_0/K,\ r_\eta\sqrt{\Delta_0/(a'_\eta K)}\big)$, the $O(K^{-1/2})$ shape of the deterministic Corollary 4.1; the constant is inflated only because the Young split replaces a_η by the smaller a'_η (with no error the split is unnecessary and a_η is recovered).* (b) Fixed noise: *if $\bar{V}\ge\sigma^2>0$ for all K (e.g. plain SGD), then $\varepsilon \ge \sqrt{\bar{V}} \ge \sigma$ regardless of K — an irreducible noise floor that only $\bar{V}\to 0$ removes. The admissible range tightens from $\eta\in(0,2/L_f)$ to $\eta\in(0,1/L_f)$, with $\eta = 1/L_f$ the degenerate boundary $a'_\eta = 0$; the ε-constant $r_\eta^2/a'_\eta = 2(\eta^{-1}+L_f)^2/(\eta^{-1}-L_f)$ is minimized at $\eta = 1/(3L_f)$, giving $r_\eta^2/a'_\eta = 16L_f$ (versus the deterministic optimum $6L_f$).*

5.4 Variance-Reduced Instantiations and Existing Rates

Corollary 5.1 collapses the entire stochastic analysis onto a single scalar, the averaged mean-squared error $\bar{V}$: the certificate triple $(\delta_\phi,\delta_g,\varepsilon)$ is an explicit, fixed function of $\bar{V}$ and the horizon K, and a concrete gradient oracle enters only by fixing how fast $\bar{V}$ decays. We first isolate the resulting budget (Corollary 5.2), then meet it with two oracles at opposite costs: a bounded-variance minibatch, which pays the $O(\varepsilon^{-4})$ rate (Corollary 5.3), and a variance-reduced estimator whose error couples to the iterate drift, which targets $O(\varepsilon^{-3})$ (Proposition 5.1). Throughout we fix $\eta = 1/(2L_f)$, so that $a'_\eta = L_f/2$ and $r_\eta = 3L_f$ as in Remark 5.4.

Corollary 5.2 (Modular slot budget)**.** *Under the assumptions of Corollary 5.1 with $\eta = 1/(2L_f)$, fix $\varepsilon>0$ and set*

$$c_1 := \min\Big\{\tfrac{1}{4},\ \tfrac{a'_\eta}{4\eta r_\eta^2}\Big\} = \frac{1}{36},\qquad c_2 := \frac{8r_\eta^2}{a'_\eta} = 144\,L_f.$$

If

$$\bar{V} \le c_1\varepsilon^2 \qquad \text{and} \qquad K \ge c_2\,\frac{\Delta_0}{\varepsilon^2}, \tag{5.14}$$

then the output $\bar{\boldsymbol{x}}$ of Corollary 5.1 is a $(\delta_\phi,\delta_g,\varepsilon)$-critical point in expectation with $\delta_\phi \le \varepsilon^2/(144L_f)$ and $\delta_g \le \varepsilon^2/(72L_f)$; in particular $\delta_\phi,\delta_g = O(\varepsilon^2)$ ride free at the ε^2 scale while the mismatch slot is driven to ε.

Proof. Write the mismatch slot of Corollary 5.1 as $\varepsilon_{\mathrm{slot}} = r_\eta\sqrt{\frac{1}{a'_\eta}\big(\frac{\Delta_0}{K}+\frac{\eta}{2}\bar{V}\big)} + \sqrt{\bar{V}}$. By (5.14), $\sqrt{\bar{V}} \le \sqrt{c_1}\,\varepsilon \le \varepsilon/2$ since $c_1 \le \frac{1}{4}$. For the first term, $K \ge c_2\Delta_0/\varepsilon^2$ gives $\Delta_0/K \le a'_\eta\varepsilon^2/(8r_\eta^2)$, while $\bar{V}\le c_1\varepsilon^2 \le a'_\eta\varepsilon^2/(4\eta r_\eta^2)$ gives $\frac{\eta}{2}\bar{V} \le a'_\eta\varepsilon^2/(8r_\eta^2)$; hence $\frac{\Delta_0}{K}+\frac{\eta}{2}\bar{V} \le a'_\eta\varepsilon^2/(4r_\eta^2)$ and $r_\eta\sqrt{\frac{1}{a'_\eta}\cdot a'_\eta\varepsilon^2/(4r_\eta^2)} = \varepsilon/2$, so $\varepsilon_{\mathrm{slot}} \le \varepsilon$. The slot

values follow from $\delta_\phi = \frac{\eta}{2}\bar{V} \le \frac{1}{4L_f}c_1\varepsilon^2 = \varepsilon^2/(144L_f)$ and $\delta_g = \frac{\Delta_0}{K} + \frac{\eta}{2}\bar{V} \le \varepsilon^2/(144L_f) + \varepsilon^2/(144L_f)$, using $\varepsilon^2/c_2 = \varepsilon^2/(144L_f)$ and $\eta = 1/(2L_f)$. □

Assumption 5.1 (Unbiased bounded-variance oracle)**.** *The estimator $d^k = \frac{1}{b}\sum_{i=1}^b \boldsymbol{g}_i^k$ averages b i.i.d. unbiased samples, $\mathbb{E}[\boldsymbol{g}_i^k \mid \mathcal{F}_k] = \nabla f(\boldsymbol{x}^k)$, each of variance $\mathbb{E}[\|\boldsymbol{g}_i^k - \nabla f(\boldsymbol{x}^k)\|^2 \mid \mathcal{F}_k] \le \sigma^2$ for a fixed $\sigma^2 > 0$.*

Under Assumption 5.1 the total error is pure variance, $V_k = \mathbb{E}\|e^k\|^2 \le \sigma^2/b$ for every k, hence $\bar{V} \le \sigma^2/b$. Unlike Lemma 5.1, which controls only the total error V_k and needs no unbiasedness, the identification $V_k \le \sigma^2/b$ uses $\mathbb{E}[d^k \mid \mathcal{F}_k] = \nabla f(\boldsymbol{x}^k)$.

Corollary 5.3 (Bounded-variance rate: $O(\varepsilon^{-4})$)**.** *Let Assumption 5.1 hold, fix $\eta = 1/(2L_f)$, and let $\varepsilon > 0$. Taking*

$$b = \left\lceil \frac{\sigma^2}{c_1\varepsilon^2} \right\rceil = \left\lceil \frac{36\sigma^2}{\varepsilon^2} \right\rceil, \qquad K = \left\lceil c_2 \frac{\Delta_0}{\varepsilon^2} \right\rceil = \left\lceil \frac{144 L_f \Delta_0}{\varepsilon^2} \right\rceil,$$

the output $\bar{\boldsymbol{x}}$ is a $(\delta_\phi, \delta_g, \varepsilon)$-critical point in expectation with $\delta_\phi, \delta_g = O(\varepsilon^2)$, and the total number of stochastic-gradient evaluations is

$$K \cdot b = O\Big(\frac{\sigma^2 L_f \Delta_0}{\varepsilon^4}\Big).$$

Proof. The choice of b gives $\bar{V} \le \sigma^2/b \le c_1\varepsilon^2$, and $K \ge c_2\Delta_0/\varepsilon^2$; thus (5.14) holds and Corollary 5.2 applies. The cost is $K \cdot b = \Theta(L_f\Delta_0/\varepsilon^2) \cdot \Theta(\sigma^2/\varepsilon^2)$. □

Assumption 5.2 (MVR variance recursion [10])**.** *The estimator d^k is unbiased and satisfies averaged L_f-smoothness; for a momentum weight $\alpha \in (0, 1]$ and constant batch b its error obeys, in expectation,*

$$V_{k+1} \le (1-\alpha)V_k + \frac{2L_f^2}{b}\,\mathbb{E}\|s^k\|^2 + \frac{2\alpha^2\sigma^2}{b}. \tag{5.15}$$

Proposition 5.1 (MVR coupling: variance decay and its floor)**.** *Let Assumption 5.2 hold, fix $\eta = 1/(2L_f)$ and $\beta = \eta/\alpha$, and impose the drift-domination batch condition*

$$b \;\ge\; \frac{4\eta L_f^2}{\alpha\, a'_\eta} \;=\; \frac{4}{\alpha}. \tag{5.16}$$

Then the coupled potential $\Phi_k := (F(\boldsymbol{x}^k) - F^\star) + \beta V_k$ is nonincreasing in expectation up to the additive noise $2\beta\alpha^2\sigma^2/b$ per step, and

$$\bar{V} \;\le\; \frac{4L_f\Delta_0}{K} + \frac{2V_0}{\alpha K} + \frac{4\alpha\sigma^2}{b}. \tag{5.17}$$

Two readings follow. (a) Constant batch: *for α, b held fixed (with $\alpha b \ge 4$) the $O(1/K)$ transients vanish, but $\bar{V} \to 4\alpha\sigma^2/b > 0$, an irreducible $O(\alpha/b)$ variance floor; constant-batch MVR therefore does* not *send $\bar{V} \to 0$, and Corollary 5.2 does not apply.* (b) ε-tied schedule: *initializing with a batch of size $\Theta(\sigma^2/\varepsilon^2)$ so that $V_0 = O(\varepsilon^2)$, and scaling $\alpha = \Theta(\varepsilon/\sigma)$, $b = \Theta(\sigma/\varepsilon)$—so (5.16) holds and the floor becomes $4\alpha\sigma^2/b = \alpha^2\sigma^2 = O(\varepsilon^2)$—one has $\bar{V} \le c_1\varepsilon^2$ after $K = \Theta(L_f\Delta_0/\varepsilon^2)$ steps, whence Corollary 5.2 returns an ε-slot at total first-order cost*

$$K \cdot b + \Theta(\sigma^2/\varepsilon^2) = O\Big(\frac{\sigma L_f \Delta_0}{\varepsilon^3}\Big),$$

the online rate of Chayti and Jaggi [10].

Proof. Taking expectations in (5.8) and writing $S_k := \mathbb{E}\|s^k\|^2$,

$$\mathbb{E}[F(\boldsymbol{x}^k) - F(\boldsymbol{x}^{k+1})] \ge \mathbb{E}[\delta_g^{k+1}] + a'_\eta S_k - \tfrac{\eta}{2}V_k.$$

From (5.15), $V_k - V_{k+1} \geq \alpha V_k - (2L_f^2/b)S_k - 2\alpha^2\sigma^2/b$, so

$$\Phi_k - \Phi_{k+1} \geq \mathbb{E}[\delta_g^{k+1}] + \Big(a'_\eta - \tfrac{2\beta L_f^2}{b}\Big)S_k + \Big(\beta\alpha - \tfrac{\eta}{2}\Big)V_k - \frac{2\beta\alpha^2\sigma^2}{b}.$$

With $\beta = \eta/\alpha$ the V_k-coefficient equals $\eta/2 \geq 0$, and (5.16) makes the S_k-coefficient at least $a'_\eta/2 \geq 0$; the three leading terms are nonnegative, giving $\Phi_k - \Phi_{k+1} \geq -2\beta\alpha^2\sigma^2/b$. Summing over $k = 0, \ldots, K-1$, using $\Phi_K \geq 0$ and $\Phi_0 = \Delta_0 + \beta V_0$, and dropping the nonnegative δ_g- and S_k-sums,

$$\tfrac{\eta}{2}\sum_{k=0}^{K-1} V_k \leq \Delta_0 + \beta V_0 + \frac{2K\beta\alpha^2\sigma^2}{b}.$$

Dividing by $\frac{\eta}{2}K$ and substituting $\beta = \eta/\alpha$, $\eta = 1/(2L_f)$ yields (5.17). Parts (a),(b) read off the three terms; in (b), $b = 4/\alpha$ makes the floor $4\alpha\sigma^2/b = \alpha^2\sigma^2$, and $\alpha = \Theta(\varepsilon/\sigma)$ sends it to $O(\varepsilon^2)$ while $K \cdot b = \Theta(L_f\Delta_0/\varepsilon^2) \cdot \Theta(\sigma/\varepsilon)$. □

We stress that (5.17) is the strongest self-contained conclusion the coupling yields: the $O(\varepsilon^{-3})$ total in part (b) follows from the ε-tied scaling $\alpha = \Theta(\varepsilon/\sigma)$, $b = \Theta(\sigma/\varepsilon)$ rather than from a constant-batch telescope, and in that regime it coincides with the online MVR rate of Chayti and Jaggi [10].

Remark 5.5 (Existing stochastic-DC rates on the two-sided certificate)**.** *The realization Lemma 5.1 places existing stochastic-DC guarantees on the single template of Corollary 5.1, so their ε-dependence is read off the mismatch slot alone. DCA-PAGE [29] reports $O(N + \sqrt{N}\,\varepsilon^{-2})$ in the finite-sum regime and $O(b + \sqrt{b}\,\varepsilon^{-2})$ online, measured in a one-sided gap; MDCA [10] reports $O(\sigma^2\varepsilon^{-4})$ with a Polyak estimator and $O(\sigma\varepsilon^{-3})$ with an MVR estimator under averaged smoothness, in a critical-distance measure. Up to the square-root unit change between value-typed gaps and the norm-typed residual, these are exactly the ε-slot rates of Corollary 5.3 and Proposition 5.1, with $\delta_g = O(\varepsilon^2)$ riding free in the second slot. Remark 5.3 names the mechanism: the estimator error surfaces as $\sqrt{\bar{V}}$ in the mismatch slot but only as $\frac{\eta}{2}\bar{V}$ in δ_ϕ, δ_g, so the $\sqrt{\bar{V}}$ term governs ε. A bounded-variance oracle shrinks it only through $\bar{V} \leq \sigma^2/b$, forcing $b = \Theta(\sigma^2/\varepsilon^2)$ per step and the $O(\varepsilon^{-4})$ total; a variance-reduced oracle instead couples $\bar{V}$ to the drift and amortizes it across iterations, which is what separates the $O(\varepsilon^{-3})$ rate from the bounded-variance $O(\varepsilon^{-4})$ one.*

5.5 Smoothing the concave side

A third template smooths the nonsmooth part instead of linearizing it, and it is the closest of our methods to the single-loop momentum scheme of Chayti and Jaggi [10]. Fix $\mu > 0$, replace g by its Moreau envelope g_μ from (4.36), and run one stochastic prox-gradient step per iteration,

$$\boldsymbol{x}^{k+1} = \mathrm{prox}_{\eta h}\big(\boldsymbol{x}^k - \eta(\boldsymbol{d}^k - \nabla g_\mu(\boldsymbol{x}^k))\big), \qquad \nabla g_\mu(\boldsymbol{x}^k) = \tfrac{1}{\mu}\big(\boldsymbol{x}^k - \mathrm{prox}_{\mu g}(\boldsymbol{x}^k)\big). \tag{5.18}$$

Because g_μ is $1/\mu$-smooth and deterministic, its gradient is exact, so the only stochastic object is again $\boldsymbol{d}^k$. The surrogate $F_\mu := f + h - g_\mu$ is $(L_f + 1/\mu)$-smooth, and the smoothing route of Section 4.5 identifies stationarity of F_μ with a $(\,0,\, O(\mu),\, \varepsilon\,)$-certificate of the original F: the positive side stays exact, and the smoothing bias enters only the negative-side tolerance at scale $O(\mu)$.

The distinctive feature here is that the convex side carries no stochastic noise. In the single-loop smoothed method of Chayti and Jaggi [10], the inner step draws a stochastic gradient of the convex-side term, whose irreducible variance M^2 forces the $O(\varepsilon^{-4})$ rate regardless of variance reduction on the concave-side estimator. In our model h and g are deterministic, so $M = 0$ and that obstruction is absent. This suggests the variance-reduced instantiation should reach $O(\varepsilon^{-3})$; the open point is whether the surrogate smoothness $L_\mu = L_f + 1/\mu$, once μ is tied to the target accuracy, re-enters the step-size budget.

Proposition 5.2 (Smoothed single-loop certificate; rate conjectural)**.** *Let f be L_f-smooth convex and h, g be deterministic proper closed convex, and run* (5.18) *with $\eta \leq 1/(L_f + 1/\mu)$. Then the certificate machinery*

of Lemma 5.1 and Corollary 5.1 applies to F_μ and returns, for the original F, a $(0, O(\mu), \varepsilon)$-critical point in expectation whose mismatch ε obeys the averaged-error budget of Corollary 5.1. A bounded-variance oracle therefore attains $O(\varepsilon^{-4})$. At the criticality-tied schedule $\mu = \Theta(\varepsilon)$, and provided the L_μ-dependence does not re-enter the drift-domination condition (5.16)*, the variance-reduced instantiation attains $O(\varepsilon^{-3})$. We state the variance-reduced rate as a* conjecture*: the noise-free convex side removes the M^2 obstruction of Chayti and Jaggi [10], but a full Lyapunov analysis tracking $L_\mu = L_f + 1/\mu$ under $\mu = \Theta(\varepsilon)$ remains to be carried out.*

Smoothing g also resolves the incomparability left open in Section 5.2: with g_μ smooth the decomposition is *faithful*, and both the convex-side and concave-side certificates project onto the same decomposition-free stationarity of F.

Corollary 5.4 (Decomposition-free Clarke readout)**.** *Suppose the returned point $\bar{\boldsymbol{x}}$ is faithful, i.e. g (or its smoothing g_μ) is differentiable at $\bar{\boldsymbol{x}}$, so that $\partial_C F(\bar{\boldsymbol{x}}) = \partial\phi(\bar{\boldsymbol{x}}) - \nabla g(\bar{\boldsymbol{x}}) = \partial\Phi(\bar{\boldsymbol{x}}) - \nabla G(\bar{\boldsymbol{x}})$. Then the convex-side certificate $(0, \delta_g, \varepsilon)$ and the concave-side certificate $(0, \delta_G, \varepsilon)$ of the same iterate both yield*

$$\operatorname{dist}\big(\mathbf{0}, \partial_C F(\bar{\boldsymbol{x}})\big) \;\le\; \varepsilon + \sqrt{2L_\bullet\, \delta_\bullet},$$

with $(L_\bullet, \delta_\bullet) = (L_g, \delta_g)$ on the convex side and $(L_\bullet, \delta_\bullet) = (L_G, \delta_G)$, $L_G \le \rho + L_f + L_g$, on the concave side. At the ε-tied schedule both give $\operatorname{dist}(\mathbf{0}, \partial_C F(\bar{\boldsymbol{x}})) = O(\varepsilon)$ at $O(\varepsilon^{-3})$ oracle cost; the convex-side constant is the smaller of the two. Without faithfulness no such reduction exists (Example 3.2), so the two certificates are genuinely incomparable rather than one dominating the other.

Proof. Since g is L_g-smooth at $\bar{\boldsymbol{x}}$, g^* is $1/L_g$-strongly convex, so $\Phi_g(\bar{\boldsymbol{x}}, \boldsymbol{v}) \ge \frac{1}{2L_g}\|\boldsymbol{v} - \nabla g(\bar{\boldsymbol{x}})\|^2$; applied to $\boldsymbol{v} \in \partial_{\delta_g} g(\bar{\boldsymbol{x}})$ this gives $\|\boldsymbol{v} - \nabla g(\bar{\boldsymbol{x}})\| \le \sqrt{2L_g\delta_g}$. With $\boldsymbol{u} \in \partial\phi(\bar{\boldsymbol{x}})$ and $\|\boldsymbol{u} - \boldsymbol{v}\| \le \varepsilon$, the triangle inequality together with $\boldsymbol{u} - \nabla g(\bar{\boldsymbol{x}}) \in \partial_C F(\bar{\boldsymbol{x}})$ gives the convex-side bound. The concave-side bound is identical with G in place of g, using $\nabla G = \rho\boldsymbol{x} - \nabla f + \nabla g$ and hence $L_G \le \rho + L_f + L_g$. The schedule substitution uses $\delta_g, \delta_G = O(\varepsilon^2)$ from Corollary 5.1 and Proposition 5.1 and (5.5). □

6 Complexity of Nonconvex Constrained DC Optimization

In this section, we analyze the complexity of the proposed algorithmic framework for solving the constrained DC problem (**P**). Beyond the assumptions stated in (**P**), we impose the following additional assumptions throughout this section:

6.1 Assumptions

Given nonconvex constraints, we impose the following nonsingularity condition on the constraints to ensure the feasibility of solutions. Constraint qualifications are often required in solving constrained optimization problems; see [13, 23, 25, 32].

Assumption 6.1. *For the iterate sequence $\{\boldsymbol{x}^k\}_{k\in\mathbb{N}}$ generated by the algorithms, there exists a constant $\gamma_c > 0$ such that*

$$\|\nabla \boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^k)\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^k) + \nabla \boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^k)[\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^k)]_+\| \ge \gamma_c \left(\|\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^k)\|^2 + \|[\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^k)]_+\|^2\right)^{1/2}, \quad \forall\, k \ge 0.$$

6.2 Convergence Criterion

The goal of this subsection is to establish the certificate generated by the algorithmic framework below. For each iteration, we define the transported same-point tolerances

$$\delta_\phi^{k+1} := \epsilon_\phi^{k+1},$$

$$\delta_g^{k+1} := \epsilon_g^k + g(\boldsymbol{x}^{k+1}) - g(\boldsymbol{x}^k) - \langle \boldsymbol{b}^k, \boldsymbol{x}^{k+1} - \boldsymbol{x}^k\rangle,$$

and the certificate residual

$$R_{k+1} := \frac{1}{\gamma_k}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\| + \alpha_{k+1}.$$

The main result of this subsection is the bound

$$\operatorname{dist}\left(\partial_{\delta_\phi^{k+1}}\phi(\boldsymbol{x}^{k+1}), \partial_{\delta_g^{k+1}} g(\boldsymbol{x}^{k+1})\right) \le R_{k+1}.$$

This certificate separates three effects: the ϕ-side oracle error, the transported g-side Fenchel error, and the proximal optimality residual.

7 Conclusion

7.1 Future Works

A Finite-Time Limitation of Exact Same-Point DC Criticality

A.1 An explicit hard instance and the role of the DC oracle

[41] prove that finding an ε-stationary point of a nonsmooth nonconvex Lipschitz function is impossible in finite time, and introduce (δ, ε)-stationarity as a relaxed target. We use their resisting-oracle construction. Our proof strategy is similar to Theorem 1.1.2 in (Nesterov, 2018) and Theorem 5 in [41], where the resisting strategy is used to prove a lower bound.

Throughout the appendix, let $\{x^k\}_{k=1}^K$ be the chronological query sequence generated by the method and let $Q = \{x_1 < \cdots < x_K\}$ denote these points sorted in ascending order (relabeling if necessary). We assume without loss of generality (WLOG) that the initial point is queried and belongs to Q; otherwise, one may query the initial point before running the algorithm.

A.2 Background

We first recall the resisting instance at the level of the weak transcript $(F(x), u(x) - v(x))$, which suffices to establish that every queried point has DC residual bounded away from zero. Whenever the algorithm queries a point x, the oracle returns

$$F(x) = 0, \qquad u(x) \in \partial\phi(x), \qquad v(x) \in \partial g(x),$$

chosen so that $u(x) - v(x) = L$.

Lemma A.1. *Given a finite sequence of real numbers $\{x_k\}_{k=1}^K \subset \mathbb{R}$, there exists a family of DC pairs $F_\theta = \phi_\theta - g_\theta$ such that, for any $k \in [1, K]$,*

$$F_\theta(x_k) = 0, \qquad u_\theta(x_k) - v_\theta(x_k) = L, \quad u_\theta(x_k) \in \partial\phi_\theta(x_k),\ v_\theta(x_k) \in \partial g_\theta(x_k).$$

Moreover, for sufficiently small $\varepsilon < L$, the DC ε-critical sets of two different choices of θ can be chosen disjoint.

Proof. Without loss of generality, we assume $x_1 < x_2 < \cdots < x_K$, reordering the indices if necessary. We construct $F_\theta = \phi_\theta - g_\theta$ as a family of sawtooth functions and show that it simultaneously satisfies the resisting-transcript property and the disjointness of ε-critical sets.

Let

$$\delta = \min\left\{\min_{k\in[1,K-1]}(x_{k+1} - x_k),\ \frac{\Delta}{L}\right\},$$

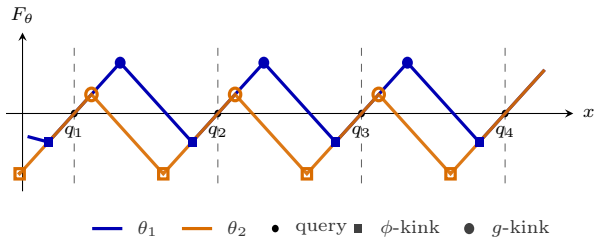


Figure 5: DC resisting family. Distinct hidden shifts θ preserve the same query transcript $F(q_k) = 0$, $u_k - v_k = L$.

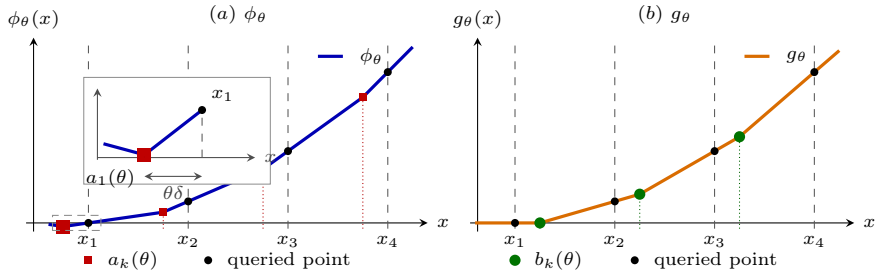


Figure 6: Illustration of the two convex components ϕ_θ and g_θ, drawn in separate panels to avoid overplotting. The representative parameters are $L = 1$, $x_k \in \{0, 2, 4, 6\}$, $\delta = 2$, and $\theta = 1/4$. The kinks of ϕ_θ occur at $a_k(\theta)$, the kinks of g_θ occur at $b_k(\theta)$, and the two components agree at the queried points x_k.

with the first term omitted when $K = 1$. For $\theta \in (0, 1/2)$, define

$$a_k(\theta) = x_k - \theta\delta, \quad k \in [1, K], \qquad b_k(\theta) = \frac{x_k + x_{k+1}}{2} - \theta\delta, \quad k \in [1, K-1],$$

and set $\alpha_\theta = Lx_1 - 2L\theta\delta$. We define the two convex components by

$$\phi_\theta(x) = \alpha_\theta - Lx + 2L\sum_{k=1}^{K}(x - a_k(\theta))_+, \qquad g_\theta(x) = 2L\sum_{k=1}^{K-1}(x - b_k(\theta))_+.$$

Since $(\cdot)_+$ is convex and all hinge coefficients above are nonnegative, ϕ_θ and g_θ are proper, closed, convex, piecewise-linear functions, with kinks located at $\{a_k(\theta)\}_{k=1}^{K}$ and $\{b_k(\theta)\}_{k=1}^{K-1}$, respectively (see Figure 6).

By construction, the breakpoints interlace as

$$a_k(\theta) < x_k < b_k(\theta) < a_{k+1}(\theta) < x_{k+1}, \qquad k \in [1, K-1],$$

so that exactly k kinks of ϕ_θ and $k-1$ kinks of g_θ precede the query point x_k. Consequently,

$$\phi'_\theta(x_k) = (2k-1)L, \qquad g'_\theta(x_k) = 2(k-1)L,$$

and therefore

$$\phi'_\theta(x_k) - g'_\theta(x_k) = L \qquad \text{for every } k \in [1, K],$$

independently of k. Setting $u_\theta(x_k) = \phi'_\theta(x_k)$ and $v_\theta(x_k) = g'_\theta(x_k)$ therefore realizes the claimed resisting transcript.

We now verify that $F_\theta(x_k) = 0$ for every $k \in [1, K]$. Expanding the hinge terms active at x_k gives

$$\begin{aligned} F_\theta(x_k) &= \alpha_\theta - Lx_k + 2L\sum_{j=1}^{k}(x_k - a_j(\theta)) - 2L\sum_{j=1}^{k-1}(x_k - b_j(\theta)) \\ &= \alpha_\theta - Lx_1 + 2L\theta\delta = 0, \end{aligned}$$

where the second equality follows by telescoping the interlacing differences $a_j(\theta) - b_j(\theta)$ and $b_j(\theta) - a_{j+1}(\theta)$, and the last equality holds by the choice of α_θ. Together with the preceding display, this shows that the oracle returns the identical transcript $\big(F_\theta(x_k),\, u_\theta(x_k) - v_\theta(x_k)\big) = (0, L)$ at every query point, for every $\theta \in (0, 1/2)$.

It remains to characterize the ε-critical set of F_θ and establish its disjointness across θ. Away from the kinks, $F'_\theta = \phi'_\theta - g'_\theta \in \{-L, L\}$ by the same interlacing argument, so F_θ is L-Lipschitz. Moreover, since $\mathrm{kink}(\phi_\theta) \cap \mathrm{kink}(g_\theta) = \emptyset$, no cancellation occurs in the subdifferential of the sum, and

$$\partial\phi_\theta(x) - \partial g_\theta(x) = \partial_C F_\theta(x), \qquad \forall x \in \mathbb{R}.$$

Since $\mathrm{dist}\big(0, \partial_C F_\theta(x)\big) = L$ at every non-kink point, it follows that for any $\varepsilon < L$ the ε-critical set coincides exactly with the kink set,

$$\mathcal{S}^\varepsilon_\theta = \{a_k(\theta) : k \in [1, K]\} \cup \{b_k(\theta) : k \in [1, K-1]\}.$$

By definition, every element of $\mathcal{S}^\varepsilon_\theta$ equals a θ-independent quantity minus $\theta\delta$; that is, $\mathcal{S}^\varepsilon_\theta$ is a rigid translate, by $-\theta\delta$, of a fixed configuration determined solely by $\{x_k\}_{k=1}^K$. Since $\theta \mapsto \theta\delta$ is injective on $(0, 1/2)$, it follows that $\mathcal{S}^\varepsilon_{\theta_1} \cap \mathcal{S}^\varepsilon_{\theta_2} = \emptyset$ whenever $\theta_1 \neq \theta_2$, which completes the proof. □

Corollary A.1 (Affine tail of the DC pair)**.** *Fix $\theta = \theta_0 \in (0, 1/2)$ and write $(\phi_0, g_0) := (\phi_{\theta_0}, g_{\theta_0})$ for the pair of Lemma A.1. Then ϕ_0, g_0 are affine on $[x_K, \infty)$, with*

$$\phi'_0(x) = (2K-1)L, \qquad g'_0(x) = 2(K-1)L, \qquad x \geq x_K,$$

so that $u_0 - v_0 := \phi'_0(x) - g'_0(x) = L$ throughout this ray.

Proof. By the interlacing $a_k(\theta_0) < x_k < b_k(\theta_0) < a_{k+1}(\theta_0) < x_{k+1}$ established in the proof of Lemma A.1, the largest kink of ϕ_0 is $a_K(\theta_0) < x_K$ and the largest kink of g_0 is $b_{K-1}(\theta_0) < a_K(\theta_0) < x_K$. Hence both components are affine on $[x_K, \infty)$, with slopes equal to the values computed at x_K in the proof of Lemma A.1. □

A.3 Invisible common kink under the strong oracle

The DC pair of Lemma A.1 was designed so that its *difference* $F_0 = \phi_0 - g_0$ carries no information about θ_0 at the queried points, which suffices under the weak oracle (F, u, v). Under the strong oracle $(\phi, g, \partial\phi, \partial g)$, however, $\phi_0(x_k)$ and $g_0(x_k)$ are each individually informative: querying two points on the affine tail of Corollary A.1 reveals the intercept of ϕ_0, and hence θ_0, directly. We therefore introduce a second layer whose hidden parameter is a *location* rather than a slope offset, and which is invisible to the strong oracle by construction.

Lemma A.2 (Invisible common kink)**.** *Let (ϕ_0, g_0) be affine on $[x_K, \infty)$ with slopes u_0, v_0 there. Fix $M > 0$, $\beta \in \mathbb{R}$, and a hidden location $s > x_K$. Define*

$$c_s(x) = M|x - s| + \beta - Ms, \qquad \phi_s := \phi_0 + c_s, \qquad g_s := g_0 + c_s.$$

Then ϕ_s, g_s are proper, closed, convex, and:

(i) ***Objective invariance.*** *$\phi_s - g_s = \phi_0 - g_0$ everywhere. In particular the Lipschitz constant of $\phi_s - g_s$, its optimality gap $(\phi_s - g_s)(\boldsymbol{x}^0) - \inf_x(\phi_s - g_s)(x)$, and its Clarke subdifferential $\partial_C(\phi_s - g_s)(x)$ at every x are all independent of s.*

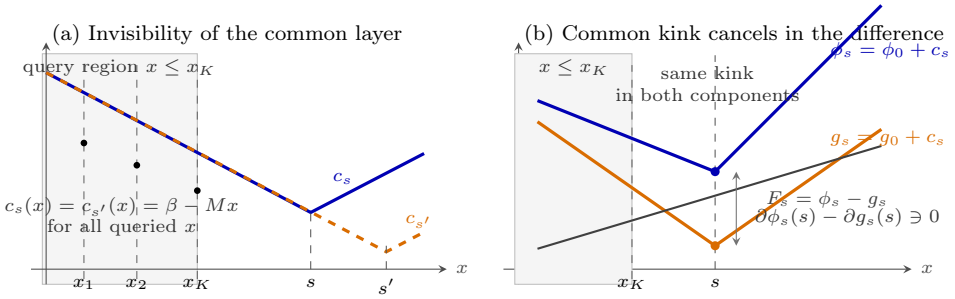


Figure 7: Invisible common-kink construction. Panel (a) shows that different hidden locations $s, s' > x_K$ induce the same values and subgradients on all queried points $x \le x_K$. Panel (b) shows that adding the same V-shaped layer to both convex components creates a common kink at s, while the objective difference $F_s = \phi_s - g_s$ remains unchanged and has no Clarke kink there.

(ii) ***Invisibility at queried points.*** *For every $x \le x_K$,*

$$c_s(x) = \beta - Mx, \qquad \partial c_s(x) = \{-M\},$$

independent of s. Consequently, for every admissible $s > x_K$,

$$\phi_s(x_k) = \phi_0(x_k) + \beta - Mx_k, \quad g_s(x_k) = g_0(x_k) + \beta - Mx_k, \quad \partial\phi_s(x_k) = \partial\phi_0(x_k) - M, \quad \partial g_s(x_k) = \partial g_0(x_k) - M,$$

none of which depends on s. In particular the DC residual at each queried point equals the background residual, $\operatorname{dist}(0, \partial\phi_s(x_k) - \partial g_s(x_k)) = \operatorname{dist}(0, \partial\phi_0(x_k) - \partial g_0(x_k))$.

(iii) ***Spurious DC-criticality at s.*** *At $x = s$,*

$$\partial\phi_s(s) - \partial g_s(s) = \{u_0 - v_0\} + M[-1, 1] - M[-1, 1] = \{u_0 - v_0\} + [-2M, 2M].$$

If $M \ge |u_0 - v_0|/2$, then $0 \in \partial\phi_s(s) - \partial g_s(s)$. Since $\partial_C(\phi_s - g_s)(s) = \{u_0 - v_0\}$ by (i) and the affineness of $\phi_0 - g_0$ at s, we have $0 \notin \partial_C(\phi_s - g_s)(s)$ whenever $u_0 \ne v_0$.

Proof. (i) Immediate from $\phi_s - g_s = (\phi_0 + c_s) - (g_0 + c_s) = \phi_0 - g_0$.

(ii) For $x \le x_K < s$ we have $|x - s| = s - x$, so

$$c_s(x) = M(s - x) + \beta - Ms = \beta - Mx,$$

which does not involve s; differentiating gives $\partial c_s(x) = \{-M\}$ on this ray. The displayed identities follow by adding c_s (respectively its subgradient) to ϕ_0, g_0 (respectively $\partial\phi_0, \partial g_0$) at x_k. The residual identity holds because subtracting the same constant $-M$ from both $\partial\phi_0(x_k)$ and $\partial g_0(x_k)$ leaves their Minkowski difference, and hence its distance to 0, unchanged.

(iii) Since ϕ_0, g_0 are affine at $s > x_K$ with slopes u_0, v_0 and c_s has a kink at s with $\partial c_s(s) = M[-1, 1]$, the subdifferential sum rule gives $\partial\phi_s(s) = \{u_0\} + M[-1, 1]$ and $\partial g_s(s) = \{v_0\} + M[-1, 1]$; their Minkowski difference is $\{u_0 - v_0\} + [-2M, 2M]$, which contains 0 once $M \ge |u_0 - v_0|/2$. The Clarke-subdifferential claim follows from (i), since $\phi_0 - g_0$ is affine, hence smooth, at s. □

A.4 Strong-transcript reduction

Lemma A.3 (Strong-transcript reduction)**.** *Let (ϕ_0, g_0), M, and β be as in Lemma A.2, and let $s, s' > x_K$ with $s \ne s'$. Let (ϕ_s, g_s) and $(\phi_{s'}, g_{s'})$ be the corresponding pairs. Then:*

(i) *The two instances return an identical strong-oracle transcript at $x_1, \dots, x_K$:*

$$\big(\phi_s(x_k), g_s(x_k), \partial\phi_s(x_k), \partial g_s(x_k)\big) = \big(\phi_{s'}(x_k), g_{s'}(x_k), \partial\phi_{s'}(x_k), \partial g_{s'}(x_k)\big), \qquad k = 1, \dots, K.$$

(ii) *Consequently, a deterministic algorithm interacting with $\mathcal{O}_{\mathrm{DC}}$ produces the same query sequence $x_1, \dots, x_K$ against both instances; a randomized algorithm does so as well when conditioned on its internal randomness.*

(iii) *The two spurious critical points s, s' are distinct, and by Lemma A.2(ii), neither instance has an ε-critical point among $x_1, \dots, x_K$ once $\varepsilon < L$.*

Consequently, no algorithm making K queries under $\mathcal{O}_{\mathrm{DC}}$ can guarantee, on both instances simultaneously, that some queried point is ε-critical. Following the same two-instance, disjoint-target argument used by Zhang et al. [41, Theorem 5] (and reproduced for the θ-family in Lemma A.1), its success probability against the uniform mixture over $\{s, s'\}$ is at most $1/2$.

Proof. Part (i) is Lemma A.2(ii) applied to both s and s'. Part (ii) holds because the algorithm's next query is a (possibly randomized) function of the oracle responses observed so far; if these coincide at every step for the two instances, so does the induced distribution over query sequences. Part (iii) follows from $s \neq s'$ together with Lemma A.2(ii)–(iii). The final probability bound follows the identical-transcript, disjoint-target reduction already used to separate $\mathcal{S}^{\varepsilon}_{\theta_1}$ from $\mathcal{S}^{\varepsilon}_{\theta_2}$ in the proof of Lemma A.1, applied here with the singleton target sets $\{s\}$ and $\{s'\}$. □

A.5 Proof of Theorem 3.1

Proof of Theorem 3.1. Let $x_1 < \cdots < x_K$ denote the (possibly random) points queried by the algorithm. Fix $\theta_0 \in (0, 1/2)$, e.g. $\theta_0 = 1/4$, and let (ϕ_0, g_0) be the disjoint-kink background pair of Lemma A.1 at $\theta = \theta_0$; by Corollary A.1 it is affine on $[x_K, \infty)$ with slope gap $u_0 - v_0 = L$. Set $M = L/2$, fix any $\beta \in \mathbb{R}$, and choose two admissible locations $x_K < s < s'$, e.g. $s = x_K + 1$, $s' = x_K + 2$. Form (ϕ_s, g_s) and $(\phi_{s'}, g_{s'})$ as in Lemma A.2.

Class membership. By Lemma A.2(i), the Lipschitz constant and optimality gap of $\phi_s - g_s$ (resp. $\phi_{s'} - g_{s'}$) equal those of $\phi_0 - g_0$, which lie within Δ, L by construction of the background pair. The component subgradient bound increases by at most M over the background bound $O(KL)$ established in Lemma A.1's proof, giving $G_K = O(KL + M) = O(KL)$; choosing the constant c in $K \le cG/L$ small enough ensures $G_K \le G$, so both instances lie in $\mathcal{D}(\Delta, L, G)$.

Failure at queried points. By Lemma A.2(ii), every queried point x_k has DC residual exactly $L > \varepsilon$ on both instances.

Spurious DC-critical point. By Lemma A.2(iii), with $M = L/2$ the instance (ϕ_s, g_s) contains a point s with $0 \in \partial\phi_s(s) - \partial g_s(s)$ and $0 \notin \partial_C(\phi_s - g_s)(s)$, and likewise $(\phi_{s'}, g_{s'})$ contains such a point at s'.

Indistinguishability and the probability bound. By Lemma A.3, the two instances are indistinguishable to the algorithm at every one of its K queries, and no algorithm can guarantee success against both simultaneously. Hence, with probability at least $1/2$, the realized instance is one on which none of the first K queried points is ε-critical, and this instance carries its own spurious DC-critical, non-Clarke-stationary point. This proves the theorem. □

B Special ase: Smooth DCA Lower Bound

The preceding upper bound has an unavoidable δ_g^{-1} dependence in worst-case first-order complexity. The cleanest way to see this is a smooth embedding reduction.

Lemma B.1 (Approximate Fenchel criticality implies smooth stationarity)**.** *Let h be convex and L-smooth. If $v \in \partial_\delta h(x)$, then*

$$\|v - \nabla h(x)\| \le \sqrt{2L\delta}. \tag{B.1}$$

Consequently, if ϕ and g are convex L-smooth and

$$a \in \partial_{\delta_\phi}\phi(x), \quad b \in \partial_{\delta_g} g(x), \quad \|a-b\| \le \eta, \tag{B.2}$$

then

$$\|\nabla(\phi - g)(x)\| \le \eta + \sqrt{2L\delta_\phi} + \sqrt{2L\delta_g}. \tag{B.3}$$

Proof. For convex L-smooth h, the conjugate h^* is $1/L$-strongly convex on its effective domain. Since

$$\Phi_h(x,v) = h(x) + h^*(v) - \langle v, x\rangle = D_{h^*}(v, \nabla h(x)), \tag{B.4}$$

we have

$$\Phi_h(x,v) \ge \frac{1}{2L}\|v - \nabla h(x)\|^2. \tag{B.5}$$

The implication (B.1) follows from $\Phi_h(x,v) \le \delta$. Applying this to ϕ and g and using the triangle inequality proves (B.3). □

Theorem B.1 (Smooth embedding lower bound)**.** *Fix a component oracle that returns*

$$(\phi(x), g(x), \nabla\phi(x), \nabla g(x)). \tag{B.6}$$

There exists a smooth DC subclass with both components convex and smooth such that any algorithm that guarantees a point satisfying

$$\delta_\phi, \delta_g \le c\varepsilon^2/L, \qquad \eta \le c\varepsilon \tag{B.7}$$

for a sufficiently small numerical constant $c > 0$ requires

$$\Omega(\Delta L \varepsilon^{-2}) = \Omega(\Delta/\delta) \tag{B.8}$$

component-oracle calls, where $\delta \asymp \varepsilon^2/L$.

Proof. Let $\widehat{F}$ be a smooth nonconvex hard instance with L_0-Lipschitz gradient and initial gap at most Δ, for which any first-order method needs $\Omega(\Delta L_0 \varepsilon^{-2})$ queries to find a point satisfying $\|\nabla\widehat{F}(x)\| \le \varepsilon$; see Carmon et al. [8, 9]. Define

$$q(x) := \frac{L_0}{2}\|x\|^2, \qquad \phi(x) := \widehat{F}(x) + q(x), \qquad g(x) := q(x). \tag{B.9}$$

Since $\widehat{F}$ is L_0-smooth, $\widehat{F} + q$ is convex. Moreover, ϕ is $2L_0$-smooth and g is L_0-smooth. Set the component smoothness budget $L = 2L_0$. The difference satisfies $\phi - g = \widehat{F}$.

The component oracle is equivalent to the standard first-order oracle for $\widehat{F}$ because

$$\widehat{F}(x) = \phi(x) - g(x), \qquad \nabla\widehat{F}(x) = \nabla\phi(x) - \nabla g(x), \tag{B.10}$$

while the reverse simulation is obtained by adding the known quadratic q. Finally, Lemma 1 implies that any returned point with $\delta_\phi, \delta_g \le c\varepsilon^2/L$ and $\eta \le c\varepsilon$ is an ε-stationary point of $\widehat{F}$, after decreasing c if needed. A faster algorithm would contradict the smooth lower bound. □

Remark B.1 (Condition-number phrasing)**.** *If the paper states the relaxed-to-smooth conversion through a strong convexity constant γ of g and a smoothness constant L, a conservative regime is*

$$\delta_g \lesssim \gamma\varepsilon^2/L^2. \tag{B.11}$$

Under that normalization, the same smooth embedding gives a lower bound often written as

$$\Omega\left(\frac{\gamma}{L}\cdot\frac{\Delta}{\delta_g}\right), \tag{B.12}$$

which matches the $O(\Delta/\delta_g)$ upper bound up to the condition-number factor L/γ. If the smooth DC subclass allows g to be a quadratic with curvature of order L, the clean $\Omega(\Delta/\delta)$ form applies.

C ACKNOWLEDGMENTS

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References are important to the reader; therefore, each citation must be complete and correct. If at all possible, references should be commonly available publications.

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